Mastering the TOGAF Standard: Compiled Study Guide

Compiled Study Guide, Chapters 1-22, based on Mastering the TOGAF Standard by Eric Jager (2025)

About This Compilation

This document collects the chapter study guides for all 22 chapters of *Mastering the TOGAF Standard* in one place. Every guide follows the same structure: Overview, Detailed Section Breakdown, Key Terms & Definitions, Figures & Tables Referenced, and Self-Check Questions, with book page references throughout. Use the PDF bookmarks to jump to a chapter.

Two companion guides sit alongside this file: the **Book Summary Guide** (a bird's-eye view of the whole book) and **Connecting the Concepts** (how the ideas link across chapters).

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Chapter 1: Introduction

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 1 (pp. 1-12)

Overview

Chapter 1 (pp. 1-12) explains why Jager wrote the book, who it is for, and how to read it. The opening pages (pp. 1-5) frame the book's core claim: the TOGAF Standard is a very applicable framework that has not been applied well, because people find it too large, too abstract, or too vague about *how* to do things (p. 2). The opening also gives a one-paragraph summary of the whole Standard: at its heart it is about establishing and implementing an **Architecture Capability**, carried out by architecture roles that use skills and competencies (p. 4). Section 1.1 answers the main criticisms of TOGAF (pp. 5-7). Section 1.2 names the two audiences, those who have and have not read the Standard (pp. 7-9). Section 1.3 introduces the four colour-coded **reading tours** (pp. 9-12). Leave with this idea: *tailoring is a must*. Not everything in the Standard has to be used (p. 5). Chapter 1 belongs to the PINK reading tour (p. 1).

> **Reading note:** This chapter is almost entirely Jager's own voice and motivation. The few statements that come from the Standard are quoted with reference numbers: the "essential scaffolding" quote (p. 5) and the note that ADM phase descriptions say *what* to do, with the *how* in the Series Guides (p. 6). Everything else, including his rebuttal of the criticisms, is his opinion.

Detailed Section Breakdown

Chapter Opener (pp. 1-5)

- **What architects can do.** Enterprise architects help organizations translate strategy into an executable whole, show the implications of that strategy, and visualize the impact of change for every level of stakeholder (p. 1).
- **Frameworks as reference works.** Architecture frameworks capture the tested knowledge of many practitioners, so architects "don't have to reinvent the proverbial wheel". They only need to follow and adapt them (pp. 1-2).
- **TOGAF's position.** The TOGAF Standard is described as the world's leading architecture framework, used by tens of thousands of architects for decades (p. 2).
- **Why the book exists.** Jager does not think TOGAF is incomplete. If anything it is "overly complete", somewhat cumbersome, and a little inconsistent. His main reason is that many architects give up on it because of the volume of information and its sometimes "fuzzy" or academic explanation (p. 2).
- **Scope of the book.** It is not a one-to-one description of the Standard. It covers the most important parts, plus extra topics and context he felt were needed (pp. 2-3).
- **Fundamental Content vs. Series Guides.** Since the 10th Edition (April 2022), the Standard is split into the durable **Fundamental Content** (the TOGAF Framework) and the faster-changing **Series Guides**, which give industry-, style-, purpose-, and problem-specific guidance. The book focuses on the Fundamental Content, plus the Series Guides that have proven essential (pp. 3-4; see Ch. 3, Sections 3.2-3.3).
- **TOGAF in a nutshell.** Establish and implement the Architecture Capability, carried out by architecture roles that use architecture skills and competencies (p. 4).
- **The work in four steps.** Understand business needs (goals and objectives); design the architecture (data, integration, technology choices); plan implementation as a flexible, iterative roadmap that

delivers value early; and govern and execute, keeping alignment with the vision while the architecture evolves (pp. 4-5).

- **Tailoring is a must.** Not everything in the Standard needs to be used in its entirety. Keeping this in mind makes the framework "a lot less overwhelming" (p. 5).

1.1 Why This Book (pp. 5-7)

- **TOGAF as de facto standard.** It is the most widely used EA framework. The Standard calls the TOGAF Framework "the essential scaffolding across industries, domains, and styles" (p. 5).
- **Three criticisms he has collected.** (1) Real examples of practical usage are missing. (2) Its prescriptions are vague, because it says the ADM should be adapted without saying how. (3) "Using TOGAF can be best explained as studying TOGAF and then doing something else instead" (p. 5).
- **His response.** The book gives practical examples from daily practice to show TOGAF's added value and to interpret the "vague" prescriptions (p. 6).
- **The criticism he accepts.** The Standard mostly describes *what*, not *how*. ADM phase descriptions recommend what may be done, and the how is left to the Series Guides, which Jager says also under-explain it. He calls this "an omission" (p. 6).
- **Aim.** To show how the Standard is used "practical and pragmatic", with powerful examples, and so debunk the "study TOGAF then do something else" quote (p. 7).

Jager's commentary (author's view): He considers most criticism of TOGAF "mostly unfounded", and he has raised the what-vs-how gap with The Open Group without it changing the framework (pp. 5, 7).

1.2 Who This Book Is For (pp. 7-9)

Audience	Their likely situation	What the book offers
Those who **have** read the Standard	Probably do not apply it daily, or think they don't, because the theory lacks meaningful examples or is too vague to apply	Practical, example-led application (p. 8)
Those who **have not** read the Standard	May be put off by the amount of theory or by negative talk about the framework	An accessible starting point that explains the theory with real-world scenarios (p. 8)

- **Breadth is necessary.** The Standard is extensive because EA's scope is broad, and every aspect must be described with little ambiguity. The book translates that theory into an accessible form (p. 8).

1.3 How to Read This Book (pp. 9-12)

- **Goal: the how, not just the what.** Each major topic is given clear steps and illustrations (p. 9).
- **Reading tours.** Besides reading cover to cover, four alternative routes exist (pp. 9-10, Figure 1-1 on p. 11):

Tour	Purpose	Chapters in this tour (from each chapter's opener)
PINK	Learn the very basics	1, 2, 3, 4, 11, 12, 22
GREEN	Learn the major concepts	3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 17, 20
BLUE	Dive into the details	5, 12, 13, 14, 16, 17, 18, 19
GOLD	Content you won't easily find elsewhere	2, 5, 6, 9, 10, 15, 17, 20, 21

- **Recommended route for newcomers.** Those who have not read the Standard should read the whole book, to see "all the ingenious relationships" between parts of the framework (p. 10).
- **Conventions.** References appear as bracketed numbers. Personal experiences use special formatting. "Further Reading" boxes point to the TOGAF Library or online resources. A steps icon marks short

step-by-step approaches. High-resolution figures can be downloaded from the author's website (p. 12).

- **Diagrams are not strict ArchiMate.** They are illustrative but use the ArchiMate colour scheme for recognizability. All names in examples are fictitious (p. 12).

Key Terms & Definitions

Term	Definition	Page
Architecture framework	A reference work describing how to map and interpret the changes an organization needs, and their impact on desired outcomes	pp. 1-2
TOGAF Framework (Fundamental Content)	The durable, stable core of the TOGAF Standard, and the focus of the book	pp. 3-4
TOGAF Series Guides	Supplementary, fast-evolving guidance (industry-, style-, purpose-, problem-specific) that is no longer a permanent part of the Framework	p. 3
Architecture Capability	What the Standard comes down to: the capability to do architecture, carried out by roles using skills and competencies	p. 4
Tailoring	Adapting the framework and using only the parts that apply to the situation. "A must"	p. 5
Reading tours	Colour-coded alternative routes through the book (PINK, GREEN, BLUE, GOLD)	pp. 9-11

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 1-1	The alternative reading tours and the chapters each one covers	p. 11

Self-Check Questions

Q: According to Jager, what does the TOGAF Standard boil down to "in a nutshell"?

A: Establishing and implementing the Architecture Capability, which is carried out by architecture roles using architecture skills and competencies (p. 4).

Q: Into which two parts did the 10th Edition divide the Standard, and which one does the book focus on?

A: Fundamental Content (the TOGAF Framework) and the Series Guides. The book focuses on the Fundamental Content, plus the Series Guides that have proven essential (pp. 3-4).

Q: Name the four broad steps of architecture work that Jager describes in the opener.

A: Understand business needs; design the architecture; implementation planning (a flexible, iterative roadmap); and governance and execution (pp. 4-5).

Q: Which criticism of TOGAF does Jager agree with?

A: That the Standard mostly describes the *what* and not the *how*. Guidance on how is pushed into the Series Guides, and even there it is weakly articulated (p. 6).

Q: What are the three quoted criticisms of TOGAF?

A: There are no real examples of practical usage; its prescriptions are vague (adapt the ADM without saying how); and "using TOGAF" means studying it and then doing something else (p. 5).

Q: Which reading tour should you follow if you only want to brush up on the basics? And for content you won't easily find elsewhere?

A: PINK for the basics; GOLD for content not easily found elsewhere (pp. 9-10).

Q: Why are the book's diagrams not strictly ArchiMate-compliant?

A: The book explains the TOGAF Standard, not ArchiMate. The diagrams are illustrative only, though they keep the ArchiMate colour scheme for recognizability (p. 12).

Q: Why does Jager say that keeping "tailoring is a must" in mind matters?

A: Because not everything in the Standard needs to be used in its entirety. Remembering this makes the framework feel far less overwhelming (p. 5).

Q: Why are the Series Guides no longer a permanent part of the TOGAF Framework?

A: They evolve much faster than the durable Fundamental Content, so The Open Group separated them (p. 3).

Q: What reading approach does Jager recommend for people who have never read the Standard, and why?

A: Read the book cover to cover, because it reveals all the relationships between the various parts of the framework (p. 10).

Chapter 2: Preliminary Notes

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 2 (pp. 13-30)

Overview

Chapter 2 (pp. 13-30) gives the historical background needed to understand where the TOGAF Standard came from and why it looks the way it does. Section 2.1 traces Enterprise Architecture from the 1960s to today, including the Zachman Framework (pp. 13-16). Section 2.2 follows TOGAF's own evolution: its early annual releases (2.2.1), the Technical/Enterprise Edition years through version 9 and the 10th Edition and its 2025 Technical Corrigendum (2.2.2), and a mapping of TOGAF's ancestor TAFIM onto today's ADM (2.2.3) (pp. 16-29). Section 2.3 summarizes (pp. 29-30). Leave with this idea: EA has moved from an IT-focused discipline to a strategic, business-centric practice, and TOGAF has grown from an enhanced TAFIM into a full EA framework, although its "technical" image has stuck (pp. 16, 22, 30). Chapter 2 is part of the PINK and GOLD reading tours (p. 13).

> **Reading note:** The history is factual narrative. Jager's own remarks are set apart in the text: that TOGAF's IT-framework reputation may come from the "Technical Edition" subtitle (pp. 19, 22), and his speculation that the 10th Edition may be starting to merge Phases B, C, and D (p. 29). Don't confuse the **Zachman Framework** (an ontology, 1987) with TOGAF (a method-based framework, 1995).

Detailed Section Breakdown

Chapter Opener (p. 13)

- **Why history matters.** Knowing TOGAF's origins gives a complete picture of how it evolved and of the moments that shaped its growth (p. 13).
- **Catalyst.** The emergence of architecture frameworks in general was the catalyst for creating and evolving the TOGAF Standard (p. 13).

2.1 A Brief History (pp. 13-16)

Period	What happened in EA	Page
1960s-1970s	EA emerges as large organizations need formal methods to manage complex IT systems and align them with business goals	p. 13
1980s	The term "Enterprise Architecture" takes hold, focused on documenting the structure of information systems. Zachman publishes his framework (1987)	p. 14
1990s	Zachman grows popular. Interest in aligning IT with business grows. The Open Group introduces what is now the TOGAF Standard (1995)	p. 14
Early-to-late 2000s	EA becomes a strategic management discipline. The focus becomes holistic: business processes, people, and organizational structures, not just IT	p. 14
2010s	EA aligns with BPM, data management, and security, and guides digital transformation, cloud adoption, and agile practices	p. 15
Recent years	EA becomes more adaptive and agile in response to digitization, data-driven decisions, AI, IoT, and blockchain	p. 16

- **Zachman Framework.** Published in 1987, it is "actually an ontology". It was the first framework to focus on structuring IT infrastructure and business processes, and it was later extended and also criticized for its practical value (pp. 14-15).

- **Value of EA.** EA aligns IT and business processes with business goals, increases the value of IT, and reduces the complexity, cost, and risk of IT projects (p. 15).
- **Overall arc.** EA has evolved "from primarily IT-focused to strategic and business-centric" (p. 16).

2.2 The Evolution of the TOGAF Standard (pp. 16-29)

- **Today's components.** The ADM, a set of standards, and a set of tools and techniques, aimed at a holistic EA aligned with business objectives (pp. 16-17).
- **Origin.** TOGAF evolved from an enhanced version of **TAFIM** (Technical Architecture Framework for Information Management) into a full EA framework (p. 17).

2.2.1 The Early Versions (pp. 17-18)

Version	Year	Significance
TOGAF 1	1995	Published as the "Open Architecture Framework", then quickly renamed. A proof of concept following a 1994 proof of need (p. 17)
TOGAF 2.0	1996	Annual release cycle. Proof of concept becomes proof of application (p. 17)
TOGAF 3.0	1997	Reaffirms relevance to practical architectures (p. 17)
TOGAF 4	1998	Content accessible via the Internet, which sharply increases users (p. 17)
TOGAF 5	1999	Largely rewritten around a refined, expanded ADM (p. 18)
TOGAF 6	2000	Building blocks incorporated (p. 18; see Ch. 7, Section 7.5.2)

2.2.2 The More Recent Versions (pp. 18-26)

Version	Year	Significance
TOGAF 7 "Technical Edition"	2001	More structured ADM, architecture views, an expanded Business Scenarios topic, and a comparison with other frameworks. Framework = ADM + Foundation Architecture + Resource Base (pp. 18-19)
TOGAF 8 "Enterprise Edition"	2002	Broader EA application. The **Preliminary Phase** was added (then called "Preliminary Framework and Principles"), and the other phases got their current names (pp. 21-22)
TOGAF 8.1.1	2006	Still "Enterprise Edition". TOGAF became a registered trademark a year earlier (p. 23)
TOGAF 9	2009	The last version with a subtitle. Called "revolutionary": modular structure, the **Architecture Content Framework** introduced, and the **Architecture Repository** introduced (p. 24)
TOGAF 9.2	2018	The first significant expansion of the Business Architecture domain (p. 24)
TOGAF Standard, 10th Edition	2022	First version with "Standard" in its name. It leans on other bodies of knowledge, e.g., refers to the BIZBOK Guide for Business Architecture via Series Guides (p. 24)
Technical Corrigendum 1	15 May 2025	Mostly textual clarifications. It removed all arrowheads from the ADM diagram to emphasize iteration (p. 25)

- **Foundation Architecture (TOGAF 7).** Technical Reference Model + Standards Information Base + Building Block Information Base (p. 19, Figure 2-4).
- **Resource Base (TOGAF 7).** Advice on views, governance, business scenarios, principles, and reference models. It is comparable to today's TOGAF Library (p. 20; see Ch. 3, Section 3.4).
- **Direction of travel.** Flexibility, usability, accessibility, business-IT alignment, agile methods, and support for digital transformation (p. 25).

Jager's commentary (author's view): The "Technical Edition" subtitle may be why many still call TOGAF an IT framework. "Undoubtedly, it did not help." Switching to "Enterprise Edition" a year later was a "futile

hope" to erase that memory (pp. 19, 22).

2.2.3 Comparison of TAFIM and TOGAF (pp. 26-29)

TAFIM phase	TOGAF ADM phase
Initiation and Architecture Framework	Preliminary Phase
Baseline Characterization	Architecture Vision
Target Architecture	Business Architecture; Information Systems Architectures; Technology Architecture
Opportunity Identification	Opportunities and Solutions
Migration Options	Migration Planning
Implementation Planning	Implementation Governance
SBA Administration	Architecture Change Management

(Table 2-1, p. 28.) Note that the ADM Requirements Management phase has no TAFIM counterpart in the table.

- **Target Architecture split three ways.** TAFIM's single Target Architecture phase became Phases B, C, and D (p. 28).
- **Planning vs. governance.** TAFIM's Implementation Planning corresponds to TOGAF's Implementation Governance, so the emphasis moved from planning to governance. Planning itself sits in Migration Planning (p. 29).
- **SBA Administration.** "Standards-Based Architecture" administration maps to Architecture Change Management, which keeps the architecture current (p. 29).
- **10th Edition convergence.** Phases B, C, and D are now described identically: create Baseline and Target Architectures for that domain (p. 28).

Jager's commentary (author's view): This may be "a first step toward reversing the division" of B, C, and D. "Whether this will actually happen... remains to be seen" (pp. 28-29).

2.3 Summary (pp. 29-30)

- Frameworks were the catalyst for TOGAF. EA arose in the 1960s-70s from IT complexity. Zachman (1987) came first, and TOGAF followed in 1995 (pp. 29-30).
- TOGAF derives from TAFIM, whose phases are logically similar to the ADM (p. 30).
- EA moved to a holistic, then strategic and business-centric, practice. TOGAF matured into a full EA framework (10th Edition 2022, Corrigendum 2025) (p. 30).

Key Terms & Definitions

Term	Definition	Page
Zachman Framework	John Zachman's 1987 framework ("actually an ontology"), the first to structure IT infrastructure and business processes	p. 14
TAFIM	Technical Architecture Framework for Information Management, the framework TOGAF was originally based on	p. 17
Foundation Architecture	TOGAF 7 component: Technical Reference Model, Standards Information Base, Building Block Information Base	p. 19
Resource Base	TOGAF 7 advice collection (views, governance, scenarios, principles, reference models), comparable to today's TOGAF Library	p. 20

Term	Definition	Page
Preliminary Phase	ADM phase added in TOGAF 8 (2002), originally "Preliminary Framework and Principles"	p. 22
Architecture Content Framework	Introduced in TOGAF 9, connecting all artifacts of the framework	p. 24
Technical Corrigendum 1	15 May 2025 update: textual clarifications, and removal of arrowheads from the ADM diagram	p. 25
SBA Administration	TAFIM's "Standards-Based Architecture" administration phase, the ancestor of Architecture Change Management	p. 29
BIZBOK Guide	The Business Architecture Guild's body of knowledge, which the 10th Edition references for Business Architecture	p. 24

Figures & Tables Referenced

Figure / Table	What it shows	Page
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Figure 2-2	Early versions of the TOGAF Standard	p. 18
Figure 2-3	More recent versions of the TOGAF Standard	p. 18
Figure 2-4	The Foundation Architecture in context of the framework	p. 20
Figure 2-5	The ADM in TOGAF 7 (2001)	p. 21
Figure 2-6	The ADM in TOGAF 8 (2002), with the Preliminary Phase	p. 23
Figure 2-7	The ADM with and without arrowheads	p. 25
Figure 2-8	TAFIM phases and TOGAF ADM phases mapped	p. 27
Table 2-1	Mapping of the TAFIM and TOGAF ADM phases	p. 28

Self-Check Questions

Q: Which framework is TOGAF originally based on?

A: TAFIM, the Technical Architecture Framework for Information Management (pp. 17, 30).

Q: In which version was the Preliminary Phase added, and what was it first called?

A: TOGAF 8 (2002). It was called "Preliminary Framework and Principles" (p. 22).

Q: Which version introduced the Architecture Content Framework and the Architecture Repository?

A: TOGAF 9, released in 2009 (p. 24).

Q: What did the 2025 Technical Corrigendum change visually in the ADM, and why?

A: It removed all arrowheads, to emphasize that the ADM is iterative (p. 25).

Q: TAFIM's Target Architecture phase became which ADM phases?

A: Three phases: Business Architecture (B), Information Systems Architectures (C), and Technology Architecture (D) (p. 28).

Q: Why does Jager think many people still call TOGAF an IT framework?

A: In his view, the "Technical Edition" subtitle given to TOGAF 7 (2001) planted that image, and renaming it "Enterprise Edition" in 2002 did not erase it (pp. 19, 22).

Q: What were the three components of TOGAF 7's Foundation Architecture?

A: The Technical Reference Model, the Standards Information Base, and the Building Block Information Base (p. 19).

Q: How does the 10th Edition handle Business Architecture differently from 9.2?

A: 9.2 tried to build the domain out within TOGAF. The 10th Edition again refers to other bodies of knowledge, notably the BIZBOK Guide, through Series Guides (p. 24).

Q: Why is Zachman described as "actually an ontology"?

A: The book labels it that way to distinguish it from method-based frameworks. It is a classification structure, not a step-by-step method (p. 14). (The book gives the label but no further explanation.)

Q: Summarize how EA's focus has shifted over its history.

A: From documenting IT systems (1980s), to holistic coverage of processes, people, and structure (2000s), to a strategic, business-centric tool for digital transformation (2010s onward) (pp. 14-16, 30).

Chapter 3: The TOGAF Standard

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 3 (pp. 31-52)

Overview

Chapter 3 (pp. 31-52) explains what the TOGAF Standard actually consists of. Section 3.1 defines it as an open, vendor-neutral EA framework whose structure mirrors an organization's Architecture Capability (pp. 32-36). Section 3.2 walks through the six documents of the **Fundamental Content** (pp. 36-40). Section 3.3 covers the faster-moving **Series Guides** (pp. 40-42). Section 3.4 describes the **TOGAF Library** and its categories (pp. 43-46). Section 3.5 lists the key advantages of using the Standard (pp. 46-50), and Section 3.6 summarizes (pp. 50-52). Leave with this idea: the Standard is modular by design. You can adopt the parts you need, starting with the ADM, and tailoring is "more of a necessity than an option" (pp. 39-40, 52). Chapter 3 is part of the PINK and GREEN reading tours (p. 31).

> **Reading note:** Most of this chapter describes the Standard's structure, which is factual. The Netflix/video-store capability example (pp. 34-35) is Jager's own illustration. The advantages in 3.5 are presented as the book's view, supported by one cited study (O'Higgins) (p. 48). Keep three similar names apart: the **TOGAF Library** (guidance portfolio), the **Reference Library** (a class of the Architecture Repository that the TOGAF Library supports), and **The Open Group Library** (which contains the TOGAF Library) (pp. 43-44).

Detailed Section Breakdown

Chapter Opener (p. 31)

- **"Standard" in the name.** The 10th Edition is the first with "Standard" in its title, which fits The Open Group's role as a vendor-neutral standards body and sends a clear message: "this is a standard, for Enterprise Architecture" (p. 31).
- **Composition.** About six core documents, supplemented by Series Guides. The complete set, together with the TOGAF Reference Library, is part of the TOGAF Library (p. 31).

3.1 What Is the TOGAF Standard (pp. 32-36)

- **Definition.** An open EA framework, developed and maintained by The Open Group, describing the general approach to establishing an Enterprise Architecture as a series of independent but closely related documents (p. 32).
- **Why a standard.** The Open Group works with customers, suppliers, and consortia to capture requirements and share best practice. Standards ensure "openness, interoperability, and consensus" (p. 32).
- **Product- and vendor-independent.** It can be applied to any type of architecture in almost any context. The TOGAF Library adds practical, context-specific guidance (p. 33).
- **Structure mirrors the Architecture Capability.** In Figure 3-2, the ADM sits at the centre as "the heart of the framework", surrounded by Architecture Content, techniques, and applications of the method, all inside the Architecture Capability (pp. 33-34; ADM in Ch. 12, capability in Ch. 9).
- **Inputs and outputs.** Market-driven business needs feed the Standard as problems and challenges. The Standard returns business solutions and refined business needs. Both streams update the business vision and drivers (p. 34).

- **Business capabilities.** Changes to the business or operating model often require new business capabilities, which represent the change and provide business value (p. 34).
- **Example: video store to streaming (Figure 3-3).** Store Management and Physical Product Management become Bandwidth Management and Digital Product Management. Contract Management keeps its name but changes content (suppliers become studios). Invoice Management barely changes (pp. 34-35).
- **Versatility.** It supports different architecture styles, integrates with other standards, adapts to industries and use cases, and is highly scalable. Reduced to essentials, it is Fundamental Content plus Series Guides (p. 36).

3.2 Fundamental Content (pp. 36-40)

Document	What it covers	Where the book expands it
Introduction and Core Concepts	Core terms and definitions: the EA definition, architecture domains, a brief ADM overview, deliverables, artifacts, building blocks, the Enterprise Continuum, Architecture Repository, Content Framework	Ch. 4-7, 11 (p. 37)
Architecture Development Method (ADM)	The ADM cycle, adapting it, scoping an architecture problem, integration. "The heart of the TOGAF Standard": flexible, iterative, adaptable	Ch. 12 (p. 37)
ADM Techniques	Techniques for applying the ADM, e.g., Architecture Principles, Stakeholder Management, Architecture Alternatives and Trade-Offs	Ch. 14 (p. 38)
Applying the ADM	Adapting the ADM to process styles (iteration) and specialized architectures such as Security	Ch. 13 (p. 38)
Architecture Content	The Architecture Content Framework: building blocks and artifacts (catalogs, matrices, diagrams, maps), plus the Enterprise, Architecture, and Solutions Continua	Ch. 5-7 (pp. 38-39)
EA Capability and Governance	The Architecture Capability Framework (structure, roles, processes) and a Governance Framework that serves as an adaptable starting point	Ch. 8-10 (p. 39)

- **Building blocks vs. artifacts.** Building blocks are the core components (processes, information concepts, data entities, application systems). Artifacts are the deliverables produced during the ADM, such as catalogs, matrices, diagrams, and maps (pp. 38-39).
- **Why separate documents.** So that specialist areas can be studied and addressed in isolation. They form a cohesive whole but "don't have to be adopted all at once". You can cherry-pick, e.g., use the ADM but skip the Architecture Capability components (pp. 39-40).
- **Two scenarios for gradual adoption.** A new adopter starts with core areas such as the ADM. An organization with an existing framework blends TOGAF parts into what already works (p. 40).

3.3 Series Guides (pp. 40-42)

- **Purpose.** Better guidance for developing *useful* EA that supports stakeholders' decisions and organizational change (p. 40).
- **Range of topics.** Setting up an EA Capability, Business and Security Architecture, agile methods, the digital enterprise, business models, business capabilities, business scenarios, agile delivery via time-boxed increments (Ch. 18), reference models, EA maturity assessment, and project-management guidance (pp. 40-42).
- **Not all are relevant.** "Not all the TOGAF Series Guides will be relevant in every situation", but architects should know what exists (p. 41).

- **Faster evolution.** They are driven by The Open Group Architecture Forum through incremental delivery, so they respond to industry trends faster than the core (p. 41).

3.4 The TOGAF Library (pp. 43-46)

- **What it is.** A portfolio of guidelines, templates, patterns, and reference material that accelerates new architectures. The TOGAF Documentation Set is part of it, and it is itself part of The Open Group Library (p. 43).
- **Link to the Architecture Repository.** It supports the repository's **Reference Library** class (p. 44; see Ch. 7).

Library category	Examples	Page
Fundamental Content (domain-specific guides)	Business Intelligence & Analytics; Metadata Management; Business Capability Planning; Applying the ADM using Agile Sprints	p. 44
Different architecture styles and methodologies	The TOGAF Leader's Guide to Establishing and Evolving an EA Capability; Microservices Architecture	p. 45
Architecture areas/domains	Environmentally Sustainable Information Systems; Architecture Maturity Models; Business Models	p. 45
Use with other standards and frameworks	TOGAF Content Framework with ArchiMate; TOGAF and SABSA integration	p. 45
Specific verticals or use cases	Cloud Ecosystem Reference Model	p. 45
Practical application	"Template Deliverables" set: Capability Assessment, Communications Plan, Architecture Roadmap templates	p. 45

- All content is downloadable at no additional cost (p. 45).

3.5 Key Advantages of the TOGAF Standard (pp. 46-50)

Advantage	What it means	Page
Comprehensive coverage of domains	Business, Data, Application, and Technology are all addressed, giving a unified, coherent approach	pp. 46-47
Adaptability and tailoring	The ADM is meant to be adapted to the enterprise and to architecture styles. It is iterative (Ch. 13.3) and customizable (Ch. 17)	pp. 47-48
Business and IT alignment	Structured steps bridge the gap, e.g., mapping processes to find redundancies and gaps. Stakeholder engagement (Ch. 14.3) reduces misalignment	p. 48
Facilitating digital transformation	Assess the current state, envision the future state, and roadmap between them. O'Higgins' study links good Business Architecture to transformation success	pp. 48-49
Process innovation	Clear insight into existing processes reveals areas ripe for innovation	p. 49
Common language	Shared vocabulary reduces misunderstanding and supports decision-making	p. 49
Scalability and resource optimization	Modular, so it applies from small projects to enterprise-wide initiatives and avoids ad hoc planning	p. 49

- **The four domains in brief.** Business (strategy, governance, organization, key capabilities); Data (logical and physical data assets and their management); Application (blueprint of systems, their interactions, and their relation to business capabilities); Technology (hardware, software, network infrastructure) (p. 47).

> **Cross-reference note:** p. 48 points to "Chapter 18" for digital transformation. In this book, digital transformation is Chapter 19 and Chapter 18 is Agile Architecture.

3.6 Summary (pp. 50-52)

- The Standard is an open, product- and vendor-independent framework of Fundamental Content (six documents) plus Series Guides, structured to mirror an Architecture Capability (p. 50).
- The TOGAF Library, part of The Open Group Library, helps organizations customize the framework (p. 51).
- Frameworks save organizations from reinventing the wheel, but every organization is unique, so tailoring "is more of a necessity than an option". Selecting the parts that add value saves time, money, and resources (p. 52).

Key Terms & Definitions

Term	Definition	Page
TOGAF Standard	An open, vendor-neutral EA framework by The Open Group, describing the general approach to establishing an EA	p. 32
Fundamental Content	The six core documents of the Standard	p. 36
Series Guides	Supplementary, faster-evolving guidance on specific concerns, styles, and use cases	pp. 40-41
TOGAF Library	Portfolio of guidelines, templates, patterns, and reference material supporting practical application	p. 43
Reference Library	A class of information in the Architecture Repository that the TOGAF Library supports	p. 44
Building block	A core component of the architecture (e.g., business process, data entity, application system)	p. 38
Artifact	A deliverable component produced during ADM phases: catalog, matrix, diagram, or map	p. 39
Enterprise / Architecture / Solutions Continua	Classification systems for categorizing architectural assets	p. 39
Architecture Capability Framework	Organizational structure, roles, and processes needed to run an EA Capability	p. 39
Business capability	What a business does, often created or changed when the business or operating model changes	p. 34

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 3-1	The TOGAF Standard as a set of documents	p. 32
Figure 3-2	Structure of the TOGAF Standard in relation to the Architecture Capability	p. 33
Figure 3-3	Capabilities before and after a video store's digital transformation	p. 35
Figure 3-4	The TOGAF Library	p. 43
Figure 3-5	The TOGAF Reference Library categories	p. 44

Self-Check Questions

Q: Name the six documents of the Fundamental Content.

A: Introduction and Core Concepts; Architecture Development Method; ADM Techniques; Applying the ADM; Architecture Content; EA Capability and Governance (p. 36).

Q: What sits at the centre of the Standard's structure (Figure 3-2), and what surrounds it?

A: The ADM, "the heart of the framework". It is supported by Architecture Content, techniques, and applications of the method, within the Architecture Capability (pp. 33-34).

Q: Must an organization adopt all six documents at once?

A: No. They can be cherry-picked, e.g., the ADM without the Architecture Capability components. A new adopter can start with the ADM, and an organization with an existing framework can blend TOGAF parts in (pp. 39-40).

Q: What is the difference between building blocks and artifacts?

A: Building blocks are the core components that make up the architecture (processes, data entities, applications). Artifacts are the catalogs, matrices, diagrams, and maps produced during ADM phases to describe it (pp. 38-39).

Q: Why do Series Guides change faster than the Fundamental Content?

A: They are produced by The Open Group Architecture Forum through a streamlined, incremental process that responds to industry trends and market demand (p. 41).

Q: Which Architecture Repository class does the TOGAF Library support?

A: The Reference Library (p. 44).

Q: In the video-store example, why is Contract Management drawn with a dashed line?

A: Its name stays the same but its content changes: agreements with DVD/Blu-ray suppliers become agreements with film and series studios (p. 35).

Q: Which TOGAF Library category holds the "Template Deliverables" set, and what does it include?

A: Practical Application. It includes templates for a Capability Assessment, a Communications Plan, and an Architecture Roadmap (p. 45).

Q: What are the four architecture domains, and what does each describe?

A: Business (strategy, governance, organization, capabilities), Data (logical and physical data assets), Application (systems, their interactions, and their relation to capabilities), and Technology (hardware, software, network) (p. 47).

Q: Why does the chapter say tailoring is "more of a necessity than an option"?

A: Because every organization is unique. Selecting the framework parts that add value lets the organization apply the approach to its own situation, saving time, money, and resources (p. 52).

Q: What do the business-needs input and output streams around the Standard contribute to?

A: Business needs from the market go in as problems and challenges. Business solutions and refined needs come out. Both streams form and update the business vision and drivers (p. 34).

Chapter 4: Architecture Definition

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 4 (pp. 53-62)

Overview

Chapter 4 (pp. 53-62) asks what Enterprise Architecture actually *is*, and shows that even the TOGAF Standard does not settle on one answer. Section 4.1 gives the Standard's ISO/IEC/IEEE 42010:2022-based definition and its supplementary second sentence, then argues that a key element is missing: the link to business objectives (pp. 54-56). Section 4.2 separates Enterprise Architecture from IT Architecture (p. 57). Section 4.3 shows that the Standard effectively uses three definitions: ISO, Gartner (via the Leader's Guide), and FEAPO (via the BIZBOK Guide) (pp. 58-61). Section 4.4 summarizes (pp. 61-62). Leave with this idea: because no single definitive definition exists, each organization must define how *it* practices EA, so there is no internal ambiguity (pp. 61, 62). Chapter 4 is part of the PINK and GREEN reading tours (p. 53).

> **Reading note:** The quoted definitions come from the Standard, the Leader's Guide (Gartner), and BIZBOK (FEAPO). Jager's own views are: the Standard's definition lacks the business-objective link (p. 55), his own description of EA as "a framework for understanding and managing the overall structure and strategy" (p. 56), the claim that there is "absolutely nothing technical" to EA (p. 57), and his worry that the 2008 Gartner definition is outdated (p. 58).

Detailed Section Breakdown

Chapter Opener (p. 53)

- **Why definitions matter.** A definition gives an unambiguous description. Leaving room for interpretation creates different perceptions "with all the consequences that entails" (p. 53).

4.1 What Is Enterprise Architecture (pp. 54-56)

- **The Standard's definition (ISO/IEC/IEEE 42010:2022, adopted with Technical Corrigendum 1, 2025).** "The fundamental concepts or properties of an entity in its environment and governing principles for the realization and evolution of this entity and its related lifecycle processes" (p. 54).
- **The supplementary "second definition".** "The structure of components, their interrelationships, and the principles and guidelines governing their design and evolution over time" (p. 54).
- **What they imply together.** Architecture defines the characteristics of a system, and frameworks surround the system to oversee its development (p. 54).
- **"System" is broad.** In TOGAF, "system" is not limited to an IT system. It can be an enterprise, something larger, or something as small as a project, which leaves plenty of room for interpretation (p. 54).
- **Greenslade's 2003 definition.** Chris Greenslade (then Chairman of The Open Group) defined (IT) architecture as "the fundamental organization of a system, embodied in its components, their relationships to each other and the environment, and the principles governing its design and evolution", practically represented in Architectural Descriptions from stakeholders' viewpoints (p. 55).
- **The missing component.** The current definition lacks the relation to achieving business objectives. EA must ensure that objectives and intended results are achieved, and it must ensure coherence from organizational structures to information systems and from capabilities to applications (p. 55).

Jager's commentary (author's view): EA "is perhaps best described as a framework for understanding and managing the overall structure and strategy of an organization". It creates a holistic view of business processes, information systems, and technology, keeps them coherent, aligns them to goals, resolves inconsistencies, and enables planning for growth (p. 56).

4.2 Confusion Over Definitions (p. 57)

	Enterprise Architecture	IT Architecture
Purpose	Enables the organization to achieve its goals and objectives, realizing its strategy	Far more technical in nature
Answers	Stakeholder concerns and questions about the portfolio or specific projects	Technical design questions
Diagrams and matrices	Answer stakeholders' questions. They do not illustrate technical solutions	Illustrate technical solutions
Role	Oversees all domains (Technology is one) and coordinates. A "strategic business management tool"	Executes the Technology Architecture (IT Architects)

- **Why the confusion exists.** EA originated "in the technology corner" (see Ch. 2, Section 2.2), and the two are still lumped together (p. 57).
- **No ultimate definition.** The Standard's definition is one of many, which is perhaps why even the Standard uses more than one (p. 57).

Jager's commentary (author's view): The two are "as different as night and day", and "there is absolutely nothing technical to Enterprise Architecture" (p. 57).

4.3 Multiple Definitions (pp. 58-61)

#	Source in TOGAF	Original author	Definition (abridged)	Page
1	Core Concepts (Fundamental Content)	ISO/IEC/IEEE 42010:2022	Fundamental concepts/properties of an entity in its environment, plus governing principles for its realization and evolution	pp. 54, 58
2	Leader's Guide (Series Guide)	Gartner (2008)	"The process of translating business vision and strategy into effective enterprise change by creating, communicating, and improving the key principles and models that describe the enterprise's future state and enable its evolution"	p. 58
3	BIZBOK Guide (referenced body of knowledge)	FEAPO	"The holistic planning, analysis, design, and implementation for the development and execution of strategy by applying principles and practices to guide organizations through the integration and interoperation of all other architecture domains"	p. 60

- **Figure 4-1, how to read it.** EA at the centre. Around it are the organizations that define it (The Open Group, Business Architecture Guild, FEAPO, Gartner, ISO). Blue = sources where definitions are found. Yellow = specific references. Purple = the definitions of EA, Business Architecture, or Architecture (pp. 59).
- **What the FEAPO definition gets right.** It links EA to developing and executing strategy, and therefore to business goals and objectives, which the Standard's own definition lacks (p. 60).
- **The apple analogy.** "A round piece of fruit with a stem" could equally describe a cherry or a grape. Definitions keep us from "comparing apples and oranges" (pp. 60-61).
- **Conclusion.** EA manifests at different levels and in different ways, so it resists a single definition. Organizations must define how they practice EA (p. 61).

Jager's commentary (author's view): Using a 2008 definition in a framework updated in 2025 "might raise a few eyebrows". It may be outdated or superseded (p. 58).

4.4 Summary (pp. 61-62)

- **Three definitions in one standard:** ISO/IEC/IEEE 42010:2022-based (the Standard adds a second sentence to complete it), Gartner via the Leader's Guide, and FEAPO via the BIZBOK Guide (pp. 61-62).
- The Standard's own definition misses the realization of business goals. The Gartner definition risks being outdated. The BIZBOK/FEAPO definition is adopted indirectly because TOGAF refers to BIZBOK for Business Architecture (p. 62).
- **Takeaway.** Define how EA is practiced in your organization to avoid ambiguity (p. 62).

Key Terms & Definitions

Term	Definition	Page
Architecture (TOGAF / ISO 42010:2022)	Fundamental concepts or properties of an entity in its environment and governing principles for its realization and evolution and related lifecycle processes	p. 54
Architecture (TOGAF supplement)	The structure of components, their interrelationships, and the principles and guidelines governing their design and evolution over time	p. 54
System	In TOGAF, any entity: an enterprise, something larger, or a project. Not just an IT system	p. 54
Enterprise Architecture (Gartner)	Translating business vision and strategy into effective enterprise change via principles and models of the future state	p. 58
Enterprise Architecture (FEAPO/BIZBOK)	Holistic planning, analysis, design, and implementation for developing and executing strategy across all architecture domains	p. 60
IT Architecture	A far more technical discipline that executes Technology Architecture, coordinated by EA	p. 57
FEAPO	Federation of Enterprise Architecture Professional Organizations	p. 59

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 4-1	The many definitions of EA and their source organizations and documents	p. 59

Self-Check Questions

Q: Which external standard does the TOGAF Standard's definition of architecture come from, and since when?

A: ISO/IEC/IEEE 42010:2022, adopted with Technical Corrigendum 1 (2025) (p. 54).

Q: What key element does Jager say is missing from the Standard's definition?

A: Its relation to achieving business objectives, i.e., EA ensuring that goals and intended results are realized (pp. 55, 62).

Q: What can "system" mean in the TOGAF definition?

A: An enterprise, something even larger, or something smaller such as a project. It is not limited to IT systems (p. 54).

Q: List the three EA definitions used within the TOGAF Standard and where each comes from.

A: (1) The ISO/IEC/IEEE 42010:2022-based definition in Core Concepts; (2) Gartner's 2008 definition in the Leader's Guide; (3) FEAPO's definition via the BIZBOK Guide (pp. 61-62).

Q: Why does the FEAPO definition appeal to Jager?

A: It ties EA explicitly to the development and execution of strategy, and so to business goals, which the Standard's definition lacks (p. 60).

Q: How does EA differ from IT Architecture?

A: EA enables the organization to achieve its goals and strategy, oversees and coordinates all domains, and answers stakeholder concerns. IT Architecture is technical and executes the Technology Architecture (p. 57).

Q: What is the point of the apple analogy?

A: A loose description ("round fruit with a stem") fits cherries and grapes too. Precise definitions are needed to know what something is and is not (pp. 60-61).

Q: Given that no definitive definition exists, what does the chapter recommend?

A: Define how EA is practiced or represented in your own organization so that there is no internal ambiguity (pp. 61-62).

Q: Why does Jager question the Gartner definition?

A: It dates from 2008 yet appears in a framework updated in 2025, so it may be outdated or superseded (p. 58).

Q: How did Greenslade's 2003 definition say architecture is represented in practice?

A: In Architectural Descriptions from the viewpoints of the stakeholders (p. 55).

Chapter 5: Enterprise Metamodel

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 5 (pp. 63-110)

Overview

Chapter 5 (pp. 63-110) explains the **Enterprise Metamodel**: the set of entities and relationships in which every artifact of the Content Framework is rooted (p. 63). Section 5.1 describes the metamodel's five sections and all 36 of its entities: 8 General Entities (5.1.1), 19 Business Architecture entities (5.1.2), and 3 entities each for Data (5.1.3), Application (5.1.4), and Technology (5.1.5) (pp. 63-101). It illustrates them with a running example of a fictional healthcare organization. Section 5.2 shows how the metamodel, the Content Framework, and the ADM are "inextricably linked", and gives five steps for tailoring the metamodel in the Preliminary Phase (pp. 101-107). Section 5.3 summarizes (pp. 107-110). Leave with this idea: the TOGAF metamodel is a *Foundation* Enterprise Metamodel, a starting point that must be tailored before you build the Content Framework (pp. 63, 106). Chapter 5 is part of the GREEN, BLUE, and GOLD reading tours (p. 63).

> **Reading note:** Entity definitions in Tables 5-1 to 5-5 come from the Standard. The longer explanations, the healthcare/HR recruitment examples, and two points are Jager's own: his strong suggestion to add a **stakeholder** entity (p. 104), and his observation that the **business information** entity may signal the return of an Information Architecture domain (p. 87). Easy to confuse: **capability** (general entity: an ability) vs. **business capability** (an ability *with a purpose*); **value stream** vs. **process**; **function** vs. **process**; **actor** vs. **organization unit**.

Detailed Section Breakdown

Chapter Opener (p. 63)

- **Root of all content.** Every entity in the (Extended) Content Framework is rooted in the Enterprise Metamodel, making it fundamental to the Standard (p. 63).
- **Foundation, not final.** No two organizations are alike, so the metamodel is a Foundation Enterprise Metamodel to be tailored, i.e., a starting point (p. 63).
- **Sections, not layers.** The Content Framework categorizes the metamodel into five interrelated sections. This is not an architecture diagram of domains or layers (p. 63).

5.1 Sections of the Enterprise Metamodel (pp. 63-65)

- **What it contains.** The most common entities and relationships likely to occur in most organizations. It gives context to the artifacts named in the ADM phase descriptions (pp. 63-64).
- **Kept in sync with the Content Framework.** The metamodel supplies the entities, the Content Framework defines work products from them, and the Content Framework evolves the metamodel (p. 64).
- **Five sections.** General Entities, Business Architecture, Data Architecture, Application Architecture, Technology Architecture (p. 64, Figure 5-1 on p. 65).

5.1.1 General Entities (pp. 65-73)

Eight entities so generic that they can be related to any domain (pp. 65-67):

Entity	Definition (Table 5-1)	Key explanation and example
Principle	Qualitative statement of intent the architecture should meet, with a name, rationale, and implications	Like a country's constitution: high-level, strategic, binding, rarely changed, based on business goals. Architecture and Enterprise Principles (Ch. 14.2) (p. 68)
Constraint	An (mostly external) factor preventing particular approaches to goals	Example: a staffing agency's standard job profiles prevent creating a new role (p. 68)
Requirement	Quantitative statement of a business need that a particular architecture or work package must meet	Specifies the objective. Guides design, development, and validation (p. 68)
Assumption	Statement of probable fact not yet fully validated, due to external constraints	Lets decisions proceed despite uncertainty, e.g., assuming an existing app supports certain requirements (pp. 69-70)
Location	A place where activities occur. Can be composed and decomposed	Physical sites or regions. Also data centres in IT distribution diagrams (p. 70)
Gap	Statement of difference between two states	Result of gap analysis in Phases B, C, D. Input to migration planning (p. 70, Figure 5-4)
Work package	Set of actions to achieve business objectives. Can be part of a project, a project, or a program	Also called an *initiative*. Closes gaps from Baseline through Transition to Target (pp. 71-72)
Capability	An ability that an organization, person, or system possesses	Central to capability-based planning: what an organization *can do*, not what it *has*. Airplane (flies), motorcycle (weaves through traffic) (p. 72)

- **Principle-requirement-constraint relationship (Figure 5-3).** Two principles frame an objective. One is detailed by a requirement and implemented by an initiative. The other is limited by a constraint, and an initiative is linked to the constraint to try to remove it (p. 69).
- **Gap example (Figure 5-4).** Application A lacks needed functionality. Application B is introduced alongside it, data and interfaces migrate, then A is decommissioned. This illustrates gaps across Baseline, Transition, and Target (p. 71).
- **Capability vs. business capability.** A business capability is an ability "to achieve a specific purpose or outcome". The added purpose makes it actionable and more meaningful, because "simply having an ability says nothing about its usefulness" (p. 73).
- **Target is not definitive.** The "desired future state" is a state at a point in time, not a final state (p. 70, footnote).

5.1.2 Business Architecture (pp. 74-92)

The largest section: 19 entities (p. 74, Table 5-2 on pp. 74-76).

Entity	Definition (abridged)	Explanation / example	Page
Driver	External or internal condition motivating the organization to define goals	Market, legislation, tech, leadership changes. Stakeholder-related drivers are called *concerns*	p. 77
Goal	High-level statement of intent, used to measure success	Qualitative: "increase", "improve"	p. 77
Objective	Aim declared in a SMART way	Details a goal, e.g., "comply with legislation by Q3 2028"	p. 77
Measure	Indicator tracked to determine success or alignment	Three types: KPIs, metrics, target values. Example: the Q3 Legal Department Report	p. 78

Entity	Definition (abridged)	Explanation / example	Page
Course of action	Direction from strategic goals and objectives, often to deliver the value proposition	What the organization *decided to do*. Strategies (long-term, broad) vs. tactics (short-term, narrow)	p. 79
Business capability	Ability a business may possess or exchange to achieve a purpose	See 5.1.1	p. 80
Value stream	End-to-end activities creating an overall result for a customer, stakeholder, or end user	Made of stages and supported by business capabilities. Example: Acquire Talent	pp. 80-83
Function	Business behaviours grouped by chosen criteria, close-coupled to org units	Logical grouping of activities. *What* is needed, broader than process	p. 83
Process	Sequence of activities achieving an outcome, decomposable	*How* things get done step by step	pp. 83-84
Organization unit	Self-contained unit of resources with goals, objectives, measures	Like a department. With goals and objectives it behaves like a *stakeholder*	pp. 84-85
Actor	Person, organization, or system with a role that initiates or interacts with activities	Can be a whole organization but not a department (that's an org unit)	p. 85
Role	Usual or expected behaviour of an actor. An actor may have several roles	Example: two HR staff with the role HR Business Partner	pp. 85-86
Product	Outcome offered to customers (materials and/or services)	Example: an employment agreement	p. 86
Contract	Agreement between consumer and provider setting functional and non-functional parameters	Formalizes roles, responsibilities, and deliverables	p. 87
Business information	A concept and its semantics used within the business	Links to BIZBOK's *information concept*. Realized by the data entity	pp. 87-89
Control	Decision step with logic to set a process's execution approach or ensure compliance	Used for governance, compliance, monitoring, risk, and change management	pp. 89-90
Event	Organizational state change that triggers processing	Example: "candidate selected" triggers Employee Hiring	p. 90
Service quality	Non-functional requirements assigned to a business, application, or technology service	Reliability, responsiveness, satisfaction. Tied to IT service management	p. 91
Business service	Encapsulates a unique element of business behaviour	The link between Business and the Data/Application/Technology architectures	p. 91

- **Driver to course of action chain (Figure 5-6).** Driver: new legislation. Goal: implement measures. Objective: by Q3. Measure: legal department report. Course of action: acquire staff (pp. 79-80).
- **Value stream vs. business process.** A value stream reflects the *business model and value proposition*: overall value-creating behaviour, not time-ordered tasks. A process reflects the *operating model*: a time-ordered sequence for an individual case, with decision points (pp. 80-81).
- **Acquire Talent example (Figures 5-7 to 5-9).** Stages: Sourcing and Attraction, Selection and Assessment, Offer and Onboarding. Each is supported by process steps from the parent processes Acquiring Talent, Evaluating Candidates, and Employee Hiring (pp. 81-84).
- **Value is intangible.** The value stream delivers *hired talent*, which is intangible. The process delivers a signed employee agreement, which is tangible. "Value streams never deliver a tangible end product. They deliver value", and value is "in the eye of the beholder" (p. 82).
- **Function vs. process (Figure 5-10).** Functions are broader and more abstract and contain processes. They are tied to activities, not departments (pp. 83-84).

- **Business service as hinge.** It sits between the strategic and tactical architectures and the operational ones (p. 91).

Jager's commentary (author's view): The business information entity's link to BIZBOK's information concept suggests a possible return of the Information Architecture domain, which disappeared after TOGAF 8 (p. 87). From a TOGAF perspective, an organization unit related to goals, objectives, and measures is best represented as a stakeholder (pp. 84-85).

5.1.3 Data Architecture (pp. 92-95)

Entity	Definition	Explanation
Data entity	Data the business recognizes as a distinct concept	Modelled in ERDs or class diagrams. Accessed by a business service, consumed by an actor, and realizes business information (p. 93)
Logical data component	Data structure of logically related data entities	Technology-agnostic: *what* the data is (p. 94)
Physical data component	Realizes logical data components in a technology's format or schema	Databases, warehouses, cloud storage (p. 95)

- **Data to information (Figure 5-15).** The numbers 22, 23, 22, 25, 21 become information only with context ("next five days' temperatures"). Linking a data entity to business information turns data into value (p. 94).
- **Equivalence.** According to the Standard, the data entity corresponds to BIZBOK's information concept (p. 89).

5.1.4 Application Architecture (pp. 96-98)

Entity	Definition	Explanation
Application service	The automated elements of a business service	Delivers or supports all or part of one or more business services (p. 97)
Logical application component	Application functionality defined by services offered and data maintained, independent of technology	E.g., "HR recruitment system" without choosing a product (pp. 97-98)
Physical application component	Realization of logical functionality via applications that may be hired, procured, or built	Installed software, configuration, deployment (p. 98)

- **Relationship (Figure 5-18).** The logical component implements the application service, which supports a business service and may use data entities. The product choice happens in physical design (pp. 97-98).

5.1.5 Technology Architecture (pp. 98-101)

Entity	Definition	Example
Technology service	Technical capability providing infrastructure that supports applications	Cloud storage service (p. 99)
Logical technology component	Implementation-independent encapsulation of technology services	"DBMS" without naming a product (p. 100)
Physical technology component	Realization using a particular deployable technology product	MySQL realizing the DBMS (pp. 100-101)

- **Logical-to-physical pattern.** The same pattern recurs in Data, Application, and Technology: service, then logical component (what), then physical component (which product) (pp. 94-101).

5.2 Tailoring the Enterprise Metamodel (pp. 101-107)

- **The triangle (Figure 5-21).** ADM, Enterprise Metamodel, and Content Framework are inextricably linked. The ADM (Preliminary Phase) creates the organization-specific metamodel. That metamodel feeds the Content Framework. The Content Framework's deliverables are then produced in the ADM phases (pp. 101-105).
- **Add and remove.** Architects may drop irrelevant entities and relationships or add new ones. The Standard "does not aim to constrain an enterprise's selection of artifacts" (p. 103).
- **Timing.** This happens in the Preliminary Phase, the first ADM iteration, while the EA Capability is established. Many architects overlook this step (pp. 103-104).
- **Suggested additions.** Stakeholder, business collaboration, business interaction, business interface, resource, risk, and value. Stakeholder is used in the Stakeholder Catalog, and most of the others appear in the Business Model Diagram (Ch. 12.4.2) (p. 105).

Step	What to do	Page
1. Understand the Enterprise Metamodel	Learn the Foundation Enterprise Metamodel's standard entities and relationships	p. 106
2. Customize the metamodel	Add exceptional entities unique to the organization	p. 106
3. Integrate the customized metamodel	Use it as input to the Content Framework, adding or removing deliverables	p. 106
4. Align deliverables	Produce ADM deliverables according to the customized Content Framework	p. 106
5. Continuously evolve	Regularly review and update both the metamodel and the Content Framework	p. 106

Jager's commentary (author's view): He "strongly" suggests adding a **stakeholder** entity. The Standard implicitly uses the organization unit for this, while ArchiMate has an explicit stakeholder element, and adding it improves TOGAF-ArchiMate integration (p. 104). "Tailoring the Enterprise Metamodel is an absolute necessity" (p. 106).

5.3 Summary (pp. 107-110)

- Five sections, 36 entities: 8 general, 19 business, and 3 each for data, application, and technology (pp. 107-108).
- **TOGAF vs. ArchiMate.** ArchiMate classes driver, goal, objective, and measure as *motivation elements*. TOGAF puts them all under Business Architecture because it has no separate motivation or strategy layer, only the four domains (p. 108).
- The metamodel is input to both the Content Framework and the ADM, and it takes time and care to tailor (p. 110).

Key Terms & Definitions

Term	Definition	Page
Enterprise Metamodel	Model of the common entities and relationships in which all architecture content is rooted	pp. 63-64
Foundation Enterprise Metamodel	TOGAF's standard starting set of 36 entities, to be tailored	pp. 63, 108
Entity	An architecture concept with its own definition, belonging to one metamodel section	p. 65

Term	Definition	Page
General Entities	Eight cross-domain entities: principle, constraint, requirement, assumption, location, gap, work package, capability	pp. 65-67
Business capability	An ability a business may possess or exchange to achieve a specific purpose or outcome	p. 73
Value stream	End-to-end activities creating an overall result for a customer, stakeholder, or end user, made of stages	p. 80
Concern	A stakeholder-related driver	p. 77
SMART	Specific, Measurable, Achievable, Realistic, Timebound	p. 77
Business service	Entity linking the Business Architecture to the Data, Application, and Technology Architectures	p. 91
Information concept (BIZBOK)	High-level abstraction of information the business uses, which corresponds to the TOGAF data entity	pp. 88-89
Logical vs. physical component	Technology-independent description of *what* vs. the concrete realization by a product	pp. 94-101

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 5-1	The Foundation Enterprise Metamodel	p. 65
Figure 5-2 / Table 5-1	General Entities	pp. 66-67
Figure 5-3	Principle, requirement, constraint, and initiatives framing an objective	p. 69
Figure 5-4	Gaps between Baseline, Transition, and Target Architectures	p. 71
Figure 5-5 / Table 5-2	Business Architecture entities	pp. 74-76
Figure 5-6	Driver, goal, objective, measure, course of action	p. 79
Figures 5-7 to 5-10	Acquire Talent value stream, processes, and functions	pp. 81-84
Figure 5-11	Actors in an organization unit with an assigned role	p. 86
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Figure 5-19 / Table 5-5, Figure 5-20	Technology Architecture entities and relationships	pp. 98-100
Figure 5-21	Relation between ADM, Enterprise Metamodel, and Content Framework	p. 102

Self-Check Questions

Q: Name the five sections of the Enterprise Metamodel and how many entities each has.

A: General Entities (8), Business Architecture (19), Data (3), Application (3), Technology (3), for 36 in total (pp. 64, 107-108).

Q: How does a business capability differ from a capability?

A: A business capability is an ability "to achieve a specific purpose or outcome". The purpose makes it actionable and valuable, whereas a plain capability is just an ability (p. 73).

Q: How does a value stream differ from a business process?

A: A value stream shows overall value creation from the stakeholder's perspective and reflects the business model. It is not time-ordered and delivers intangible value. A process is a time-ordered sequence for an individual case, reflects the operating model, and can produce tangible outputs (pp. 80-82).

Q: Trace the Figure 5-6 example from driver to course of action.

A: Driver: new legislation. Goal: implement measures. Objective: done by Q3. Measure: a Q3 Legal Department Report. Course of action: acquire the staff needed (pp. 79-80).

Q: Which entity connects the Business Architecture to the other three domains?

A: The business service (pp. 91-92).

Q: What are the three types of measures?

A: KPIs (outcome-focused, tied to CSFs), metrics (broader, operational), and target values (desired performance levels) (p. 78).

Q: When and where is the organization-specific metamodel created?

A: In the Preliminary Phase, during the first ADM iteration, while the Architecture Capability is established (pp. 101, 103-104).

Q: What entities does Jager suggest adding to the metamodel, and why stakeholder in particular?

A: Stakeholder, business collaboration, business interaction, business interface, resource, risk, and value. Stakeholder is used in the Stakeholder Catalog, the Standard only covers it implicitly through organization unit, and adding it improves alignment with ArchiMate (pp. 104-105).

Q: Why does TOGAF put driver, goal, objective, and measure under Business Architecture when ArchiMate calls them motivation elements?

A: TOGAF has no separate motivational or strategic elements. It organizes everything by the four architecture domains (p. 108).

Q: List the five steps for tailoring the Enterprise Metamodel.

A: Understand the metamodel; customize it; integrate it into the Content Framework; align deliverables during the ADM; continuously evolve (p. 106).

Q: In the DBMS example, what are the logical and physical technology components?

A: Logical: "Database Management System", with no product named. Physical: MySQL, the product that realizes it (p. 100).

Q: What is the difference between an actor and an organization unit?

A: An actor can represent a person, system, or an entire organization, but not a department within it. Departments are organization units (p. 85).

Chapter 6: Content Framework

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 6 (pp. 111-126)

Overview

Chapter 6 (pp. 111-126) explains the **Content Framework**, best thought of as a *categorization framework* that structures the architecture's work products and models and maps them to the ADM phases (p. 111). Section 6.1 places it alongside the Enterprise Metamodel and lists its five purposes (pp. 111-114). Section 6.2 lists its advantages in the current Standard: modular, core and extended content models, artifacts, views and viewpoints, agile alignment, tailoring guidance, outcome focus (pp. 114-117). Section 6.3 maps its sections to the ADM phases and explains the **Extended Content Framework** (pp. 117-122). Section 6.4 shows how the ADM refines the framework over time, with a recommended extra artifact, the Work Package Portfolio Map (pp. 123-125). Section 6.5 summarizes (pp. 125-126). Leave with this idea: the metamodel says *which entities* exist, the Content Framework says *which work products* use them and in *which ADM phase*. It describes what the architecture should look like when complete (pp. 113, 119). Chapter 6 is part of the GREEN and GOLD reading tours (p. 111).

> **Reading note:** The five purposes and the section-to-phase mapping are the Standard's. Jager's own points are: the placement of driver, goal, objective, and measure in a separate "Motivation" section is "one of the inconsistencies in the TOGAF Standard" (p. 121), and his recommendation to add a **Work Package Portfolio Map** (p. 124). Don't confuse the **Content Framework** (compact, phase-level) with the **Extended Content Framework** (shows the entities per section).

Detailed Section Breakdown

Chapter Opener (p. 111)

- **Purpose in one line.** Structure the work products and models that make up the architecture, promote consistent deliverables across ADM phases, provide a checklist of possible deliverables, reduce gaps, and standardize concepts, terms, and deliverables (p. 111).

6.1 A Categorization Framework (pp. 111-114)

- **ADM recap.** Eight main phases preceded by a Preliminary Phase, with Requirements Management at the centre. Requirements Management is "not a separate phase, but a mechanism" for continuously testing the architecture. Every phase except Requirements Management describes work products (pp. 111-112).
- **Tailor the Standard.** It should represent the organization in which it is used (p. 112).
- **Three things to define in the Preliminary Phase:**

#	What to define	Represented by
1	Understanding of the types of entities and their relationships	The Enterprise Metamodel (formal model)
2	A categorization framework for descriptions, work products, and models	The Content Framework
3	The specific artifacts to develop	Chosen once 1 and 2 exist

(p. 112)

- **Relation to the metamodel.** The metamodel is a detailed representation of entities. The Content Framework is a high-level summarization, i.e., a categorization framework (p. 113).

The Content Framework is intended to...	Explanation	Page
Describe a detailed model of architecture products	They differ per organization, so tailor	p. 113
Promote consistency and coherence of deliverables	The same deliverable is always developed the same way, with entities meaning the same thing	p. 113
Provide a comprehensive checklist of deliverables	Per entity category, what can or should be produced	p. 114
Reduce the risk of gaps in final deliverables	Which models use which entities is decided in advance, provided it is filled in carefully	p. 114
Help use standard concepts, terms, and deliverables	Predefined, defined entities remove ambiguity	p. 114

6.2 Advantages of the Content Framework (pp. 114-117)

Aspect	What it means	Page
Modular structure	Adopt only the relevant parts	p. 115
Core content model	Defines the building blocks and artifacts an effort should produce (actors, processes, applications, data entities, technology components), consistent across the four views	p. 115
Extended content model	Extra depth in areas such as security, governance, data management	p. 115
Architecture artifacts	Catalogs, matrices, diagrams (e.g., Business Process Catalog, Application Portfolio Matrix) aligned to ADM phases	p. 116
Views and viewpoints	Address different stakeholders' concerns. Viewpoints set the conventions for views	p. 116
Agile and digital transformation	Designed to work with DevOps and cloud architecture	p. 116
Guidance for tailoring	Select, adapt, and use the valuable parts. Gives tools and a common vocabulary	p. 116
Outcome-focused	Valuable artifacts that deliver business outcomes, not a rigid structure	p. 117

- **Versus previous versions.** More flexible, better integration (e.g., with agile), and more focus on actionable outputs that achieve business objectives (p. 117).

6.3 Mapping to the Architecture Development Method (pp. 117-122)

Content Framework section	What its models capture	ADM phase(s)
Architecture Principles, Vision, Motivation, and Requirements	Context around formal models: principles, strategic context, requirements. The business context behind the Request for Architecture Work	Preliminary, A (and Requirements) (p. 118)
Business Architecture	What motivates the organization, its structure, its capabilities	B (p. 118)
Information Systems Architectures	Models of applications and data	C (p. 119)
Technology Architecture	Technology assets used to realize IS solutions	D (p. 119)
Architecture Realization	Change roadmaps showing transitions between states	E, F, G (p. 119; Figure 6-2 p. 122)
Architecture Change Management	Internal and external events affecting the EA, generating action requirements	H (p. 119)

- **Complement to the ADM.** The Content Framework "describes what the architecture should look like when it is complete" (p. 119).
- **Extended Content Framework (Figure 6-2).** The entities from the Enterprise Metamodel, mapped onto the sections above (p. 120):
  - **Preliminary**: focuses on developing Architecture Principles (much more happens in the phase) (p. 120).
  - **Architecture Requirements**: requirements, constraints, assumptions, gaps, locations, i.e., most General Entities (p. 121).
  - **Architecture Definition**: the most entities, split into Motivation, Business, Information Systems, and Technology Architecture (p. 121).
  - **Architecture Realization**: Opportunities, Solutions, and Migration Planning (one-to-one with E and F; capabilities and work packages) and Implementation Governance. The latter, unusually, uses standards, guidelines, and specifications from the Architecture Repository rather than metamodel entities (p. 122).
  - **Architecture Change Management**: runs top to bottom on the right side (Phase H) (p. 122).

Jager's commentary (author's view): Putting driver, goal, objective, and measure in a separate Motivation section, although the metamodel puts them in Business Architecture, is "one of the inconsistencies in the TOGAF Standard". It possibly crept in because many people worked on the Standard (p. 121). He praises the direct link from the Content Framework to the ADM as "very clever" (p. 122).

6.4 Tailoring the Content Framework (pp. 123-125)

When	How the content may change	Page
Phase A	Initial content set and aligned with business objectives. Additional content or viewpoints may be identified	p. 123
Phases B, C, D	Detailed artifacts may add to or modify Content Framework components	p. 123
Phase H	Managing change may require updated standards, models, or deliverables	p. 123
Throughout the cycle	New insights, stakeholder feedback, or requirements may change the Content Framework and the metamodel	p. 123

- **Work packages are underused.** The work package entity appears in both models but the Standard uses it only in the Project Context Diagram (Ch. 12.8.2) (p. 124).
- **Recommendation: Work Package Portfolio Map** (Ch. 14.8). It lists all initiatives and links them to goals and objectives. Useful in Phases E and F as the bridge between business goals and the Architecture Roadmap (p. 124).
- **Living guides.** Iterative ADM execution refines both models as part of governance and development (p. 125).

6.5 Summary (pp. 125-126)

- The Content Framework is a categorization framework and "a one-to-one translation of the phases" of the ADM. The Extended form maps all entities, derived from the metamodel, onto the phases (p. 125).
- It is more modular, defines catalogs, matrices, and diagrams, emphasizes views and viewpoints, and supports agile and digital transformation, with a shift toward outcomes (p. 126).

Key Terms & Definitions

Term	Definition	Page
Content Framework	Categorization framework structuring architecture descriptions, work products, and models, mapped to ADM phases	pp. 111, 115
Extended Content Framework	Version of the Content Framework showing the metamodel entities per section/phase	p. 120
Categorization framework	High-level summarization of the metamodel's entities into sections	p. 113
Requirements Management	Not a separate phase but a continuous mechanism for testing the architecture	pp. 111-112
Architecture Realization	Content Framework section for change roadmaps (Phases E-G)	p. 119
View / Viewpoint	A way of looking at the architecture for stakeholders' concerns / the conventions for building views	p. 116
Work Package Portfolio Map	Jager-recommended artifact linking initiatives to goals and objectives	p. 124

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 6-1	The Content Framework by ADM phase	p. 118
Figure 6-2	The Extended Content Framework with entities	p. 120

Self-Check Questions

Q: What three things must be defined during the Preliminary Phase when establishing the Architecture Capability?

A: The types of entities and relationships (Enterprise Metamodel), a categorization framework (Content Framework), and the specific artifacts to develop (p. 112).

Q: List the five purposes of the Content Framework.

A: Describe a detailed model of architecture products; promote consistent deliverables; provide a checklist of deliverables; reduce gaps; help use standard concepts, terms, and deliverables (pp. 113-114).

Q: How does the Content Framework differ from the Enterprise Metamodel?

A: The metamodel is a detailed representation of the organization's entities. The Content Framework is a high-level categorization that structures work products and maps them to ADM phases (p. 113).

Q: Which Content Framework section corresponds to Phases F and G?

A: Architecture Realization, which holds the change roadmaps (p. 119).

Q: Which section of the Extended Content Framework does not take its entities from the metamodel, and where do they come from?

A: Implementation Governance. Its standards, guidelines, and specifications come from the Architecture Repository (p. 122).

Q: What inconsistency does Jager point out in the Extended Content Framework?

A: Driver, goal, objective, and measure sit in a separate Motivation section, even though the metamodel places them in Business Architecture (p. 121).

Q: In which phases can the ADM update the Content Framework?

A: Phase A, Phases B-D, Phase H, and throughout the cycle when new insights or requirements arise (p. 123).

Q: Why does Jager recommend a Work Package Portfolio Map, and in which phases is it used?

A: The Standard barely uses the work package entity (only in the Project Context Diagram). The map links all initiatives to goals and objectives and connects them to the Architecture Roadmap. It is used in Phases E and F (p. 124).

Q: Why is Requirements Management not described as producing its own work products?

A: It is not a separate phase but a continuous mechanism for testing the architecture (pp. 111-112).

Q: What is the main shift in the current Content Framework compared to earlier versions?

A: A focus on outcomes rather than processes: valuable artifacts that deliver business outcomes instead of a rigid structure (pp. 117, 126).

Chapter 7: Architecture Repository

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 7 (pp. 127-187)

Overview

Chapter 7 (pp. 127-187) explains the **Architecture Repository**, the Standard's *intangible* central hub where all architecture assets are kept (pp. 127-128). Section 7.1 positions it as a concept, not a tool (pp. 128-129). Section 7.2 walks through its components: Architecture Metamodel, Architecture Landscape (Strategic, Segment, Capability levels), Solutions Landscape, Architecture Requirements Repository, Reference and Standards Libraries, Governance Repository, and Architecture Capability (pp. 129-136). Section 7.3 explains the three **continua** as lenses on the repository (pp. 136-149). Section 7.4 distinguishes catalogs, diagrams, maps, and matrices (pp. 150-154). Section 7.5 covers deliverables, the per-phase artifact list (Table 7-4), and **building blocks** (ABBs and SBBs) (pp. 154-175). Section 7.6 covers views and viewpoints (pp. 176-179). Section 7.7 covers making the repository tangible with a tool (pp. 179-183). Section 7.8 summarizes (pp. 183-187). Leave with this idea: deliverables contain artifacts, artifacts contain building blocks, and building blocks are made of metamodel entities. All of it is stored, classified, and reused through the repository (p. 150). Chapter 7 is part of the GREEN reading tour (p. 127).

> **Reading note:** Repository components, continua, building block types, and views/viewpoints are from the Standard. Jager adds: where ABBs and SBBs are stored, which the Standard does not say (pp. 172, 174-175); the LEGO analogy (p. 164); the binoculars and EHR analogies for views (pp. 177-178); his ZiRA hospital experience (p. 146); a critique of Figure 7-6 (p. 142); and practical tool advice (pp. 179-183). Easy to confuse: the **Architecture Landscape** vs. the **Solutions Landscape**, and a **view** (what you show) vs. a **viewpoint** (the rules for making it).

Detailed Section Breakdown

Chapter Opener (p. 127)

- The repository is "an intangible concept in which all aspects of the architecture are collected and stored". The chapter covers its parts, the continua, building blocks, views and viewpoints, and making it tangible with tools (p. 127).

7.1 The Central Hub (pp. 128-129)

- **Not a tool.** Organizations often treat their tools (from spreadsheets to full applications) as the repository, but in TOGAF the repository is an intangible concept (p. 128).
- **What it should contain.** Only reusable building blocks, an organization-specific metamodel, and architecture governance information. "A well-designed repository does not contain diagrams of specific solutions" (p. 128).
- **Backbone of the practice.** It brings frameworks, templates, and reference material together, for finished and in-progress work, and feeds the perspectives of the three continua (p. 128, Figure 7-1 on p. 129).

7.2 Key Components of the Architecture Repository (pp. 129-136)

Component	What it holds	Page
Architecture Metamodel	How artifacts are organized and how information types relate. Based on the Enterprise Metamodel (Ch. 5)	pp. 129-130
Architecture Landscape	Views of the current, planned, and aspirational architecture at three levels (below), plus reusable ABBs	pp. 130-131
Solutions Landscape	Models and documents of individual Solution Architectures, and reusable SBBs. Not the content/data the solutions produce	pp. 131-132
Architecture Requirements Repository	All requirements created or updated during the ADM, at Strategic, Segment, and Capability levels	pp. 132-134
Reference Library	Best practices, guidelines, templates, patterns, reference architectures, and the Viewpoint Library	p. 134
Standards Library	Standards and policies the architecture must conform to (legal/regulatory, industry, organizational)	pp. 134-135
Governance Repository	Decisions, rules, principles, approvals, waivers, compliance assessments. Provides accountability and traceability	p. 135
Architecture Capability	Skills, processes, structures, and roles that support EA	pp. 135-136

- **Three levels of granularity (Figure 7-2):**

Level	Purpose	Used by
Strategic Architecture	Long-term summary view of the whole organization, for direction setting	Executives (p. 130)
Segment Architecture (also "Domain Architecture")	More detailed operating models for areas of the organization	Program or portfolio level (p. 130; Ch. 10.3.2)
Capability Architecture	Detail on supporting a unit of capability: current, target, and increments. Groups work packages into portfolios and programs	Project/work package level (p. 130)

- These three levels matter for agile architecture (Ch. 18) (p. 131).
- **Requirements by level.** Strategic (executive, long-term direction), Segment (portfolio/project, tactical breakdown of strategic requirements), Capability (needs of individual initiatives or work packages) (pp. 132-133).
- **External inputs.** Both libraries are fed by external reference models (e.g., BIZBOK) and external standards (e.g., the TOGAF Standard itself) (p. 135).
- **Architecture Board.** Gives visibility to the Governance Repository and Architecture Capability, serves as an escalation mechanism, and directs the Architecture Capability (p. 136).

7.3 The Three Continua (pp. 136-149)

- **What a continuum is.** A lens or conceptual framework for viewing and classifying the repository's assets, scaling detail, complexity, and governance to need. Left to right gives an increasingly detailed view (pp. 136-137).

Continuum	Role	Page
Enterprise Continuum	Umbrella over the other two. Bridges conceptual architectures and practical solutions. Enables reuse	p. 138
Architecture Continuum	Classifies architecture assets from generic to specific. Applies to the Architecture Landscape, Reference Library, and Standards Library	pp. 138, 141

Continuum	Role	Page
Solutions Continuum	Parallel to the Architecture Continuum. Shows how architectures become deployable solutions. Applies to the Reference Library and Solutions Landscape	pp. 138, 147

7.3.1 The Enterprise Continuum (pp. 139-141)

- Shows evolution "from generic to specific, from abstract to concrete, and from logical to physical". It is the outermost continuum, classifying contextual assets (policies, standards, strategic initiatives, structures, organization-level capabilities) and also solutions (pp. 139-140).
- Its scale runs from foundation (most general) to organization-specific (Figure 7-5, p. 140).

7.3.2 The Architecture Continuum (pp. 141-146)

Type	Description	Healthcare example	Page
Foundation Architectures	Most generic. Universal services and technologies (network protocols, OS, security frameworks, SOA)	n/a	pp. 142-143
Common Systems Architectures	Generic across industries. Complete for one problem domain (security, networking, etc.)	Patient management, scheduling, billing systems	p. 143
Industry Architectures	Reference models and standards for one industry	EHR, telemedicine, HIE, HL7, ICD-11	pp. 143-144
Organization-Specific Architectures	Most tailored: one organization's goals, processes, culture	A hospital's customized EHR with a patient portal and local pharmacy integration	p. 144

- **Four steps (Figure 7-7):** start with Foundation, adopt Common Systems, tailor with Industry, customize with Organization-Specific (p. 145).

Jager's commentary (author's view): Figure 7-6 is hard to read, because Foundation and Common Systems parts are not visibly reused in the later models (p. 142). In Dutch hospitals he reused **ZiRA** (the Dutch Hospital Reference Architecture), an Industry Architecture, which saved enormous time (p. 146).

7.3.3 The Solutions Continuum (pp. 147-149)

Type	Description	Page
Foundation Solutions	Highly generic concepts, tools, products, services (including professional and support services)	p. 148
Common Systems Solutions	Implementations of a Common Systems Architecture, e.g., an enterprise management or security product	pp. 148-149
Industry Solutions	Reusable industry-specific packages, e.g., a physical database schema or point-of-service device	p. 149
Organization-Specific Solutions	Implementations of the Organization-Specific Architecture, with the most unique content	p. 149

- **Two directions (Figure 7-9).** Left to right delivers solution value. Right to left addresses business needs (p. 148).
- **Solutions inventory.** A populated Solutions Continuum is a solutions inventory. Connecting the two continua aims to build organization-specific solutions on generic ones (pp. 147, 149).

7.4 Catalogs, Diagrams, Maps, and Matrices (pp. 150-154)

- **Hierarchy (Figure 7-10):** a deliverable contains one or more artifacts, an artifact contains one or more building blocks, and a building block contains one or more entities (p. 150).

Form	Definition	Example	Page
Catalog	List of building blocks of one entity type, for governance or reference. Entities are *not* linked	Stakeholder Catalog	p. 151
Map	Textual representation of architecture content	Business Capability Map, Organization Map	pp. 151-153
Diagram	Graphical representation of the same kind of content	Process Flow Diagram	pp. 151-153
Matrix	Grid of relationships between two or more entities ("cross-mapping")	Entity A vs. Entity B	p. 153

- **Less is more.** Better to create more views than to overload one artifact (p. 154).

7.5 Architecture Deliverables (pp. 154-175)

- Deliverables support conversations about architecture issues. Example: an Architecture Definition Document containing an Organization Map, a Process Flow Diagram, and an Application Landscape (pp. 154-155, Figure 7-12).

7.5.1 Architecture Artifacts (pp. 155-163)

- **Not all are required.** "Creating all the artifacts listed in the TOGAF Standard is by no means a necessity." Use those that add value (pp. 155, 157).
- Artifacts are derived from the tailored metamodel and created in the Preliminary Phase, Requirements Management, and Phases A-E (p. 156, Figure 7-13).

Phase	Recommended artifacts (Table 7-4)
Preliminary	Principles Catalog (p. 157)
A: Architecture Vision	Stakeholder Catalog; Value Chain Diagram; Solution Concept Diagram; Business Model Diagram; Business Capability Map; Value Stream Map (p. 157)
B: Business	Catalogs: Organization/Actor; Driver/Goal/Objective; Role; Business Service/Function; Location; Process/Event/Control/Product; Contract/Measure; Business Capabilities; Value Stream; Value Stream Stages; Business Glossary. Matrices: Business Interaction; Actor/Role; Value Stream/Capability; Strategy/Capability; Capability/Organization. Diagrams and maps: Business Footprint; Business Service/Information; Functional Decomposition; Product Lifecycle; Goal/Objective/Business Service; Business Use-Case; Organization Decomposition; Process Flow; Business Event; Business Capability Map; Value Stream Map; Organization Map; Information Map (pp. 158-160)
C: Data	Data Entity/Data Component Catalog; Data Entity/Business Function Matrix; Application/Data Matrix; Conceptual Data, Data Dissemination, Data Security, Data Migration, and Data Lifecycle Diagrams (p. 160)
C: Application	Application Portfolio and Interface Catalogs; Application/Organization, Role/Application, Application/Function, and Application Interaction Matrices; Application Communication, Application and User Location, Application Use-Case, Enterprise Manageability, Process/Application Realization, Software Engineering, Application Migration, and Software Distribution Diagrams (pp. 161-162)
D: Technology	Technology Standards and Technology Portfolio Catalogs; Application/Technology Matrix; Environments and Locations, Platform Decomposition, Processing, Networked Computing/Hardware, and Network and Communications Diagrams (pp. 162-163)
E: Opportunities and Solutions	Project Context Diagram; Benefits Diagram (p. 163)

7.5.2 Building Blocks (pp. 163-175)

- **Definition.** "A potentially reusable component that can be combined with other building blocks to deliver architectures and solutions". They are packages of functionality corresponding to metamodel entities, with clear boundaries, and should not be tightly bound to an implementation (pp. 165-166).
- **LEGO analogy (Figure 7-14).** Generic bricks combine into bigger sets. Three SBBs stack into an ABB (pp. 164-165).
- **Level of detail grows.** A block may begin as a name and become detailed later. Sometimes less detail is more effective, e.g., one Business Capability Map instead of a 100-page document (pp. 165-167).

7.5.2.1 Building blocks in architecture design (pp. 167-168). An architecture is "a set of building blocks depicted in an architectural model and a specification of how those building blocks are connected". Only include blocks that address the problem. Blocks interact in complex ways, and one can support many or partially support one. Blocks start general and become precise, and must follow standards and Enterprise Principles (pp. 167-168).

7.5.2.2 Specification process (pp. 168-169, Figure 7-16).

Phase	What happens to building blocks
A	High-level model of candidate blocks, e.g., a Business Capability Map
B, C, D	Specify the entities that flesh out that map. Reuse existing blocks from the repository, create missing ones via catalogs/matrices/diagrams, and remove obsolete ones
E	Work packages (initiatives) are defined to create the missing blocks
Overall	ABBs are identified for goals, then iteratively refined into SBBs that are bought or built

(pp. 169)

7.5.2.3 Types (p. 170). ABBs = the *what and why* (technology-agnostic). SBBs = the *how* (specific technologies and products). SBBs normally *implement* ABBs, but when the ABB is a capability and the SBB a data, application, or technology entity, the SBB can also *automate* it (Figure 7-17).

7.5.2.4 Architecture Building Blocks (pp. 171-173). Created in the initial ADM iteration. They capture requirements and capabilities, specify fundamental functionality, attributes, interoperability, and relationships, and map to business or organizational entities. Example: a Human Resources Management capability. Stored (per Jager) in the Architecture Landscape, Reference Library, and Standards Library (Figure 7-18).

7.5.2.5 Solution Building Blocks (pp. 173-175). Detailed, product- or vendor-aware, procured or developed. They map to IT topology and operational policies, with attributes such as security, manageability, scalability, performance, and configurability. Example: the Candidate Evaluation Service. Stored (per Jager) in the Solutions Landscape (Figure 7-19).

	ABB	SBB
Answers	What and why	How
Technology	Agnostic	Product/vendor-aware
Created	Early (initial iteration, A-D)	Later, refined from ABBs
Stored (Jager)	Architecture Landscape, Reference Library, Standards Library	Solutions Landscape

7.6 Views and Viewpoints (pp. 176-179)

- **View.** A representation of a system from the perspective of specific stakeholders, addressing their concerns, e.g., a security view (p. 176).
- **Viewpoint.** The perspective and rules (a template) for creating a view, e.g., a performance viewpoint (p. 176).
- **Relationship (7.6.1).** A viewpoint is a blueprint, and many views can come from one viewpoint. "A view is what you actually present... a viewpoint is the set of rules" for creating it. Both live in the Reference Library (p. 177).
- **Binoculars analogy (Figure 7-20).** Two people on two peaks look at one village. People = stakeholders, binoculars = viewpoints, village = view (pp. 177-178).
- **EHR example (Figure 7-21).** Physician and patient see the same EHR system differently, so multiple viewpoints are needed (p. 178).

7.7 Using the Architecture Repository (pp. 179-183)

- **Make it tangible with a tool.** Office tools soon fail: no unified modelling language, and every artifact must be tracked manually. A good tool must relate entities so changes propagate (pp. 179-180).

7.7.1 Deciding on a Tool (pp. 180-183)

- Budget drives the choice. Gartner's yearly report helps. A tool is "an absolute must" to make the repository tangible (p. 180).
- **Table 7-5 considerations (p. 181):** modelling language support (full ArchiMate, BPMN, user-defined properties, coloured views); metamodel customization; flexibility and stability (change one concept into another while keeping relationships, handle large models); import/export (share with non-users, publish large PDFs as posters); support (help desk, user community).
- Publishing artifacts to a website is "an easy step to higher maturity" (p. 182).
- **Minimum components to represent (Figure 7-22):** Architecture and Solutions Landscapes, Reference and Standards Libraries, Architecture Requirements Repository (p. 182).
- "A well-managed repository is characterized by ruthlessly minimizing information collected and maintained" (p. 183).
- **Reality check.** Nine out of ten tools don't separate the Architecture and Solutions Landscapes. Use an intuitive folder structure to do so (p. 183).

7.8 Summary (pp. 183-187)

- Three continua (Enterprise as umbrella, Architecture, Solutions), each with four types that become more detailed from left to right (pp. 183-184).
- Catalogs, diagrams/maps, and matrices visualize building blocks. Artifacts consist of blocks, and deliverables consist of artifacts (p. 184).
- Building blocks come in ABB and SBB flavours, are defined mainly in A-D, and are stored in the repository. Views and viewpoints communicate them. A tool is essential (pp. 185-187).

Key Terms & Definitions

Term	Definition	Page
Architecture Repository	Intangible central hub storing reusable building blocks, the organization-specific metamodel, and governance information	p. 128

Term	Definition	Page
Architecture Landscape	Current, planned, and aspirational architecture at Strategic, Segment, and Capability levels	p. 130
Solutions Landscape	Solution Architectures and reusable SBBs	p. 131
Governance Repository	Record of decisions, approvals, waivers, and compliance assessments	p. 135
Enterprise Continuum	Umbrella lens showing evolution from generic to specific, abstract to concrete, logical to physical	pp. 138-140
Architecture Continuum	Foundation, Common Systems, Industry, Organization-Specific Architectures	pp. 141-144
Solutions Continuum	Foundation, Common Systems, Industry, Organization-Specific Solutions	pp. 147-149
Catalog / Map / Diagram / Matrix	List of one entity type / textual representation / graphical representation / relationship grid (cross-mapping)	pp. 151-153
Architecture deliverable	Product of architecture work made of one or more artifacts	pp. 150, 154
Building block	Potentially reusable component combined with others to deliver architectures and solutions	p. 165
ABB / SBB	Technology-agnostic what-and-why block / product-aware how block that implements (or automates) an ABB	pp. 170-174
View / Viewpoint	Representation for stakeholders' concerns / rules or template for creating views	p. 176

Figures & Tables Referenced

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Figure 7-2	Three levels of architecture granularity	p. 131
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Figures 7-20, 7-21	Viewpoints and views (village, EHR)	pp. 177-178
Table 7-5, Figure 7-22	Tool considerations. Minimum repository components in a tool	pp. 181-182

Self-Check Questions

Q: Is the Architecture Repository a tool? What should it contain?

A: No, it is an intangible concept. It should contain reusable building blocks, an organization-specific metamodel, and governance information, but not diagrams of specific solutions (p. 128).

Q: Name the three levels of the Architecture Landscape and who uses each.

A: Strategic (executive direction, whole organization), Segment/Domain (program or portfolio), and Capability (units of capability, grouping work packages and projects) (p. 130).

Q: Name the four types in the Architecture Continuum, from generic to specific.

A: Foundation, Common Systems, Industry, Organization-Specific Architectures (p. 141).

Q: What do left-to-right and right-to-left movements on the Solutions Continuum represent?

A: Left to right: delivering solution value by building on more generic solutions. Right to left: addressing business needs (p. 148).

Q: What is the difference between a catalog and a matrix?

A: A catalog lists building blocks of one entity type without linking them. A matrix cross-maps relationships between two or more entities (pp. 151, 153).

Q: How are deliverables, artifacts, building blocks, and entities related?

A: A deliverable has one or more artifacts, each artifact has one or more building blocks, and each building block has one or more entities (p. 150).

Q: Distinguish ABBs from SBBs.

A: ABBs describe the technology-agnostic what and why (capabilities, requirements). SBBs describe the how: specific products and technologies that implement, or for capabilities also automate, ABBs (pp. 170-174).

Q: In which ADM phases are building blocks mainly specified, and what happens in Phase E?

A: Mainly in A, B, C, and D. In Phase E, work packages are defined to create the missing building blocks (p. 169).

Q: Explain views and viewpoints using the village analogy.

A: The people on the peaks are stakeholders, their binoculars are viewpoints (the rules and perspective), and the village is the view (what is seen) (pp. 177-178).

Q: Must every artifact in Table 7-4 be produced?

A: No. The Standard only shows possibilities. Produce the artifacts that add value and ignore the rest (pp. 155, 157).

Q: Which repository components should an architecture tool minimally support?

A: The Architecture and Solutions Landscapes, the Reference and Standards Libraries, and the Architecture Requirements Repository (p. 182).

Q: What did Jager use ZiRA for, and which continuum type is it?

A: As a reusable Dutch hospital reference architecture (business function and care process models) when doing hospital EA. It is an Industry Architecture (p. 146).

Chapter 8: Architecture Governance

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 8 (pp. 189-227)

Overview

Chapter 8 (pp. 189-227) covers how architecture is controlled and steered. At its core, architecture governance turns an organization's architectural needs into "a coherent set of policies, processes, procedures, and standards" (p. 189). Section 8.1 presents the Standard's Architecture Governance Framework, led by a CIO/CTO with development, implementation, and deployment capabilities that all use the Architecture Repository (pp. 190-193). Section 8.2 gives Jager's **ten steps for implementing governance** (pp. 193-204). Section 8.3 is about the **Architecture Board**: what it really is, CEO vs. CIO/CTO leadership, when to set one up, its responsibilities, its composition and size, and **ten steps to set it up** (pp. 204-224). Section 8.4 summarizes (pp. 224-227). Leave with this idea: an Architecture Board is a *board-level*, strategic body, not a meeting of IT architects. Without top-level sponsorship, architecture work has "little or no point" (pp. 205, 210). Chapter 8 is part of the GREEN reading tour (p. 189).

> **Reading note:** The Governance Framework (Figure 8-1), the Architecture Board definition, and its key responsibilities come from the Standard. The two ten-step lists are Jager's practical approach. His opinions include: add the **CEO** to the framework (pp. 193, 206), the CIO belongs on the business side (p. 192), boards made up only of C-level members are too strategic (p. 214), and his healthcare-regulator story (pp. 219-220).

Detailed Section Breakdown

Chapter Opener (p. 189)

- Governance is "the leadership compass" and backbone of the organization. The Architecture Governance Framework lines the organization's vision and standards up with real business demands (p. 189).

8.1 Managing the Architecture (pp. 190-193)

- **Grounded in daily business.** Governance is a shared responsibility of business leaders, not only the CIO/CTO. It reduces costs and risks and speeds up decisions and delivery (p. 190).
- **Purpose per TOGAF.** To control and manage the architecture and the Architecture Capability, the "executive body behind the architecture" (p. 190).
- **The framework is a starting point.** It shows a "possible structure" and should be adapted (p. 190).

Element of the Governance Framework (Figure 8-1)	Role	Page
CIO/CTO	Overall direction. Directly manages the Architecture Board and three capabilities	p. 191
Development capability	Chief Architect, Enterprise Architect, Segment Architect (formerly Domain Architect) (Ch. 10.2)	p. 191
Implementation capability	PMO, external vendors, implementation projects	p. 192
Deployment capability	Service management and operational systems that the architecture must fit into	p. 192

Element of the Governance Framework (Figure 8-1)	Role	Page
Architecture Repository	Used by all capabilities and continuously updated	p. 192

Jager's commentary (author's view): Information ownership lies with the business, so the CIO should sit on the business side, yet "nine out of ten" organizations (and the Standard) place the CIO in IT. The framework should add the **CEO** role (pp. 192-193).

8.2 Implementing Architecture Governance (pp. 193-204)

- Phase G (Implementation Governance) puts this approach into practice (p. 193).

Step	What it involves	Page
1. Establish the Architecture Governance Framework	Stakeholder kick-off workshops; a Governance Charter (scope, objectives, principles, what is governed, what success looks like); choose components; set up the board; formalize processes and decision criteria	pp. 194-195
2. Define governance roles and responsibilities	Role-definition workshop; role descriptions; map responsibilities in a RACI matrix	pp. 195-197
3. Develop governance policies and standards	Policy-drafting group; high-level policies broken into standards; stakeholder review; publish and train	p. 198
4. Set up an architecture compliance process	Domain compliance checklists; checkpoints in the project lifecycle; compliance reviews; exception handling with rationale and mitigation	pp. 198-199
5. Implement an architecture review process	Review templates (context, principle compliance, risks, solutions); scheduled reviews; preparation; documented decisions	pp. 199-200
6. Establish KPIs	e.g., % of compliant projects, number of reviews, number of exceptions, time to resolve issues; baselines, targets, dashboards	pp. 200-201
7. Integrate governance into the ADM	Checkpoints per phase (e.g., Preliminary: framework in place; B: alignment with goals); no progress without board sign-off	pp. 201-202
8. Develop a Communications Plan	Identify audiences; tailor materials; workshops and webinars (Ch. 14.3.3)	p. 202
9. Automate governance processes	Governance or EA tool; automated alerts and KPI reports (Ch. 7.7.1)	pp. 202-203
10. Ensure continuous improvement	Quarterly reviews; gather feedback; adjust and communicate the framework	p. 203

- **RACI (Tables 8-1, 8-2).** Responsible = carries out the activity. Accountable = ultimately answerable, and the responsible person reports to them. Consulted = gives input (stakeholder or expert). Informed = receives the output or needs to stay informed. In the example, for "Establish the Architecture Governance Framework" the Architect is Accountable, the Business Leader Responsible, and the Project Manager Consulted (p. 197).

8.3 Architecture Board (pp. 204-224)

- **What it does.** Oversees compliance, facilitates decisions, manages architecture risk, and reviews and approves deliverables. It manages architecture roles *functionally*, not hierarchically (p. 204).
- **"Board" means boardroom.** It is not a group of architects discussing content. The issues are strategic, so it is formed by senior management: C-level, key stakeholders, domain experts (p. 205).
- **Standard's definition.** "A cross-organization entity, which oversees the implementation of the strategy", representative of all key stakeholders and typically executives (p. 205).
- **The common failure.** In "nine out of ten cases" the board is formed by IT architects. Strategic decisions are then made elsewhere, leaving the Architecture Capability "in a perpetual state of

reactive action" and "structurally behind the times" (pp. 205-206).

8.3.1 Diverse Leadership (pp. 206-209)

Scenario	Scope of architecture work when new legislation hits	Page
1. CEO leadership	Broad, cross-domain: organizational model, roles, processes and the function model, information flows, aligned application landscape	pp. 206-207
2. CIO/CTO leadership	Narrow: CIO only information delivery, CTO only the technical side. A small part of the EA	p. 207

- **Figure 8-3.** The CEO directs both the CIO/CTO and the Architecture Board. The CIO/CTO manages unit managers, who manage the teams (p. 208).
- **Not just for large organizations.** Smaller and flatter organizations benefit too. The sponsor need not be the CEO, but attention must come from the highest levels (p. 209).

8.3.2 When to Set Up an Architecture Board (pp. 209-210)

- **Typical triggers:** a new CIO/CTO; mergers or acquisitions; restructuring (business change or rapid growth); recognition that strategy is not being implemented, or is implemented poorly (p. 209).
- **A sponsor is not enough.** A board is needed for organization-wide support. The board is the architecture's primary sponsor and itself needs a top-level sponsor, ideally the CEO. The company's board of directors is an often-overlooked resource (p. 210).
- Involvement should span planning through maintenance. Establish sponsorship *before* work begins (p. 210).

Jager's commentary (author's view): In nine out of ten cases, failed architecture efforts are due to "a significant lack of involvement and encouragement from senior management" (p. 210).

8.3.3 Key Responsibilities (pp. 211-213)

Category	Responsibilities	Page
Strategic	Provide a solid basis for decision-making; ensure continued flexibility (meet changing needs and exploit new technology while enforcing compliance); improve architecture maturity (never let it drop)	p. 211
Operational	Ensure consistent management and implementation (fixed membership, staggered terms); provide advice and guidance tested against the existing architecture	p. 212
Governance	Provide a process for accepting and approving architecture work; implement a control mechanism (usually in Phase G); recognize divergence from the architecture (Phase H)	pp. 212-213

- Roles and responsibilities should be formally approved by the lead sponsor, ideally the CEO (p. 213).

8.3.4 Composition and Size (pp. 213-215)

- **Size.** Four or five standing members, never more than ten. Membership can rotate, with staggered fixed terms for continuity (pp. 213-214).
- **Stakeholder representation.** Each member can represent assigned stakeholders and gather their input (p. 213).
- **Mix of levels.** Not exclusively C-level. Lower-level managers and experts join per topic (not as permanent members), so issues are covered strategically, tactically, and operationally (p. 214).

Jager's commentary (author's view): All-C-level boards he has seen were "too-strategic", leaving implementation teams without the information they need (pp. 214-215).

8.3.5 How to Set Up an Architecture Board (pp. 215-224)

Step	Key activities	Page
1. Define purpose and scope	Objectives; scope (domains, decision types, level of operation); governance principles	pp. 216-217
2. Identify key stakeholders	Business stakeholders, Enterprise and Segment Architects, security and compliance, IT leaders, project managers; define roles (chair, secretary, decision-makers)	p. 217
3. Develop the operating model	Meeting cadence; standard agenda; decision-making (consensus, majority voting, escalation)	p. 218
4. Define engagement model and workflows	Submission process and templates; review criteria; feedback loop; decision documentation	pp. 218-219
5. Establish key artifacts and tools	Principles and standards; decision templates; collaboration and tracking tools	pp. 220-221
6. Pilot the board	Low-risk pilot project; full cycle; feedback; refine	p. 221
7. Roll out organization-wide	Communicate; onboard projects; monitor metrics; iterate	pp. 221-222
8. Ensure ongoing communication and alignment	Regular updates; cross-board collaboration; continuous education	pp. 222-223
9. Measure and optimize performance	KPIs including business value delivered (often forgotten); feedback loops; continuous improvement	p. 223
10. Celebrate successes and promote value	Publicize achievements; keep communicating business value	pp. 223-224

Jager's commentary (author's view): At a healthcare regulator, the "Architecture Board" was all technicians debating backup designs. He built a new advisory group with management input, which was eventually renamed the Architecture Board, while the old one became the "Architects' Consultation", so nobody lost their forum (pp. 219-220).

8.4 Summary (pp. 224-227)

- Governance controls and manages the architecture and the Architecture Capability, using the Standard's framework as a possible structure, and is implemented in ten steps (pp. 224-225).
- The Architecture Board should be made up of C-level management, key stakeholders, and domain experts, but not exclusively C-level. It needs top-level sponsorship and is set up in ten steps. Keep communicating its business value (pp. 226-227).

Key Terms & Definitions

Term	Definition	Page
Architecture governance	Streamlining architectural needs into coherent policies, processes, procedures, and standards. Controls and manages the architecture and the Architecture Capability	pp. 189-190
Architecture Governance Framework	TOGAF's reference structure: CIO/CTO, Architecture Board, development/implementation/deployment capabilities, and the repository	pp. 190-192
Architecture Board	Cross-organization, executive-level entity overseeing implementation of the strategy, representative of key stakeholders	p. 205
Governance Charter	Document stating governance scope, objectives, principles, governed areas, and success criteria	p. 194

Term	Definition	Page
RACI matrix	Maps activities to roles as Responsible, Accountable, Consulted, Informed	pp. 196-197
Compliance process	Checklists, lifecycle checkpoints, reviews, and exception handling verifying conformance	pp. 198-199
Segment Architect	The role formerly called Domain Architect	p. 191

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 8-1	Architecture Governance Framework	p. 191
Figure 8-2	Ten steps for implementing architecture governance	p. 194
Table 8-1	Example RACI matrix for governance activities	p. 196
Table 8-2	The four RACI responsibilities explained	p. 197
Figure 8-3	Architecture Board led by the CEO	p. 208
Figure 8-4	Ten steps for setting up an Architecture Board	p. 216

Self-Check Questions

Q: What are the three capabilities in TOGAF's Governance Framework, and who directs them?

A: Development (architects), implementation (PMO, vendors, projects), and deployment (service management, operational systems). The CIO/CTO directs them (pp. 191-192).

Q: Why does the word "board" in Architecture Board matter?

A: It refers to the boardroom: issues are strategic, so the board should consist of senior management, key stakeholders, and domain experts, not a group of IT architects (p. 205).

Q: What happens when an Architecture Board is made up only of IT architects?

A: Strategic decisions are made outside the Architecture Capability, which stays reactive and "structurally behind the times" (pp. 205-206).

Q: Contrast CEO-led and CIO/CTO-led architecture in the legislation scenario.

A: CEO-led work is broad and cross-domain (organization, processes, information, applications). CIO-led work covers only information delivery and CTO-led work only the technical side (pp. 206-207).

Q: Name four events that typically trigger the creation of an Architecture Board.

A: A new CIO/CTO; mergers or acquisitions; organizational restructuring; recognition that strategy is not being implemented or is implemented poorly (p. 209).

Q: What size does Jager recommend for an Architecture Board, and how is continuity kept with rotation?

A: Four or five standing members, not more than ten. Fixed terms that end at different times give continuity (pp. 213-214).

Q: Which board responsibilities link to ADM Phases G and H?

A: The control mechanism validating execution against the proposal (usually Phase G), and recognizing divergence from the architecture (Phase H) (pp. 212-213).

Q: List the ten steps for implementing architecture governance.

A: Establish the framework; define roles and responsibilities; develop policies and standards; set up compliance; implement review; establish KPIs; integrate into the ADM; develop a Communications Plan; automate; continuously improve (pp. 194-203).

Q: In RACI, who is Accountable vs. Responsible?

A: Responsible carries out the activity. Accountable is ultimately answerable for its completion, and the responsible person reports to them (p. 197).

Q: Which KPI for the Architecture Board does Jager say is often forgotten?

A: Business value delivered. People get caught up in content and neglect the value proposition (p. 223).

Q: Why does Jager recommend adding lower-level managers and experts to the board on a per-topic basis?

A: All-C-level boards can be too strategic and hand implementation teams too little information. A mix covers strategic, tactical, and operational angles (pp. 214-215).

Chapter 9: Architecture Capability

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 9 (pp. 229-245)

Overview

Chapter 9 (pp. 229-245) is about the **Architecture Capability**: an organization's ability to develop, maintain, and govern its Enterprise Architecture (p. 229). Section 9.1 defines it simply as the ability to practice architecture (p. 230). Section 9.2 shows that the capability is built like any other capability, by running through the ADM phases, each with specific focus areas (Table 9-1), and stresses that it is an ongoing discipline, not a project (pp. 230-236). Section 9.3 explains the four **Architecture Purpose Capabilities** from the Leader's Guide (support strategy, portfolio, projects, solution delivery) and how the capability's position and reporting line determine how strategic it is (pp. 236-241). Section 9.4 summarizes (pp. 241-245). Leave with this idea: *where* the capability sits in the organization chart decides whether architecture is proactive and strategic or reactive and solution-focused (pp. 240-241). Chapter 9 is part of the GREEN and GOLD reading tours (p. 229).

> **Reading note:** That the ADM can be used to establish the capability is the Standard's view (p. 230). The four purposes come from the Leader's Guide, a Series Guide (p. 237). Jager adds: which leader suits each purpose (CEO, CIO/CTO, unit manager) (pp. 237-239), the question of whether "solution delivery" EA would be better done by a Solution Architect (p. 238), and the practical note that the ADM steps for the capability often happen in one or two meetings (p. 236).

Detailed Section Breakdown

Chapter Opener (p. 229)

- **Definition and scope.** The ability to develop, maintain, and govern the EA. It includes governance, organizational structures, processes, roles, responsibilities, and skills, so that architecture work is standardized, repeatable, and aligned with strategy (p. 229).
- **Benefits.** Responding to change, managing complexity, better decisions, and properly governed and resourced initiatives (p. 229).

9.1 The Ability to Practice Architecture (p. 230)

- **A loose definition of capability:** "The ability of an organization to do something", where *something* here is "to practice architecture" (p. 230).

9.2 Creating the Architecture Capability (pp. 230-236)

- **Same as any capability.** "As with any business capability, the establishment of an Enterprise Architecture Capability can be supported by the TOGAF Architecture Development Method", especially Phase A (p. 230).
- **Ongoing discipline.** Not a separate phase or stand-alone project, but an ongoing discipline that provides context, environment, and resources (p. 231).
- **All four domains.** Like any capability, it needs Business, Data, Application, and Technology design (p. 231).

ADM phase	Focus areas for creating the Architecture Capability (Table 9-1)	Page
A: Architecture Vision	Define the project (stakeholders, which largely decides positioning); identify stakeholder concerns and business requirements; identify business goals and drivers (also shapes positioning); define scope (a high-level plan); define organization-wide constraints; perform an Architecture Maturity Assessment	pp. 231-233
B: Business Architecture	Agree on a common language; define the process (an adapted ADM, using only the useful parts); identify views and viewpoints (don't forget governance); decide on the framework (deliverables and their requirements); define performance metrics; agree on the Governance Framework (draw and document it)	pp. 231, 233-234
C and D: Data, Application, Technology	Specify and govern the Architecture Repository structure (the metamodel); define requirements for deliverables (create, maintain, publish, distribute, govern; toolset; publishing); define the technology and infrastructure	pp. 231, 234-235
E: Opportunities and Solutions	Determine the organizational change needed	pp. 232, 235
F and G: Migration Planning, Implementation Governance	Govern the organizational change	pp. 232, 235
H: Architecture Change Management	Manage change	pp. 232, 235
Requirements Management	Manage requirements: articulate and align all of them	pp. 232, 236

- **Maturity first.** How the capability can be implemented depends heavily on the organization's maturity level (p. 233; see Ch. 15).
- **Outcome.** The business goals and desired outcomes defined here largely determine the capability's position in the organization (p. 236).

Jager's commentary (author's view): In practice this is less formal. The steps are often agreed in one or two meetings, and the agreements should be recorded and, if possible, signed off by the main sponsor (p. 236).

9.3 Positioning the Architecture Capability (pp. 236-241)

- **Strategic tool advantage.** Organizations that see EA as a strategic business management tool, not a technical instrument, use it to translate strategy into implementation (p. 236).

Purpose (Leader's Guide, Figure 9-1)	What the capability does	Typical scope	Suggested leader / reporting line	Page
Architecture to Support Strategy	Target architectures and roadmaps for changes over 3-10 years. Shapes drivers, goals, objectives, initiatives. "Architecture in its most comprehensive form"	Organization-wide, many programs	CEO or strategy executive	pp. 237, 240
Architecture to Support Portfolio	Supports change initiatives spanning multiple functions, phases, and projects, e.g., a cloud migration portfolio	One portfolio	CIO (or CTO)	pp. 237-238, 240
Architecture to Support Projects	Fits the project delivery method. Clarifies purpose and value, identifies requirements, monitors governance, aligns projects, e.g., moving a CRM to the cloud	One project	CIO/CTO; reporting often to an IT unit manager	pp. 238, 240

Purpose (Leader's Guide, Figure 9-1)	What the capability does	Typical scope	Suggested leader / reporting line	Page
Architecture to Support Solution Delivery	Defines how the change is designed and delivered, with its constraints, controls, and requirements	One project or part of one	Unit manager; IT domain	pp. 238-240

- **Maturity assessment guides positioning.** An Architecture Maturity Assessment indicates where the capability should sit to achieve business goals. Position shows how the organization views architecture (p. 239).
- **Reporting line (Figure 9-2).** "The higher the direct reporting line... the more the Architecture Capability moves toward the strategic side" (pp. 239-240).
- **No right or wrong.** Position depends on the organization's needs (p. 240).
- **The risk of positioning too low.** Toward the operational side, work becomes solution-oriented, reactive, and disconnected from goals and objectives. Architecture risks becoming "a stand-alone entity" (p. 241).
- **Link to roles.** The capability relates directly to TOGAF's architecture roles, which relate to the domains (p. 241; see Ch. 10).

Jager's commentary (author's view): EA that only supports solution delivery "is actually similar to Solution Architecture", so the organization might be better served by a Solution Architect (p. 238). Strategy-level architecture "would benefit greatly from being led by a CEO" (p. 237).

9.4 Summary (pp. 241-245)

- The capability is a must, built via the ADM, and is an ongoing discipline (pp. 241-242).
- The creation steps are listed per phase (pp. 242-244).
- Four ways to position it. Agree on how the organization views it, use a maturity assessment to focus it, and remember that a higher reporting line means more strategic work (pp. 244-245).

Key Terms & Definitions

Term	Definition	Page
Architecture Capability	The organization's ability to practice (develop, maintain, govern) Enterprise Architecture	pp. 229-230
Capability (loose definition)	The ability of an organization to do something	p. 230
Architecture Maturity Assessment	An assessment of the current maturity of architecture, done in Phase A when creating the capability, which guides positioning	pp. 233, 239
Architecture Purpose Capabilities	The Leader's Guide's four uses: support strategy, portfolio, projects, solution delivery	p. 237
Positioning / reporting line	Where the capability sits in the organization. Higher means more strategic	pp. 239-240

Figures & Tables Referenced

Figure / Table	What it shows	Page
Table 9-1	ADM phases and focus areas for creating the Architecture Capability	pp. 231-232
Figure 9-1	Architecture Purpose Capabilities	p. 237

Figure / Table	What it shows	Page
Figure 9-2	Positioning of the Architecture Capability on an organization chart	p. 240

Self-Check Questions

Q: How does the TOGAF Standard say the Architecture Capability should be created?

A: Like any other business capability, by working through the ADM phases, particularly Phase A (p. 230).

Q: Is establishing the Architecture Capability a one-time project?

A: No. It is an ongoing discipline that provides the context, environment, and resources to govern and deliver architecture (p. 231).

Q: Which Phase A steps most influence where the capability is positioned?

A: Defining the project (its stakeholders) and identifying business goals and drivers (pp. 232-233).

Q: Name the six Phase B focus areas for creating the capability.

A: Agree on a common language; define the process; identify views and viewpoints; decide on the framework; define performance metrics; agree on the Governance Framework (pp. 233-234).

Q: What do Phases C and D focus on when creating the capability?

A: Specifying and governing the Architecture Repository's structure (metamodel), defining requirements for deliverables (including tools and publishing), and defining the technology and infrastructure (pp. 234-235).

Q: Name the four Architecture Purpose Capabilities.

A: Architecture to support strategy, portfolio, projects, and solution delivery (p. 237).

Q: What time horizon does "Architecture to Support Strategy" cover?

A: Three to ten years of organizational change (p. 237).

Q: What happens when the capability is positioned toward the operational side of the organization?

A: Work becomes solution-oriented and reactive, is less connected to goals and strategy, and architecture risks becoming a stand-alone entity (p. 241).

Q: Why does Jager suggest a Solution Architect for "support solution delivery"?

A: EA used only for solution delivery is effectively Solution Architecture (p. 238).

Q: What tool helps decide where the capability's focus should be?

A: An Architecture Maturity Assessment (pp. 233, 239).

Chapter 10: Architecture Roles, Skills, and Competencies

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 10 (pp. 247-278)

Overview

Chapter 10 (pp. 247-278) covers the people side of the Architecture Capability: which roles exist, how to tell jobs from roles, and what skills and competencies architects need. Section 10.1 distinguishes **job**, **role**, and **assignment** (pp. 248-249). Section 10.2 describes the core roles from the "Architecture Roles and Skills" Series Guide: Chief Architect (a specialization of Enterprise Architect), Enterprise, Segment, and Solution Architect (pp. 249-255). Section 10.3 examines the Segment Architect from a **horizontal** (architecture-domain) and a **vertical** (business-segment) perspective (pp. 255-263). Section 10.4 covers TOGAF's skill categories and proficiency levels, then adds Jager's own list of **ten competencies**, which the Standard lacks (pp. 263-277). Section 10.5 summarizes (pp. 277-278). Leave with this idea: each of the four Architecture Capabilities from Chapter 9 maps to a role, and architects need competencies (skills plus abilities plus knowledge), not just skills (pp. 259, 266). Chapter 10 is part of the GREEN and GOLD reading tours (p. 247).

> **Reading note:** Roles, skill categories, and proficiency levels come from the Standard and its Series Guides (G198, G249). The ten competencies and Table 10-3 are Jager's own addition, because "a list of competencies... is exactly what is missing from the TOGAF Standard" (p. 266). His anecdotes (the "Tribe Architect", p. 258; leaving an IT-minded employer, p. 267) and the Governance Framework inconsistency (Domain vs. Segment Architect, p. 262) are his commentary.

Detailed Section Breakdown

Chapter Opener (pp. 247-248)

- Establishing the Architecture Capability and forming the architecture team happen early in the Preliminary Phase and depend on defining the right roles, skills, and competencies. A skills framework helps (p. 247).
- The 2024 Series Guide "Architecture Roles and Skills" gives a fuller picture of skills per role, but ambiguity about jobs and roles persists in practice (pp. 247-248).

10.1 Difference Between Jobs, Roles, and Assignments (pp. 248-249)

Term	Definition (Architecture Roles and Skills)
Job	The formal position, usually defined by an employment contract and tied to hierarchy
Role	The expected set of activities, typically tied to a skill set
Assignment	A specific task or project a person is responsible for, usually within their job and role

(p. 248)

- **Why it matters.** Unclear whether "Enterprise Architect" is a job or a role, organizations struggle to position architecture, and it stays seen as part of IT, which needs different skills from strategic work (pp. 248-249).

- **Segment Architect is a job.** TOGAF created the Segment Architect profile to end confusion over domain roles (Technical, Application, Business Architect). It is a complete package with a role description and responsibilities (p. 249).

10.2 Core Architecture Roles (pp. 249-255)

- **Starting point to tailor.** People may hold several roles and roles may be shared. The guide gives three primary viewpoints (enterprise, segment, solution), which equal the three core roles. Specialist, user, and stakeholder roles are left to other professional frameworks (pp. 249-250, Figure 10-1).
- **Tiered model.** The Chief Architect at the top, then Enterprise, then Segment, with increasing depth for distinct parts of the landscape. A minimal three-tier model that can be expanded (organizations typically have 3-7 management levels) (pp. 250-251).

Role	Essence	Key responsibilities	Page
Chief Architect	Special case of the Enterprise Architect. Integrates the EA view into the wider strategic and operational context	Unifies executives and board around the landscape vision; makes concepts accessible; leads roadmaps for transformation; accountable for landscape governance; leads EA, Segment, and Solution Architects	pp. 251-252
Enterprise Architect	Seasoned expert steering transformative change that bridges business and technology across the organization	Leads major initiatives with a seat at the senior table; builds roadmaps across business segments; defines guiding principles and policies; establishes the framework for integration, interoperability, governance; coaches and mentors	pp. 252-253
Segment Architect	Manages and champions specific business or technical capabilities within a segment	Defines, shapes, and governs systems and components; creates segment blueprints; covers policies and standards; ensures compliance; fosters integration within the segment	pp. 253-254
Solution Architect	Drives delivery of new or evolved capabilities	Crafts solutions meeting immediate requirements and goals within service levels; decomposes into systems and components; builds roadmaps via Transition Architectures; ensures adherence to principles and standards	pp. 254-255

- **Dashed line (Figure 10-2).** It shows that the Chief Architect is a specialization of the Enterprise Architect, not a separate role (p. 251).

Jager's commentary (author's view): In Europe few organizations recognize the Chief Architect, so its duties are usually merged into the Enterprise Architect role (p. 252).

10.3 The Role of the Segment Architect (pp. 255-263)

- **Same job, different roles.** A Segment Architect assigned to infrastructure takes a Technology role (likely in IT). One assigned to business issues takes a Business Architecture role. The job is the same, but roles, skills, and competencies differ. Hierarchy follows the *assignment*, not the job description (p. 255).
- **Definitions.** "A segment groups capabilities in a structured way, often as part of a larger portfolio or specific change program." A domain is "any useful grouping of capabilities", with four segment-based viewpoints: business, application, data, technology (p. 256).

Perspective	Basis	Segment Architect becomes...	Source	Page
Horizontal	The four architecture domains	Business, Data, Application, or Technology Architect	Fundamental Content	pp. 255-256

Perspective	Basis	Segment Architect becomes...	Source	Page
Vertical	Organizational pillars (business domains), each holding several architecture domains	Architect for one pillar, e.g., Research, Consultation, Treatment	Only in the Roles and Skills Series Guide	pp. 256-257, 260-261

Jager's commentary (author's view): The vertical perspective makes role assignment hard. Each pillar often combines Business, Data, and Application, while Technology sits with IT "in nine out of ten cases". This leads to "nonsensical" new roles, such as a "Tribe Architect" in an agile organization (pp. 257-258).

10.3.1 The Horizontal Perspective (pp. 258-260)

Architecture Capability (Ch. 9)	Core role	Example specializations
Support strategy	Chief Architect and/or Enterprise Architect	n/a
Support portfolio	Segment Architect	Business, Data, Application, Technology Architect
Support projects	Segment Architect	Business, Data, Application, Technology Architect
Support solution delivery	Solution Architect	Cloud, Network, Software Architect

(pp. 259-260, Figures 10-4, 10-5)

- If there is no Chief Architect, the Enterprise Architect takes the strategic side as well (p. 259).
- Combining roles into one creates ambiguity about which capability is delivered and who is responsible (p. 260).

10.3.2 The Vertical Perspective (pp. 260-263)

- **Pillars as segments.** In healthcare, the pillars might be Research, Consultation, and Treatment. Each holds a Segment Architecture spanning all four domains (pp. 260-261, Figure 10-6).
- **More operational.** Owning Application and Technology domains makes the vertical Segment Architect's work more substantive, overlapping with Solution Architect work (p. 261).
- **Segment vs. Solution Architecture (Figure 10-7).** A Solution Architecture is a discrete, focused business operation and how IT supports it, usually for a single project or release. A Segment Architecture covers a whole pillar, whose services are used across other processes (pp. 261-262).

Jager's commentary (author's view): The Governance Framework (Ch. 8) names Chief, Enterprise, and *Domain* Architects but not Segment or Solution Architects. This is an inconsistency that The Open Group told him it is gradually fixing (pp. 262-263).

10.4 Skills and Competencies (pp. 263-277)

- **Division of labour.** The Enterprise Architect does the strategic work, the Segment Architect details segments or domains, and the Solution Architect details proposed solutions (p. 264).
- **Skills frameworks.** Define roles in a work area, the skills each needs, and the depth required. The "Architecture Skills Framework" (G198) was updated in September 2024 as "Architecture Roles and Skills" (G249) (p. 264).

Skill category (Table 10-1)	Examples	Page
Application	Application management, API patterns, software design principles	p. 265

Skill category (Table 10-1)	Examples	Page
Architecture	Modelling, building block design, principle design, migration planning, Security Architecture, requirements management, governance	p. 265
Business	Enterprise organization knowledge, business cases, business process, strategic planning	p. 265
Change	Managing business change, project management methods and tools	p. 265
Data	Data analysis, exchange, management	p. 265
Generic	Leadership, teamworking, interpersonal skills	p. 265
Legal	Data protection, contract, procurement law, fraud	p. 265
Technology	Compute, storage, network infrastructure	p. 265

Proficiency level (Table 10-2)	Meaning
1 Background	Basic understanding, reliant on specialists
2 Aware	Understands and can address issues, with specialist support
3 Knowledgeable	Enough detail for effective decisions. Specialist input only in difficult cases
4 Expert	Extensive experience, practical and applied knowledge

(p. 266)

- **Competency vs. skill (Figure 10-8).** A competency combines skills, abilities, and knowledge. The Standard lists skills and proficiency levels but no competencies (p. 266).

Jager's ten competencies	Core idea	Mainly needed by (Table 10-3)	Page
Strategic thinking and vision	Turn long-term vision into strategy and roadmap initiatives. Anticipate. Think holistically	Enterprise (and Segment)	p. 268
Strategic decision-making	Decide under uncertainty. Present scenarios and trade-offs (Ch. 14.9)	Enterprise, Segment	p. 269
Leadership and influence	Inspire teams, build consensus, be a trusted advisor	Enterprise	p. 269
Change leadership	Drive change with resilience. Handle resistance. Be culturally sensitive	Enterprise, Segment	p. 269
Business and technology alignment	Bridge strategy and technology. State tech value in business terms	Enterprise, Segment	pp. 269-270
Communication and collaboration	Explain complex ideas to any audience. Listen	Segment, Solution (listening: all)	p. 270
Governance and compliance	Enforce standards. Ensure regulatory adherence. Measure effectiveness	All three	p. 270
Adaptability and resilience	Adapt to change, keep composure, keep learning	All three	p. 270
Analytical thinking and problem solving	Break problems down, find root causes, apply structured creativity	All three	p. 271
Systems thinking	Understand interconnectedness. Design scalable, sustainable solutions	Segment, Solution	p. 271

- Each competency maps to skills in the Series Guide (Figure 10-9, p. 276). The SFIA 9 skills directory is suggested for further reading (p. 277).

Jager's commentary (author's view): An employer that saw architecture as IT reduced his strategic role to IT scope, and he ended the contract. Agree upfront on the perspective from which the architecture work

will be done, so that skills and competencies match (pp. 267-268).

10.5 Summary (pp. 277-278)

- Three viewpoints equal three core roles: Enterprise (with the Chief specialization), Segment (formerly Domain), and Solution (pp. 277-278).
- The organization can be viewed horizontally (four domains) or vertically (service segments) (p. 278).
- The Standard covers only skills and proficiency. The book adds ten competencies mapped to those skills (p. 278).

Key Terms & Definitions

Term	Definition	Page
Job / Role / Assignment	Formal contractual position / expected activities tied to a skill set / specific task or project	p. 248
Chief Architect	Specialization of the Enterprise Architect that integrates the EA view into strategic and operational context	p. 251
Segment Architect	Job profile (formerly Domain Architect) for a segment of the architecture, horizontal or vertical	pp. 249, 263
Segment	A structured grouping of capabilities, often part of a portfolio or change program	p. 256
Horizontal perspective	Segments based on the four architecture domains	pp. 255-256
Vertical perspective	Segments based on organizational pillars or business domains	pp. 256, 260
Skills framework	Defines roles, skills per role, and depth of knowledge needed	p. 264
Proficiency levels	Background, Aware, Knowledgeable, Expert	p. 266
Competency	Skills plus abilities plus knowledge	p. 266

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figures 10-1, 10-2	Core architecture roles and relationships	pp. 250-251
Figure 10-3	Business domains/segments vs. architecture domains	p. 257
Figures 10-4, 10-5	Architecture Capabilities mapped to roles and specializations	pp. 259-260
Figure 10-6	Vertical perspective and associated architectures	p. 261
Figure 10-7	Segment vs. Solution Architectures	p. 262
Table 10-1	Skill categories	p. 265
Table 10-2	Proficiency levels	p. 266
Figure 10-8	Competencies vs. skills	p. 267
Table 10-3	Required competencies per core role	pp. 272-275
Figure 10-9	Competencies mapped to skills	p. 276

Self-Check Questions

Q: Define job, role, and assignment.

A: Job: the formal contractual position. Role: the expected set of activities tied to a skill set. Assignment: a specific task or project, usually within the job and role (p. 248).

Q: What are the three core architecture roles, and how does the Chief Architect relate to them?

A: Enterprise, Segment, and Solution Architect. The Chief Architect is a specialization of the Enterprise Architect, not a separate role (pp. 251, 263).

Q: What did the Segment Architect replace, and is it a job or a role?

A: It replaced the Domain Architect. It is a job profile that includes a role description and responsibilities (pp. 249, 263).

Q: Match each of the four Architecture Capabilities to a core role.

A: Strategy: Chief/Enterprise Architect. Portfolio and projects: Segment Architect. Solution delivery: Solution Architect (p. 259).

Q: Contrast the horizontal and vertical perspectives on segments.

A: Horizontal segments follow the four architecture domains. Vertical segments follow organizational pillars, each containing several or all domains. The vertical view appears only in the Series Guide (pp. 255-257, 260).

Q: Why does the vertical perspective make the Segment Architect role more operational?

A: The architect becomes responsible for the Application and Technology domains in the pillar, which is usually Solution Architect territory (p. 261).

Q: Name the four proficiency levels.

A: 1 Background, 2 Aware, 3 Knowledgeable, 4 Expert (p. 266).

Q: How does a competency differ from a skill?

A: A competency consists of skills plus abilities plus knowledge (p. 266).

Q: Name at least six of Jager's ten competencies.

A: Strategic thinking and vision; strategic decision-making; leadership and influence; change leadership; business and technology alignment; communication and collaboration; governance and compliance; adaptability and resilience; analytical thinking and problem solving; systems thinking (pp. 268-271).

Q: What inconsistency does Jager note between the Governance Framework and the roles Series Guide?

A: The Governance Framework lists Chief, Enterprise, and Domain Architects, while the Series Guide uses Segment Architect and adds Solution Architect, which the framework does not mention (p. 262).

Q: Why should architect and organization agree on perspective before work starts?

A: A mismatch between the architect's (e.g., strategic) competencies and the organization's (e.g., IT-only) view of the discipline leaves both dissatisfied (pp. 267-268).

Chapter 11: Architecture Domains

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 11 (pp. 279-294)

Overview

Chapter 11 (pp. 279-294) explains the four architecture domains that every architect works in and connects them to the roles from Chapter 10. Section 11.1 defines Business, Data, Application, and Technology Architecture, notes the history of an "Information" domain, introduces the Interoperability Model, and states that a *complete* EA must address all four domains (pp. 280-286). Section 11.2 separates **domains** (areas of concern) from **layers** (levels of abstraction inside a system), and introduces cross-cutting concerns such as security, motivation, and governance (pp. 286-289). Section 11.3 maps roles to domains (pp. 289-292). Section 11.4 summarizes (pp. 292-294). Leave with this idea: if you skip a domain, what you produce is by definition not a complete Enterprise Architecture, and management must understand the risks of that choice (pp. 285-286). Chapter 11 is part of the PINK and GREEN reading tours (p. 279).

> **Reading note:** Domain definitions and the "complete EA" rule are from the Standard. Jager's views: Information Architecture may return (pp. 282, 285); data + context = information (p. 284); his critique that ACMM treats security only as IT security (p. 288); and his role-to-domain scope figures (pp. 290-292). Small text slips: p. 285 says the "2008 eighth edition", but Ch. 2 dates TOGAF 8 to 2002 (p. 22), and the Summary calls the domains "roles" (p. 292).

Detailed Section Breakdown

Chapter Opener (p. 279)

- **Roles and capabilities recap.** Chief/Enterprise Architect supports strategy, Segment Architect supports portfolios (and sometimes large projects), Solution Architect supports solutions. Every architect deals with one or more domains (p. 279).

11.1 Four Architecture Domains (pp. 280-286)

Domain	Definition	Page
Business Architecture	Business strategy, governance, organization, and key business capabilities, so IT supports business objectives	p. 280
Data Architecture	Structure of logical and physical data assets and data management resources	p. 280
Application Architecture	Blueprint of individual systems, their interactions, and their relationships to core business processes	p. 281
Technology Architecture	Hardware, software, and network infrastructure supporting mission-critical applications	p. 281

- **Information Systems Architectures.** The collective name for Data and Application Architecture (p. 281, Figure 11-1).
- **History.** Four domains since TOGAF 8. TOGAF 7 recognized only Technology Architecture. TOGAF 8 called the data domain "Data *or Information* Architecture", and the "or Information" was later dropped (pp. 281-282).

- **Interoperability Model (Figure 11-2).** A five-layer model used in healthcare. It splits Business into organization and processes and uses Information instead of Data. Otherwise it matches TOGAF's four domains, with legislation/regulation and security as preconditional columns (pp. 282-283).
- **Data entity vs. information concept.** TOGAF's Data Architecture has the data entity but no information concept, while BIZBOK describes information concepts in much more detail. The two are not comparable (pp. 283-284).
- **Data + Context = Information.** "1 Primary Lane" means little until you add that it is the President's home address. A list of numbers becomes information once you know they are next week's temperatures (p. 284).
- **Complete EA rule.** "A complete Enterprise Architecture should address all four architecture domains." Time and resource limits often prevent this, and architecture descriptions are scoped to purpose in the first ADM phase (p. 285).
- **Partial architectures.** If business processes won't change, a full Business Architecture may not be needed, but it must still be understood, because Data, Application, and Technology build on it (p. 285).
- **Risk of incompleteness.** Inconsistency and integration problems, with later costs. The architect must spell out the risks and trade-offs so management understands them (pp. 285-286).

Jager's commentary (author's view): With growing BIZBOK integration, the Information Architecture domain may reappear in TOGAF (pp. 282, 285).

11.2 Difference in Domains and Layers (pp. 286-289)

- **Organizations' part.** Architects translate strategy into execution, but organizations must define their strategy and work with architects on feasible plans. EA helps shape drivers, goals, objectives, and initiatives (p. 286).
- **Broad scope.** Organizational design, process design, information flow visualization, and so on. EA, once technology-focused (around 1990), now covers all aspects of an organization (p. 286).

	Architecture domains	Architecture layers
What	Broad divisions: areas of concern or expertise in the overall architecture	Levels of abstraction within a system or Solution Architecture
Examples	Business, Data, Application, Technology	Presentation, business logic, data access, infrastructure (Figure 11-3)
Purpose	Categorize the concerns and perspectives to address	Organize components and functionality, communicating via interfaces, for modularity and reuse

(pp. 287-289)

- **Complementary.** Layers organize systems *within* a domain (p. 289).
- **Cross-cutting concerns (Figure 11-4).** Security (sometimes called a domain, but it cuts across all of them), motivation, and governance (pp. 288-289).

Jager's commentary (author's view): The ACMM maturity model (Ch. 15) lists IT security as one of nine elements, which ignores that security is cross-cutting (p. 288).

11.3 Mapping Roles to Domains (pp. 289-292)

Role	Domain scope	Figure / page
Chief/Enterprise Architect	The entire ecosystem: all four domains. A Chief Architect focuses on strategy. Without one, the EA does both	Figure 11-6, p. 290

Role	Domain scope	Figure / page
Segment Architect	One domain per specialization: Business, Data, Application, or Technology Architect	Figure 11-7, p. 291
Solution Architect	A defined (usually technical) project. "Nine out of ten" times within Technology. Specializations: Cloud, Network, Software	Figure 11-8, pp. 291-292

- Practice differs, but recognize the difference between roles and specializations and don't hold false expectations of a role (p. 292).

11.4 Summary (pp. 292-294)

- The four domains, the Interoperability Model alternative, and the complete-EA rule (pp. 292-293).
- Domains vs. layers. Security, motivation, and governance as cross-cutting concerns (p. 293).
- Role-domain mapping: EA covers all four, Segment Architects one each, Solution Architects mostly Technology (p. 294).

Key Terms & Definitions

Term	Definition	Page
Business Architecture	Business strategy, governance, organization, key capabilities	p. 280
Data Architecture	Logical and physical data assets and data management resources	p. 280
Application Architecture	Blueprint of systems, interactions, and relation to core processes	p. 281
Technology Architecture	Hardware, software, and network infrastructure	p. 281
Information Systems Architectures	Collective name for Data and Application Architecture	p. 281
Interoperability Model	Five-layer model (healthcare) splitting business into organization and process, with an information layer	pp. 282-283
Architecture layer	A logical division of a software system by function (presentation, logic, data access, infrastructure)	p. 287
Cross-cutting concern	A concern spanning all domains: security, motivation, governance	pp. 288-289

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 11-1	The four core architecture domains	p. 280
Figure 11-2	Interoperability Model (five-layer model)	p. 283
Figure 11-3	Architecture layers	p. 287
Figure 11-4	Domains and cross-cutting concerns	p. 289
Figure 11-5	Roles and specializations vs. domains	p. 290
Figures 11-6 to 11-8	Scope of the Chief/Enterprise, Segment, and Solution Architect	pp. 290-292

Self-Check Questions

Q: Which two domains make up the Information Systems Architectures?

A: Data Architecture and Application Architecture (p. 281).

Q: Can an architecture that covers only three domains be called a complete Enterprise Architecture?

A: No. By definition a complete EA addresses all four domains. Partial work risks inconsistency and later integration costs, which must be explained to management (pp. 285-286).

Q: If business processes won't change, can you skip Business Architecture?

A: A comprehensive Business Architecture may not be needed, but it must still be considered and understood, because the other domains build on it (p. 285).

Q: How do architecture domains differ from architecture layers?

A: Domains are broad areas of concern across the whole architecture. Layers are levels of abstraction inside a system (presentation, logic, data access, infrastructure) (pp. 287-289).

Q: Name three cross-cutting concerns.

A: Security, motivation, and governance (pp. 288-289).

Q: Explain "Data + Context = Information" with the book's example.

A: "1 Primary Lane" is meaningless raw data until you add the context that it is the President's home address. Then it becomes information (p. 284).

Q: How does the Interoperability Model differ from TOGAF's four domains?

A: It splits Business into organization and processes and uses Information instead of Data. Otherwise it matches TOGAF's four domains (pp. 282-283).

Q: Which domain does a Solution Architect usually work in, and why?

A: Technology, because the scope is usually a defined technical project (p. 291).

Q: What did TOGAF 8 call the data domain, and why does Jager find that interesting?

A: "Data or Information Architecture". The "or Information" was later dropped, but BIZBOK integration may bring Information Architecture back (pp. 282, 285).

Chapter 12: Architecture Development Method

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 12 (pp. 295-560)

Overview

Chapter 12 (pp. 295-560) is the longest chapter in the book. It walks through the **Architecture Development Method (ADM)**, "the heart of the TOGAF Standard" (p. 295). Section 12.1 introduces the ten phases and their objectives (pp. 296-300). Section 12.2 explains the four key documents created along the way: Request for Architecture Work, Statement of Architecture Work, Architecture Definition Document, and Architecture Contract (pp. 300-305). Sections 12.3 to 12.12 cover each phase in turn: its steps (in tables), Jager's short "steps" boxes, and the artifacts it can produce, all illustrated with the fictional healthcare organization. The phases are Preliminary (12.3), A Architecture Vision (12.4), B Business (12.5), C Information Systems, both Data and Application (12.6), D Technology (12.7), E Opportunities and Solutions with the Architecture Roadmap (12.8), F Migration Planning (12.9), G Implementation Governance with performance monitoring (12.10), H Architecture Change Management (12.11), and Requirements Management (12.12). Section 12.13 assembles the ADM documents, and Section 12.14 summarizes. Leave with this idea: the ADM describes everything you *can* do, not what you *must* do. It does not prescribe an order and must be tailored. It is not a waterfall (p. 295). Chapter 12 is part of the PINK, GREEN, and BLUE reading tours (p. 295).

> **Reading note:** The phase objectives, steps, and artifact names follow the Standard. The healthcare examples, the shaded "steps" boxes, and the personal stories (hospital strategy documents, p. 309; others later) are Jager's. Jager also adds artifacts the Standard lacks, such as the Work Package Portfolio Map (see Ch. 6 and 14). Each phase has the same fixed components: **key objectives, steps, outputs, and approach** (p. 300).

Detailed Section Breakdown

Chapter Opener (p. 295)

- **Iterative, not waterfall.** The ADM guides the creation and maintenance of artifacts aligned with business goals. "Don't be fooled by what seems to be a waterfall approach at first sight" (p. 295).
- **Descriptive, not prescriptive.** It "describes all the actions one can take, not what must be taken", does not prescribe an order, and is to be tailored (p. 295).

12.1 The Heart of the TOGAF Standard (pp. 296-300)

- **A method, not a process.** It is "still too often mistaken for a process". Its steps are possible steps that may be skipped or adapted (p. 296).

Phase	Key objective(s)	Page
Preliminary	Determine the Architecture Capability the organization needs, and establish it	p. 297
A: Architecture Vision	A high-level vision and stakeholder alignment, with business requirements articulated and agreed	p. 298

Phase	Key objective(s)	Page
B, C, D: Business, Information Systems, Technology	(1) Craft the Target Architecture. (2) Identify candidate Architecture Roadmap components from Baseline-Target gaps	p. 298
E: Opportunities and Solutions	Identify work packages; decide whether an incremental approach is needed; define the building blocks to complete the target	p. 298
F: Migration Planning	A detailed implementation plan	p. 298
G: Implementation Governance	Ensure adherence, manage changes during implementation, assess progress	p. 299
H: Architecture Change Management	Continuously assess changes in strategy, environment, and technology that affect the architecture	p. 299
Requirements Management	Not a separate phase: a continuous activity managing requirements throughout	p. 299

- **Artifacts must add value.** "A picture is worth a thousand words". The chapter's artifacts are healthcare examples to adapt (pp. 299-300).

12.2 Architecture Document Creation (pp. 300-305)

Document	Purpose	Created / approved	Page
Request for Architecture Work	Formal, high-level request from the sponsoring organization to the architecture team that triggers a cycle	Preliminary Phase	pp. 300, 302
Statement of Architecture Work	Scope and approach of the cycle. A benchmark for success and effectively a contract between the architecting organization and the sponsor (project description and scope, vision overview, acceptance criteria, plan, approvals)	Phase A	pp. 301, 303
Architecture Definition Document	"Deliverable container" for core artifacts across all four domains and all states (Baseline, Transition, Target)	Built in B-D. Target approved in E (with draft Transitions). Finalized in F	pp. 301, 303-304
Architecture Contract	Joint agreement between development partners and sponsors on deliverables, quality, and fitness for purpose	Phase E onward. Starts transition and governance	pp. 301, 304-305

- **Flow (Figure 12-2).** Request (Preliminary). Statement with candidate building blocks (A). Baseline then Target descriptions per domain, reusing the repository (B-D). Building blocks recorded in the Architecture Landscape, future standards in the repository. Gap analysis then Architecture Roadmap. Stakeholder reviews. ADD approved (E) and finalized (F). Missing blocks linked to work packages (E). Chosen solution in an Architecture Contract (pp. 302-304).
- **Loop back.** If change during execution needs major architecture changes, a new Request for Architecture Work starts a new ADM iteration (p. 305).

12.3 Preliminary Phase (pp. 305-323)

- **Purpose.** Get the organization ready: align stakeholders, define principles, tailor TOGAF, and put frameworks, tools, and governance in place. Skipping it risks misalignment and poor management of the initiative (p. 305).

12.3.1 Determining and Establishing the Architecture Capability (pp. 305-319)

- **Two objectives.** *Determine* the capability needed (scope, stakeholders, their requirements, current and target maturity, readiness for change) and *establish* it (pp. 305-306).

Step (Table 12-1, determining)	Description	Output	Page
Define the enterprise	Scope the impacted organization and its structure. Identify stakeholders	Scope of impacted organization(s)	pp. 306-307
Identify key drivers and organizational context	Find the factors that influence the architecture	Baseline of objectives, goals, drivers	pp. 306, 308
Manage relationships between frameworks	Understand how management frameworks interact. Set up a framework to manage architecture activities	EA Capability structure. Architecture Governance Framework	pp. 306, 310-312
Assess EA maturity	Evaluate where the organization stands	Maturity assessments, gaps, strategies	pp. 306, 312-313

- **Define the enterprise.** Stakeholders include those most affected who gain most value, those whose work changes, and external parties. Meet them, discuss the scope, recruit more stakeholders, and iterate (Stakeholder Management, Ch. 14.3) (p. 307).
- **Key drivers.** No earlier phase gives input, so use the TOGAF Library, other frameworks, and business and IT strategies, principles, goals, and drivers. Distill them from documents and verify them with stakeholders (p. 308).
- **Manage relationships between frameworks.** Reuse existing maturity assessments, roles, budgets, methods, tools, and principles, but remove what is outdated rather than copying it. Use PPM frameworks too. Set up an Architecture Governance Framework covering how architecture materials are managed, repository requirements, and governance processes (compliance, retirement), with stakeholder reviews of the models (pp. 310-312).
- **Assess maturity.** The "Architecture Maturity Models" Series Guide names ACMM (often IT-related) and CMMI (originally software and engineering, now also used for EA because of its business-goal focus). Choose what fits (Ch. 15) (pp. 312-313).
- **Readiness for change.** Use the Business Transformation Readiness Assessment (Ch. 14.7) (p. 313).

Step (Table 12-2, establishing)	Description	Output	Page
Establish the EA team	Form a team to do the architecture work	Architecture team. Defined roles and responsibilities	pp. 314-316
Set Architecture Principles	Define principles guiding all architecture efforts	Architecture Principles. Constraints for architecture efforts	pp. 314, 316-317
Choose and tailor the architecture framework	Tailor the TOGAF Standard, the ADM, and the Content Framework	Tailored framework, ADM, Content Framework	pp. 314, 317-318

- **The team.** Not only architects: project managers, senior managers, and others who grasp architecture can join. Composition follows positioning (IT-based means architects plus technical experts; strategic means EA plus senior and portfolio managers). "Architecture teams" are not the same as teams of people with the job title architect (pp. 314-315).
- **Example RACI (Table 12-3).** The Enterprise Architect is Accountable for every activity and also Responsible for creating the vision and implementing governance. Segment Architects are Responsible for Segment Architectures, opportunities and solutions, the migration plan, change, and requirements. Solution Architects are Responsible for opportunities, migration, change, and requirements, and Consulted on the rest (pp. 315-316).

- **Tailoring sequence.** Agree on and publish a glossary. Adapt the Enterprise Metamodel (new entities, deliverables, artifacts). Adapt the ADM (remove redundant tasks, add checkpoints, integrate with other processes). Customize the Content Framework. Set a tools strategy (general and specialist tools, business modelling). Formalize change control (pp. 317-318).
- **Not isolated.** EA frameworks must interoperate with broader business frameworks. Partition teams, processes, and deliverables to fit size and budget. You may need to revisit the Preliminary Phase from Phase A (p. 319).
- **In essence.** "The Preliminary Phase is about building the architecture needed to support the Architecture Capability" (p. 319).

Jager's commentary (author's view): In a hospital with little EA, he hand-mined stacks of strategy documents for drivers, goals, and objectives, built a Driver/Goal/Objective Diagram (Figure 12-3), and refined it with senior management into a Solution Concept Diagram. Today AI tools can speed up this scanning (pp. 309-310). Architecture decisions belong to stakeholders, not the architect (p. 311).

12.3.2 Preliminary Phase Artifacts (pp. 320-323)

- **Principles Catalog** (entity: principle). It holds Enterprise Principles (high-level guidelines shaping strategy, culture, decisions) and Architecture Principles (what a good solution looks like). Principles are used to decide at decision points and to govern change (p. 320).
- **Example principles (Table 12-5):** Comply with laws and regulations; Unambiguous ways of working; Store information once and use it more often (data sharing will "bump up against" data security); Standard products and services (no duplicative capabilities). Each has a Name, Statement, Rationale, and Implications (pp. 321-323).

12.4 Phase A: Architecture Vision (pp. 324-347)

- **The first actionable phase.** It sets the objectives, scope, and expected outcomes, building on Preliminary input (p. 324).

12.4.1 Creating the Architecture Vision (pp. 324-335)

- **Main goal.** A high-level vision and stakeholder alignment, with clearly agreed business requirements and initial approval to proceed. The result goes in an architecture vision document (p. 324).
- **Inputs.** Preliminary Phase reference material, business principles, goals, and drivers. Existing documentation is used to populate the first version of the Architecture Repository, e.g., starting capabilities as ABBs in the Architecture Landscape and the RACI in the Architecture Capability section (p. 324).

Step (Table 12-6)	Description	Output	Page
Establish the architecture project	Formal project recognition, leadership endorsement, line management support	Statement of Architecture Work	pp. 325-327
Identify stakeholders and determine requirements	Identify stakeholders and requirements so the work can be communicated and aligned	Stakeholder Catalog; Communications Plan	pp. 325, 327-328
Assess readiness for business transformation	Evaluate change readiness to guide scope and flag attention areas	Business Transformation Readiness Assessment	pp. 325, 329-330
Elaborate on business goals and Architecture Principles	Validate and refine goals and principles from the Preliminary Phase	Validated objectives, goals, drivers, principles	pp. 325, 330-331
Create a high-level architecture vision	High-level solution description aligned with strategy	Solution Concept Diagram	pp. 325, 331-332
Define the Target Architecture value propositions and KPIs	Business case, performance metrics, and measures	Value propositions and KPIs	pp. 325, 333

Step (Table 12-6)	Description	Output	Page
Identify business transformation risks and mitigation activities	Identify risks, assess their level and frequency	Risks; mitigating activities	pp. 326, 333-334
Develop Statement of Architecture Work	Draft it and secure formal approval	Statement of Architecture Work	pp. 326, 334-335

- **Run it as a project.** Each ADM iteration is ideally a project in the existing project management framework, standalone or part of broader programs (p. 326).
- **Stakeholders continue from Preliminary.** Use stakeholder analysis, mapping, and a Communications Plan (Ch. 14.3.1-14.3.3). Stakeholder requirements and concerns feed the Solution Concept Diagram (pp. 327-328).
- **Readiness factors (examples).** Vision, leadership (executive sponsorship), stakeholder engagement, governance, culture and communications. Assess each for risks and interventions. The results shape scope, activities, and risk areas (Ch. 14.7) (p. 329).
- **Goals and principles.** Validate them, resolve ambiguity, get senior management approval, and outline organization-wide and architecture-specific constraints (timelines, resources) (p. 330).
- **The vision is not a detailed solution.** The Solution Concept Diagram "simply shows the relationships between the organization's goals and objectives on the one hand and the stakeholder requirements and constraints on the other" (p. 332).
- **Value propositions and KPIs.** A business case. Value propositions per stakeholder group for approval. Metrics to monitor whether requirements are met. High-level business risks recorded in the Statement of Architecture Work (p. 333).
- **Business transformation risk.** Change almost always brings risk. Identify the initial level and frequency. Mitigation can wait for the Consolidated Gaps, Solutions, and Dependencies Matrix in Phase E, managed with the Risk Management technique (Ch. 14.8) (pp. 333-334).
- **Closing Phase A.** Secure formal approval of the Statement of Architecture Work plus funding, resources, and time. Communicate the benefits and keep stakeholders engaged (pp. 334-335).

12.4.2 Architecture Vision Phase Artifacts (pp. 336-347)

Artifact (Table 12-7)	Entities used	What it shows	Page
Stakeholder Catalog	None (or stakeholder, once added)	Stakeholders, their influence, key concerns, classification, relevant artifacts	pp. 336-339
Value Chain Diagram	Value stream	High-level primary and support activities, for presentation impact	pp. 339-343
Solution Concept Diagram	Objective, requirement, constraint	A "preliminary pencil sketch" of the solution to align stakeholders	pp. 343-344
Business Model Diagram	Actor, role, stakeholder, contract, function, process, business capability, business service, product	How the organization creates, delivers, and captures value (Business Model Canvas)	pp. 343-345
Business Capability Map	Business capability	Capabilities needed for the goals (high-level at this stage)	pp. 345-346
Value Stream Map	Value stream	End-to-end value-adding activities	pp. 346-347

- **Stakeholder Catalog example (Table 12-8).** CEO Maria Summers (keep satisfied / key player; strategy and goals), CIO Chris West (IT-business alignment), CFO James Noble (keep informed;

budget and cost), HR Manager Anna Carter (minimal effort; workforce adoption). Each is linked to the matrices and catalogs that interest them (pp. 338-339).

- **Why no stakeholder entity?** The Standard cites the sensitivity of stakeholder mapping and Phase A's informal modelling (p. 336).
- **Value Chain Diagram example (Figure 12-4, hospital).** Primary: Patient Admission and Triage; Diagnosis and Treatment; Patient Care and Monitoring; Discharge and Follow-Up. Support: IT and Medical Systems; HR and Training; Facilities and Equipment; Finance and Administration. The colour pairings show which support activity enables which primary one (pp. 340-343).
- **Business Model Diagram (Figure 12-6).** A hospital canvas. It uses entities missing from the foundation metamodel (business collaboration, interaction, interface, resource, value), so consider adding them when tailoring (Ch. 5.2) (p. 345).
- **Capability map gets refined in B.** Phase B adds more capability levels (p. 346).
- **Vision sets scope** for the following phases. Next comes Phase B (p. 347).

Jager's commentary (author's view): Stakeholders defined by role pose little misuse risk, so add a stakeholder entity when tailoring the metamodel (pp. 337-338). The "Business Capability Map" and "Value Stream Map" are really *diagrams* by TOGAF's own definition of map vs. diagram (Ch. 7.4.2), so his examples are drawn as a Business Capability Diagram and a Value Stream Diagram (pp. 346-347).

12.5 Phase B: Business Architecture (pp. 347-398)

- **Why first.** Business Architecture gives multi-dimensional views of capabilities, value delivery, information, and organization, and links them to goals. "It lays the groundwork for all architectural endeavors" and should ideally be the first architecture activity. It also shows stakeholders the business value and return on their support (pp. 347-348).
- **Scope from Phase A.** Strategy states goals, drivers, and metrics but not the path. Business Architecture provides the path (p. 348).

12.5.1 Developing Business Architecture (pp. 348-360)

- **Two objectives.** (1) A Target Business Architecture showing how the organization should function to meet objectives and address the drivers from Phase A. (2) Candidate roadmap components from Baseline-Target gaps. Reuse as much as possible (p. 348).
- **Inputs.** External reference material. Refined principles, goals, and drivers. Assessment of Preliminary capabilities. A Communications Plan. The organizational model (scope, maturity), the RACI, constraints, budget, and the governance and support strategy (p. 349).

Step (Table 12-9)	Description	Output	Page
Select reference models, viewpoints, and tools	Pick repository resources, viewpoints (operations, management, finance), and tools that fit	Views, viewpoints, required building blocks, requirements	pp. 350-352
Develop Baseline and Target Business Architecture description	Describe the current and future state only as far as needed for the vision	Baseline and Target descriptions	pp. 350, 352-353
Perform gap analysis	Check consistency, validate against principles, objectives, and constraints, check completeness, find gaps (Ch. 14.5)	Gap analysis results	pp. 350, 353-354
Define candidate roadmap components	Turn gaps into work packages for a draft roadmap, refined in Phase E	Draft Architecture Roadmap; Business Architecture roadmap components	pp. 350, 354-355

Step (Table 12-9)	Description	Output	Page
Resolve impacts across the Architecture Landscape	Check effects on other architectures and projects, and reuse opportunities	Validated building blocks and artifacts	pp. 351, 355-356
Conduct formal stakeholder review	Executives, process owners, and SMEs revisit the original motivation and the Statement of Architecture Work	Statement of Architecture Work	pp. 351, 356-357
Finalize the Business Architecture	Standardize and document building blocks (Ch. 7.5.2.2), document the rationale for each decision, trace requirements to elements	Documented building blocks and requirements	pp. 351, 357-358
Create the Architecture Definition Document	At minimum Baseline and Target Business Architectures, with visuals, iterated with stakeholder feedback	Architecture Definition Document	pp. 351, 358-359

- **Level of detail.** It depends on how much of the existing structure carries over and whether descriptions already exist. Explore alternative targets with Architecture Alternatives and Trade-Offs (Ch. 14.9) (pp. 352-353).
- **Out-of-scope requirements** go to the Requirements Repository through governed requirements management (Section 12.12) (p. 354).
- **Impact questions.** Does this Business Architecture affect existing architectures? Have recent changes affected it? Can it be reused elsewhere? Does it affect projects, and do projects affect it? (p. 355).
- **Repository use begins.** Artifacts are now recorded in the repository and viewed through the Enterprise Continuum (Ch. 7.3.1). The vision supplies the *why* behind the Statement of Architecture Work (pp. 359-360).

12.5.2 Business Architecture Methods and Modeling Techniques (pp. 360-366)

Technique	What it does	Page
Applying business capabilities	Link the Phase A Business Capability Map to org units, value streams, systems, and objectives	p. 360
Business Capability Heat Mapping	Colour capabilities by maturity, effectiveness, performance, value, or cost. Green = at target, yellow = one level below, red = two or more below, purple = desired but missing (or over-resourced)	pp. 361-362
Applying value streams	Value streams give the *why*, capabilities the *what*. Map stages to capabilities for gap analysis	p. 362
Applying the Organization Map	A network of units, partners, and stakeholders (who owns or uses capabilities and takes part in value streams). Not a reporting chart	pp. 362-363
Applying Information Maps	Map core business information and build matrices from information concepts to capabilities	p. 363
Other modelling	Activity models (BPMN), use-case models, logical data or class models ("is-a" relationships, e.g., sedan, car, vehicle)	pp. 364-365
Business Scenarios	Iteratively test and modify the assumptions behind a project, mainly in Preliminary, A, and B	p. 365

Jager's commentary (author's view): There is "no scientific approach" to rating capability maturity. Heat mapping relies on stakeholder alignment, gut feeling, and common sense (p. 362).

12.5.3 Phase B (Business Architecture) Artifacts (pp. 366-398)

Artifact	Purpose (healthcare example)	Page
Organization/Actor Catalog	Everyone who interacts with IT, including users and owners. Tests requirements for completeness (Board: CEO; Finance: CFO; IT: CIO; Quality and Safety; Clinical Operations)	pp. 369-370
Driver/Goal/Objective Catalog	How drivers become goals and SMART objectives. Reveals synergies between initiatives (e.g., Patient-Centered Care, improve satisfaction, 90% rating by Q4 2028)	pp. 370-371
Role Catalog	Business roles, used for RACI, change impacts, training	pp. 371-372
Business Service/Function Catalog	Filterable functional decomposition. Jager shows it as a Business Service/Function Matrix because cross-mapping is clearer	pp. 372-374
Location Catalog	Physical and virtual locations (campus, data centre, telehealth hub). Supports DR, latency, compliance	pp. 374-375
Process/Event/Control/Product Catalog	Each process with its trigger event, constraining control, and product	pp. 375-376
Contract/Measure Catalog	KPIs, SLAs, OLAs between services and their consumers or providers	pp. 376-377
Business Capabilities Catalog	Capabilities (what, not how) with maturity, owner, and related goal	pp. 377-378
Business Capability Map	Capabilities and sub-capabilities, expanded from Phase A	p. 379
Value Stream (Stages) Catalog / Value Stream Map	Value streams with stages, stakeholder, outcome (Inpatient Care Delivery, Patient Feedback Resolution, Cost Performance Review)	pp. 379-381
Value Stream/Capability Matrix	Links stages to the capabilities that perform them. Key for capability-based planning	pp. 381-382
Strategy/Capability Matrix	Capabilities needed per strategic goal, with priority	p. 382
Capability/Organization Matrix	Which units deliver or contribute to each capability	p. 383
Organization Map / Diagram	Units, partners, suppliers, and their relationships	pp. 383-385
Business Glossary Catalog	Unambiguous business terms. Bridges business and IT. Evolves through the ADM	pp. 385-386
Business Interaction Matrix	Roles vs. functions (performs, supports, informed, not involved)	pp. 386-387
Actor/Role Matrix	Which actors perform which roles (security, skills, training)	pp. 387-388
Business Footprint Diagram	Goals, units, functions, services, and technical components. A CxO communication tool	pp. 388-389
Business Service/Information Diagram	Information each service consumes and produces	pp. 389-390
Functional Decomposition Diagram	Capabilities or functions broken into sub-functions on one page	pp. 390-391
Product Lifecycle Diagram	Products through concept, development, operation, retirement (e.g., Feedback App v2 from Q3 2028)	pp. 391-393
Goal/Objective/Business Service Diagram	Traces goals through objectives to services. Derived from the Footprint Diagram	p. 393
Business Use-Case Diagram	Who (roles) uses which services or processes	p. 394
Organization Decomposition Diagram	Actors, roles, locations in a chain of command	pp. 394-395

Artifact	Purpose (healthcare example)	Page
Process Flow Diagram	Sequential flow with swimlanes, controls, events, products. Good for SME sessions	pp. 395-396
Business Event Diagram	Events triggering processes and outcomes (reactive behaviour)	p. 396
Information Map	Information concepts, categories (transactional = dynamic, event-driven; reference = stable, governed), and relationships	pp. 396-398

- **Phase B feeds the rest.** C structures data and applications around business processes, D aligns technology to business needs, and E identifies projects from business requirements. Give Phase B enough time (p. 398).

Jager's commentary (author's view): The Actor/Role Matrix is one of the least-used artifacts, because HR already holds this information and a second source adds little (p. 387).

12.6 Phase C: Information Systems Architectures (pp. 398-460)

- **Purpose.** A blueprint of the information systems that support the Business Architecture: the systems needed for business capabilities, defined through Data and Application Architectures, with gaps assessed. Key deliverables are data models and application portfolios (p. 398).
- **Two interconnected parts.** Data Architecture (data assets, management, governance: how data is stored, managed, used) and Application Architecture (applications, their interactions with each other and with users, aligned to goals) (p. 399).

Jager's commentary (author's view): BIZBOK's information concept sits between Business and Data Architecture but "is not yet sufficiently reflected" in the Fundamental Content, only in the "Information Mapping" Series Guide (p. 399).

12.6.1 Developing Data Architecture (pp. 400-410)

- **Objectives** mirror Phase B: a Target Data Architecture (entities and relationships, storage, handling, integration) and Data Architecture roadmap components from gaps. Reuse wherever possible (p. 400).
- **Same steps as Phase B (Table 12-31).** Select reference models, viewpoints, and tools; develop Baseline and Target descriptions; perform gap analysis; define candidate roadmap components; resolve impacts across the landscape; conduct a formal stakeholder review; finalize; update the Architecture Definition Document. "The steps don't really differ between the three phases" B, C, and D (pp. 400-402).
- **Differences in emphasis.** Use Data Architecture reference models and viewpoints. Take stock of data assets and flows for the baseline. Explore target options with Alternatives and Trade-Offs (Ch. 14.9). Add data components to the *draft* roadmap from Phase B. Ask whether the Data Architecture affects established architectures and projects, and vice versa (pp. 402-407).
- **Outputs.** An updated Statement of Architecture Work and an updated ADD with Baseline and Target Data Architectures, approved by senior management (pp. 401, 409-410).
- **Aim.** A data infrastructure that meets business goals and is secure and scalable (p. 410).

12.6.2 Phase C (Data Architecture) Artifacts (pp. 410-418)

Artifact (Table 12-32)	Purpose	Page
Data Entity/Data Component Catalog	All data used, with entities related to the data components that structure them. Built from the Information Map	pp. 411-412
Data Entity/Business Function Matrix	Where data is used and who owns it. Supports gap analysis for missing entities	pp. 412-413

Artifact (Table 12-32)	Purpose	Page
Application/Data Matrix	Which applications Create, Read, Update, or Delete which entities (CRUD). Reveals duplication, integration, ownership	p. 414
Conceptual Data Diagram	Relationships between critical entities for business stakeholders. Similar to the Information Map	pp. 414-415
Data Dissemination Diagram	How logical data components are realized by application components. Supports sizing and shows business criticality	p. 415
Data Security Diagram / Catalog	Which roles, units, and applications access which data. Classification and protection (RBAC, encryption, MFA)	pp. 415-416
Data Migration Diagram / Catalog	Extraction from source, transformation and cleansing, loading into target. Overview level in Phase C	pp. 416-417
Data Lifecycle Diagram	Stages of an entity: creation, modification, use, storage, archival, deletion	p. 418

- **Context.** Phase B supplied the business context. Data Architecture influences technology choices (databases, storage, platforms) and runs in parallel with Application Architecture (p. 418).

Jager's commentary (author's view): People know their processes better than their functions, so a **Data Entity/Business Process Matrix** is more useful than the Standard's Data Entity/Business Function Matrix (p. 413).

12.6.3 Developing Application Architecture (pp. 418-427)

- **Objectives** mirror Phase B: a Target Application Architecture, and Application Architecture roadmap components from gaps. Reuse wherever possible (p. 418).
- **Same steps (Table 12-38)** as Phases B and C-Data: select resources and viewpoints, develop Baseline and Target (only what the vision needs, explore options with Alternatives and Trade-Offs), gap analysis, roadmap components added to the Phase B draft, resolve impacts, stakeholder review, finalize, update the ADD (pp. 419-427).
- **Aim.** "A structured, cohesive, and efficient application landscape" whose applications are interoperable and exchange data effectively (p. 427).

12.6.4 Phase C (Application Architecture) Artifacts (pp. 427-441)

Artifact (Table 12-39)	Purpose	Page
Application Portfolio Catalog	All applications with function supported, purpose, lifecycle status, interfaces. Assesses redundancy, alignment, risk	pp. 429-432
Interface Catalog	Interfaces between applications (batch, direct DB, API) to scope dependencies early. Full version adds data, protocol, frequency, security	pp. 432-433
Application/Organization Matrix	Which units use which applications	pp. 433-434
Role/Application Matrix	Which roles use which applications. Essential for role-based access and security	p. 434
Application/Function Matrix	Which applications support which business functions	p. 435
Application Interaction Matrix	Application-to-application communications, with services in the cells (the Interface Catalog as a map)	pp. 435-436
Application Communication Diagram	Data flows between applications across interfaces and protocols	p. 436
Application and User Location Diagram	Where applications are used and hosted, relative to users and locations	p. 437
Application Use-Case Diagram	How actors use application components via services	p. 437

Artifact (Table 12-39)	Purpose	Page
Enterprise Manageability Diagram	How applications are monitored, controlled, supported (monitoring, alerting, logging). A filter of the Communication Diagram	pp. 437-438
Process/Application Realization Diagram	Which applications support or automate each process, with sequencing and integration points	pp. 438-439
Software Engineering Diagram	Breaks applications into packages, modules, services, operations (no example given)	p. 439
Application Migration Diagram	Baseline-to-target application migration, including temporary apps and staging areas, for cost estimation	pp. 439-440
Software Distribution Diagram	Physical applications across physical technology and locations	p. 440

- **Enrich the portfolio.** Add CIA (confidentiality, integrity, availability) scores for security compliance (Figure 12-31), hosting location (on-premises or cloud), and the capabilities automated, shown as coloured views (pp. 431-432).
- **Phase C hand-off.** It bridges business strategy and technology. Its output feeds Phase D, and its gaps drive planning in Phase E (p. 441).

Jager's commentary (author's view): Without an Application Portfolio Catalog, duplicate applications get approved too quickly ("triple or quadruple" functionality) and license costs stay hidden. Rationalization can then take years (pp. 430-431). Adding locations to the portfolio makes the Application and User Location Diagram and the Software Distribution Diagram mostly unnecessary (pp. 437, 440). None of the core roles produces the Software Engineering Diagram (p. 439).

12.7 Phase D: Technology Architecture (pp. 441-459)

- **Purpose.** Define the technology infrastructure (hardware, software, network) and technology standards that support the Business, Data, and Application Architectures, with consistency, scalability, security, interoperability, and future readiness (p. 441).

12.7.1 Developing Technology Architecture (pp. 442-452)

- **Objectives** mirror Phase B: a Target Technology Architecture, and technology roadmap components from gaps. Reuse wherever possible (p. 442).
- **Same steps (Table 12-46)** as B and C. Technology reference models and viewpoints; Baseline and Target (only what the vision needs, with options via Alternatives and Trade-Offs); gap analysis; components added to the Phase B draft roadmap; resolve impacts; stakeholder review (meeting, presentation, feedback, refinement, final approval); finalize building blocks with documented rationale; update the ADD (pp. 443-452).
- **A bridge.** Phase D links conceptual designs to the actual implementation of technology solutions, covering hardware, software, networking, security, and integration (p. 452).

Jager's commentary (author's view): In "nine out of ten organizations" IT has its own separate planning and strategy. TOGAF's idea is that work in *every* domain, technology included, is linked to business goals (p. 442).

12.7.2 Phase D (Technology Architecture) Artifacts (pp. 453-459)

Artifact (Table 12-47)	Purpose	Page
Technology Standards Catalog	Approved technologies, products, versions, and status (e.g., RHEL 10, PostgreSQL 17.5, HL7 FHIR R4 approved / R6 under review)	p. 454
Technology Portfolio Catalog	All technology in use (hardware, infrastructure and application software). The usual starting point of Phase D, fed by registries such as a CMDB	pp. 454-455

Artifact (Table 12-47)	Purpose	Page
Application/Technology Matrix	Which platforms support which applications. Reveals dependencies, reuse, and obsolescence risk	p. 455
Environments and Locations Diagram	Physical and virtual locations where components run. Essentially a filter of the Phase B Location Catalog	p. 456
Platform Decomposition Diagram	Structure of technology platforms: servers, OS, networks, databases grouped into logical and physical platforms	pp. 456-457
Processing Diagram	Components grouped into deployment units and how those connect	p. 457
Networked Computing/Hardware Diagram	Logical application components in a distributed network environment (DMZ, internet, intranet zones). A more detailed Processing Diagram	p. 458
Network and Communications Diagram	How servers, clients, databases, and devices connect across networks: boundaries and paths, not data formats	pp. 458-459

- **Alignment.** Phase D must align with Phase C's applications and is the foundation for Phase E's identification and prioritization of solutions (p. 459).

12.8 Phase E: Opportunities and Solutions (pp. 459-494)

- **Purpose.** Identify and prioritize the projects or programs (work packages) that will implement the Target Architecture. Evaluate buy, build, or reuse. Assess dependencies, opportunities, and constraints to produce a feasible Architecture Roadmap (pp. 459-460).

12.8.1 Defining Opportunities and Solutions (pp. 460-475)

- **Three objectives.** (1) Identify the work packages. (2) Decide whether an incremental approach, with Transition Architectures that each deliver business value, is needed. (3) Define the SBBs needed to complete the target by detailing the ABBs. Phase E produces the *first complete* Architecture Roadmap from the B-D gap analyses and candidate components (p. 460).

Step (Table 12-52)	Description	Output	Page
Determine/confirm key corporate change attributes	Align with the business culture. Assess units' transition capabilities, culture, skills	Implementation Factor Catalog	pp. 461, 463-464
Determine business constraints for implementation	Find drivers that could compromise the target. Mediate with Alternatives and Trade-Offs	n/a	pp. 461, 464-465
Review and consolidate gap analysis results from B-D	One list of gaps, solutions, dependencies. Find SBBs that close several gaps	Consolidated Gaps, Solutions, and Dependencies Matrix	pp. 461, 465-466
Consolidate and reconcile interoperability requirements	Find interoperability conflicts (reused SBBs, COTS, third parties). Either build a translating block or change specifications	n/a	pp. 461, 466-467
Refine and validate dependencies	Group related activities into projects and logical increments	Work Package Portfolio Map	pp. 461, 467-468
Confirm readiness and risk for business transformation	Revisit the Phase A readiness assessment. Classify and mitigate risks	Updated Consolidated Matrix	pp. 461, 468-469
Formulate Implementation and Migration Strategy	Choose the approach and methodology	Implementation and Migration Strategy	pp. 462, 470-471

Step (Table 12-52)	Description	Output	Page
Identify and group major work packages	Group activities to close capability gaps (new, existing, or bought). Classify systems as mainstream, contain, or replace	Work Package Portfolio Map	pp. 462, 471-472
Identify Transition Architecture(s)	Value-delivering milestones fitted to the strategy and capacity for change	Building blocks; Architecture Definition Increments Table; Transition Architectures	pp. 462, 472-473
Create Architecture Roadmap and Implementation and Migration Plan	Consolidate work packages and transitions into a timeline and plan	Architecture Roadmap; Implementation and Migration Plan	pp. 462, 473-475

- **Strategic approaches.** *Greenfield* (completely new), *Revolutionary* (radical overhaul), *Evolutionary* (gradual convergence, parallel running, phased introduction) (p. 470).
- **Implementation methodologies.** *Quick Wins*, *Achievable Targets*, and the *Value Chain Method* (p. 470).
- **Sequencing advice.** Unless there is a good reason otherwise, schedule difficult activities after those that more readily provide missing capabilities. Intervals between transitions need not be equal (p. 472).
- **Implementation and Migration Plan (Table 12-53).** Phases with timeframes, work packages, milestones, and dependencies. Consider a data-driven sequence (applications that create data first) and the lifecycle of existing SBBs (pp. 473-474).
- **Level of detail.** The draft roadmap should match the ADD's level of detail. If much more detail is needed, a different level of architecture may be required (p. 473).
- **Bridge.** Phase E links architecture definition to implementation planning (p. 475).

Jager's commentary (author's view): The Implementation Factor Catalog and the Consolidated Gaps, Solutions, and Dependencies Matrix are not true artifacts, because they contain no metamodel entities (pp. 463-464).

12.8.2 Phase E (Opportunities and Solutions) Artifacts (pp. 475-485)

Artifact (Table 12-54)	Purpose	Page
Project Context Diagram	Scope of work packages: the units, functions, services, processes, apps, data, and technology they affect. Used for PPM and mobilization	p. 477
Benefits Diagram / Catalog	Opportunities classified by size, benefit, complexity (goal, objective, capability, benefit) for prioritization	pp. 477-478
Implementation Factor Catalog	Factor category (risk, assumption, dependency, action, effect), factor, description, and *deduction* (resulting action or constraint). The deductions shape work packages	pp. 478-480
Consolidated Gaps, Solutions, and Dependencies Matrix	Gaps from B-D with domain, potential solutions, dependencies. Links SBBs to ABBs. Clusters similar gaps	pp. 481-483
Architecture Definition Increments Table	Projects and their incremental deliverables across Transition Architectures 1..n	pp. 483-484
Work Package Portfolio Map	Jager's addition (see 12.8.3)	pp. 488-490
Architecture Roadmap	Jager's addition (see 12.8.3)	pp. 490-493

- **Plateau Planning (Figure 12-42).** Baseline, Transition, and Target plateaus, filled with gaps from the consolidated matrix (e.g., the HR recruitment example: new process + Application B in the transition, consolidated information flows in the target). A variant shows projects instead of gaps (Figure 12-43)

(pp. 482-484).

12.8.3 Creating an Architecture Roadmap (pp. 485-494)

- **The roadmap.** A pivotal deliverable built incrementally in E and F from B-D components. It is both a planning and a communication tool, iterative and adaptable, and shows the business value of work packages at each stage (p. 485).
- **Plateaus, gaps, and work packages (Figure 12-44).** Each architecture state is a *plateau*. The *gap* entity sits between plateaus, and *work packages* close the gaps (p. 486).
- **Work packages vs. initiatives vs. projects.** A cluster of work packages is an initiative and often forms a project, program, or portfolio. A work package is done once and produces a defined result (unlike an ongoing process). In agile, it can model a sprint or increment (pp. 486-487).
- **Work package vs. process (Figure 12-45).** Both are series of related tasks producing a result, but a work package is a unique, one-time "process" (p. 487).
- **Work Package Portfolio Map (Table 12-59).** All work packages from strategy or architecture work, each linked to objectives and goals (e.g., WP A "Fit job profile X..." links to the objective "comply with legislation by Q3 2028", which links to the goal "Implement new regulatory requirements"). It is the foundation of the roadmap and lets stakeholders track progress against strategy. Add a column for related projects to avoid duplicate work in Phase F (pp. 488-490).
- **Architecture Roadmap (Figures 12-47, 12-48).** Goals and objectives down the left, work packages plotted over time with start and end dates, status, cost, and owner. It shows the cost of achieving objectives, helps prioritize, and visualizes transitions (pp. 490-493).
- **Never unlinked.** "A roadmap should never be created without linking business goals and objectives to the initiatives." SMART objectives make this measurable (p. 491).
- **Phase E conclusion.** Prioritize and sequence by value, dependencies, and constraints, with stakeholder engagement, ending in a formal agreement on the prioritized projects and solutions (p. 494).

Jager's commentary (author's view): The Standard has a work package entity but no artifact that uses it (other than the Project Context Diagram), so he proposes adding the **Work Package Portfolio Map** and **Architecture Roadmap** as Phase E artifacts (p. 488).

12.9 Phase F: Migration Planning (pp. 495-513)

- **Purpose.** A detailed, step-by-step plan from baseline to target, with Transition Architectures, milestones, resources, and timelines (p. 495).

12.9.1 Creating a Migration Planning (pp. 495-507)

- **Aims.** Prioritize transition projects and activities. Assess dependencies and risks. Plan resources (budget, people, technology). Ensure business continuity (p. 495).
- **Inputs.** The Target Architecture and roadmap components (gaps, opportunities, solutions from B-D, turned into work packages in E), the Implementation Governance Model, and earlier impact and risk insights. The roadmap is *finalized* in Phase F (pp. 495-496).

Step (Table 12-60)	Description	Output	Page
Confirm management framework interactions	Align the plan with Business Planning, EA, Project/Portfolio Management, and Operations Management	Updated Implementation and Migration Plan	pp. 496-499
Assign business value to work packages	Define and measure value. Map work packages to projects. Assign risks from Phase E. Estimate value	Updated Work Package Portfolio Map; Business Value and Risk Matrix	pp. 496, 499-500

Step (Table 12-60)	Description	Output	Page
Estimate resource requirements, project timings, and availability/delivery vehicle	Costs split into capital (establish) and operational (sustain). Savings from decommissioning. Mostly done by project managers, but stay involved	n/a	pp. 496, 501
Prioritize migration projects through cost/benefit assessment and risk validation	Net benefit of SBBs; risk mitigation; stakeholder consensus (architecture team, business, PMs, finance); formal risk review	Risk Management Plan	pp. 496, 501-504
Confirm Architecture Roadmap	Add Transition Architectures, set intervals by value, capability, and risk, consolidate deliverables by project	Transition Architecture State Evolution Table; finalized roadmap	pp. 497, 504-505
Complete Implementation and Migration Plan	Capture external dependencies, resource availability, detailed project plans and milestones	Finalized Implementation and Migration Plan	pp. 497, 505-506
Complete architecture development cycle	Create an Implementation Governance Model if maturity allows. Document lessons learned for Phase H	n/a	pp. 497, 506-507

- **Four management frameworks.** Business Planning (conceives, directs, resources), Enterprise Architecture (structures and contextualizes), Project/Portfolio Management (designs and builds), Operations Management (integrates, operates, maintains). The migration plan may end up inside another framework's plan, with EA participating (pp. 497-498).
- **Business value and risk indices.** Value covers principle adherence, financial contribution, strategic alignment, competitive position. Risk covers size and complexity, technology, organizational capacity, impact of failure. Weights are assigned (p. 500).
- **Risk Management Plan.** At least: document information, approvals, and the approach (tools, recorded information, classification metrics, risk tolerance) (p. 503).
- **Detail check.** The roadmap and plan should match the ADD's detail. Needing much more detail signals a new, more detailed ADM cycle (p. 506).
- **Summary.** Phase F develops the Transition Architectures and a phased, manageable plan that is input to Phase G (p. 507).

12.9.2 Phase F (Migration Planning) Artifacts (pp. 507-513)

Artifact (Table 12-61)	Purpose	Page
Transition Architecture State Evolution Table	Rows of domain plus service, with columns per Transition Architecture describing each service's state (new, in transition, retained, removed)	pp. 508-511
Business Value and Risk Matrix	Projects plotted by business value index (y) against risk index (x), with weighted criteria approved by senior management and colours for at-risk status	pp. 511-513

- **HR example (Figure 12-49).** Transition 1: Application A must be replaced by B. Transition 2: B introduced and services moved from A. Transition 3: B fully operational and A gone. The DBMS persists throughout (it supported A, then B). The cells hold SBBs because they name specific applications (pp. 509-510).
- **Project size** comes from the number of linked work packages, so update the Work Package Portfolio Map with projects (p. 512).

Jager's commentary (author's view): The State Evolution Table is demanding to build per project, but "the pros outweigh the cons" because it details every step per domain and service (p. 511).

12.10 Phase G: Implementation Governance (pp. 513-528)

- **Purpose.** Oversee execution so that the vision "transitions into operational reality". Define structure, roles, and responsibilities. Work with project management to monitor progress and address deviations. "Where the rubber meets the road" (pp. 513, 522).

12.10.1 Establishing Architecture Governance (pp. 514-522)

- **Objectives.** Ensure adherence to the architecture, manage changes during implementation, and continually assess progress. Uses the Governance Framework from Ch. 8 (p. 514).

Step (Table 12-62)	Description	Output	Page
Confirm scope and priorities	Gap analysis to identify needed SBBs (led by Solution Architects). Maximize value from existing investments using the Work Package Portfolio Map. Workshop with development management	Architecture Contract	pp. 514-516
Identify deployment resources and skills	Ensure teams understand EA deliverables and methods, using modelling tools for a shared language. Check team readiness (skills, tools, processes) and dependencies	n/a	pp. 514, 516-517
Guide development of solutions deployment	A project recommendation per implementation: scope (in and out), impact analysis, priorities (Business Value and Risk Matrix), strategic requirements, change requests, functional and non-functional scope. Update the repository	Change requests	pp. 514, 517-518
Perform architecture compliance reviews	Check each building block against standards and principles. Agree success metrics. Get formal sign-off. Create a deployment plan. Track time, cost, quality	Compliance assessments	pp. 515, 519-520
Implement business and IT operations	Deploy service delivery, training, and communication. Publish new baselines to the repository and update the CMDB	Architecture-compliant solutions	pp. 515, 520-521
Perform post-implementation review	Examine results. Raise change requests for deviations (which starts Phase H). Publish reviews and formally close projects	Publication of reviews	pp. 515, 521-522

- **Risk Management throughout G** (Ch. 14.8) (p. 521).
- **Bridge.** G connects designs to real-world deployment, ensuring compliance, change management, and stakeholder engagement (p. 522).

12.10.2 Architecture Performance Monitoring (pp. 522-528)

- **Dashboards and graphs** let all stakeholders monitor progress. There are two areas: work package completion, and goal and objective achievement (pp. 522-523).

12.10.2.1 Measuring progress in completing work packages (pp. 523-525).

Method	How it works	Example (4 activities, 3 complete)
Basic	Equal share per activity (25% each)	75%
Refined	Share proportional to activity duration (12.5%, 25%, 25%, 37.5%)	62.5%, which is more realistic

- Progress can be shown as pie charts per work package (Table 12-63) or as spider or bar charts. The choice is personal preference (p. 525).

12.10.2.2 Measuring achievement of goals and objectives (pp. 526-528).

- **Method.** Average the progress percentages of all work packages linked to a goal or objective via the Architecture Roadmap (p. 526).
- **Example.** Goal A has WP A at 100% and WP B at 75%, so Goal A is 87.5% realized. Goal B has WP C at 50% (on hold) and WP D and E at 0%, so Goal B is 16.7% realized (pp. 526-527, Figures 12-53, 12-54).

- Simple, but useful to both the architecture team and senior management (p. 528).

12.11 Phase H: Architecture Change Management (pp. 528-539)

- **Purpose.** Architecture is not static. Phase H manages change across the architecture's lifecycle: assess impacts, define change procedures, keep the architecture aligned with business goals (p. 528).

12.11.1 Managing Architecture Changes (pp. 528-539)

- **Key objectives.** Continuously assess changes in strategy, business environment, and technology. Have processes and governance to handle change. Evaluate impacts. Implement with minimal disruption. Inform stakeholders. Validate afterward (pp. 528-529).

Step (Table 12-64)	Description	Output	Page
Establish value realization process	Define when change is permitted, the procedures, and what triggers a new ADM cycle. Linked to governance (Ch. 8) and the Architecture Contract	Architecture Compliance Report	pp. 529-531
Deploy monitoring tools	Monitor business and technology change and capability maturity (change management apps, PRINCE2, ITIL, maturity models)	n/a	pp. 529, 532
Manage risks	Identify, classify, mitigate before transformation. Monitor continuously (Ch. 14.8)	Risk Management Plan	pp. 529, 532-533
Provide analysis for architecture change management	Use monitoring data, feedback, and reviews for scenario and impact analyses. Validate with stakeholders. Present as executive summaries or heat maps. Create feedback loops	n/a	pp. 529, 533-534
Develop change requirements to meet performance targets	Use B-D gap results. *Minor* changes go to the framework and principles (update metamodel and Content Framework). *Major* changes need a new Request for Architecture Work	Framework and principle changes; new Request for Architecture Work	pp. 530, 534-536
Manage governance process	Decide who approves and with what criteria. Write policies. Involve the Architecture Board. Use scorecards	Architecture Contract	pp. 530, 536-537
Activate the process to implement change	Phased rollout plan with timelines, dependencies, milestones. Communicate. Monitor and refine	n/a	pp. 530, 537-538

- **Architecture Compliance Report.** States whether a change complies. Non-compliance may get an exemption with valid rationale. High-impact changes need a strategy to manage them (p. 531).
- **Criteria mature over time.** Organizations' risk appetites differ, so each ADM iteration moves governance toward tailored criteria and higher maturity (Ch. 15) (p. 531).
- **Why it matters.** "Too often, organizations are left with an Enterprise Architecture that fit yesterday's organization." Change management lets the organization adapt quickly, reduce risk, and keep alignment with business goals (pp. 535, 539).
- **In short.** Identify drivers for change, formulate proposals (rationale, benefits, impact), analyze risk, cost, and impact, govern approval, plan, implement with coordination, then monitor and gather feedback (pp. 538-539).

12.12 Requirements Management (pp. 539-547)

- **Role.** Gather and document stakeholder requirements, and make them available to other phases. Track, categorize, prioritize, and version them. Refine them continuously, check them against the vision, and share them for transparency (pp. 539-540).

12.12.1 Capture, Manage, and Refine Requirements (pp. 540-545)

- **Not a separate phase.** A continuous activity throughout the ADM. Its key objective is that the requirements management process is available and applied in *all* phases (p. 540).

#	Requirements Management step	ADM phase step
1	n/a	Identify and document requirements (Phase A, recorded in the Requirements Repository)
2	Establish baseline requirements in the Requirements Repository (from B, C, D, and E)	n/a
3	Monitor baseline requirements (keep them relevant as business, regulation, and technology change)	n/a
4	n/a	Identify new and changed requirements (as priorities shift)
5	Identify changed requirements and record priorities (validate with stakeholders, manage conflicts)	n/a
6	n/a	Assess impact of changed requirements on current and previous phases (implement now or in a later iteration)
7	n/a	Implement requirements arising from Phase H (make them available for future iterations)
8	Update the Architecture Requirements Repository	n/a
9	n/a	Implement change in the current phase (no Requirements Management action)
10	n/a	Assess and revise gap analysis for past phases (record "gap requirements" and revise the target)

(Table 12-65, pp. 541-545)

12.12.2 Requirements Management Artifacts (pp. 545-547)

- **Requirements Catalog** (entities: requirement, objective, goal, assumption, constraint, gap). It captures what the organization must do to meet its objectives. Link each requirement to goals or objectives for traceability. Requirements are implemented through change initiatives scoped in Phase E and also serve as a quality check that an architecture is fit for purpose. It is often accompanied by a requirements management plan (pp. 545-546).
- Example: "A centralized digital feedback platform..." links to *Improve patient satisfaction* and *Enhance patient safety* (Table 12-67, p. 546).

12.13 The Architecture Development Method in Short (pp. 547-557)

- **A logical method.** It groups key activities to clarify relationships and information flow, with steps that *can* be performed (p. 547).
- **Phase-by-phase recap** of all steps for the Preliminary Phase, A, B-D, E, F, G, H, and Requirements Management (pp. 548-556), summarized as a poster in Figure 12-56 (p. 557).

Jager's commentary (author's view): The most common complaint about TOGAF, that "everything in the ADM has to be implemented", is "completely wrong and misguided". The ADM prescribes neither that all steps be done nor their order. It is iterative (Ch. 13.3), and skipping steps (e.g., in agile) "is actually supported and even encouraged" (pp. 547-548).

12.14 Summary (pp. 557-560)

- The ADM is a logical method, not a process. It is the heart of the Standard, comprehensive and iterative (p. 557).
- It has ten phases with distinct objectives (repeated from 12.1) (pp. 558-559).
- It describes what *can* be done, does not prescribe order, and must be tailored. Every phase and artifact has been illustrated (pp. 559-560).

Key Terms & Definitions

Term	Definition	Page
Architecture Development Method (ADM)	TOGAF's iterative, tailorable method of ten phases for developing and managing EA	pp. 295-296
Request for Architecture Work	Sponsor's formal request that triggers an ADM cycle	pp. 300, 302
Statement of Architecture Work	Scope and approach of the cycle. Effectively a contract between architects and sponsor	pp. 301, 303
Architecture Definition Document	Deliverable container for core artifacts across domains and states	pp. 301, 303
Architecture Contract	Joint agreement on deliverables, quality, and fitness for purpose. Starts implementation governance	pp. 301, 304
Baseline / Transition / Target Architecture	Current state / intermediate value-delivering state / desired future state	pp. 298, 472
Gap analysis	Comparison of baseline and target to find what is missing, to reuse, or to remove	pp. 304, 353
Plateau	A stable architecture state (baseline, transition, target) on the roadmap	p. 486
Work package	A set of actions to achieve one or more business objectives. Part of a project, a project, or a program	p. 487
Architecture Roadmap	Work packages plotted over time against goals and objectives, from baseline to target	pp. 485, 490-493
Implementation and Migration Plan	Activities, timeframes, milestones, and dependencies to realize the roadmap	pp. 473-474, 505
Business Transformation Readiness Assessment	Evaluation of readiness factors (vision, leadership, engagement, governance, culture)	pp. 329, 468
Greenfield / Revolutionary / Evolutionary	New build / radical overhaul / gradual convergence implementation approaches	p. 470
Architecture Compliance Report	States whether a change complies with the current architecture	p. 531
Heat mapping	Colour-coding capabilities or value stream stages by maturity, performance, value, or cost	pp. 361-362

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 12-1	The Architecture Development Method	p. 297
Figure 12-2	Document creation during the ADM	p. 302
Tables 12-1, 12-2, 12-3	Preliminary Phase steps. Example EA team RACI	pp. 306, 314, 315

Figure / Table	What it shows	Page
Figure 12-3	Driver/Goal/Objective Diagram (hospital)	p. 309
Tables 12-4, 12-5	Preliminary artifacts. Principles Catalog	pp. 320-323
Tables 12-6 to 12-8	Phase A steps, artifacts, Stakeholder Catalog	pp. 325-339
Figures 12-4 to 12-8	Value Chain, Solution Concept, Business Model, Business Capability, and Value Stream Diagrams	pp. 341-347
Table 12-9, Figure 12-9	Phase B steps. Business Capability Heat Map	pp. 350, 361
Tables 12-10 to 12-30, Figures 12-10 to 12-25	Phase B artifacts	pp. 367-398
Tables 12-31 to 12-37, Figures 12-26 to 12-30	Phase C (Data) steps and artifacts	pp. 400-418
Tables 12-38 to 12-45, Figures 12-31 to 12-36	Phase C (Application) steps and artifacts	pp. 419-440
Tables 12-46 to 12-51, Figures 12-37 to 12-40	Phase D steps and artifacts	pp. 443-459
Tables 12-52 to 12-59, Figures 12-41 to 12-48	Phase E steps, artifacts, Plateau Planning, Work Package Portfolio Map, Architecture Roadmap	pp. 461-493
Tables 12-60, 12-61, Figures 12-49, 12-50	Phase F steps. State Evolution Table. Business Value and Risk Matrix	pp. 496-512
Tables 12-62, 12-63, Figures 12-51 to 12-54	Phase G steps. Progress measurement methods	pp. 514-527
Table 12-64	Phase H steps	pp. 529-530
Tables 12-65 to 12-67, Figure 12-55	Requirements Management steps, artifacts, catalog	pp. 541-546
Figure 12-56	Summary of all ADM steps (poster)	p. 557

Self-Check Questions

Q: Is the ADM a prescriptive, sequential process?

A: No. It describes everything one *can* do, not what must be done. It does not fix the order, it is iterative, and it must be tailored. It only looks like a waterfall (pp. 295-296, 547-548).

Q: What are the two objectives of the Preliminary Phase?

A: To determine the Architecture Capability the organization needs, and to establish it (pp. 297, 305).

Q: Name the four key ADM documents and when each is created.

A: Request for Architecture Work (Preliminary); Statement of Architecture Work (Phase A); Architecture Definition Document (built in B-D, approved in E, finalized in F); Architecture Contract (from E, governing implementation) (pp. 300-305).

Q: What two objectives do Phases B, C, and D share?

A: Craft the Target Architecture for the domain, and identify candidate Architecture Roadmap components from Baseline-Target gaps (pp. 298, 348).

Q: What does the Solution Concept Diagram actually show?

A: Not a detailed solution, but the relationships between the organization's goals and objectives and the stakeholders' requirements and constraints, as an early "pencil sketch" (pp. 332, 343).

Q: Explain Business Capability Heat Mapping colours in Jager's maturity example.

A: Green is at the desired level. Yellow is one level below. Red is two or more levels below (or not functioning). Purple is not yet in place but desired (pp. 361-362).

Q: What are the three objectives of Phase E?

A: Identify the work packages, decide whether an incremental approach (Transition Architectures) is needed, and define the building blocks (SBBs) needed to complete the target (pp. 298, 460).

Q: Name the three strategic implementation approaches and three methodologies in Phase E.

A: Approaches: Greenfield, Revolutionary, Evolutionary. Methodologies: Quick Wins, Achievable Targets, Value Chain Method (p. 470).

Q: Why does Jager add a Work Package Portfolio Map and Architecture Roadmap as Phase E artifacts?

A: The Standard has a work package entity but barely uses it. These artifacts link every initiative to goals and objectives, so strategy realization can be planned and tracked (pp. 488-491).

Q: Which four management frameworks must the Implementation and Migration Plan align with in Phase F?

A: Business Planning, Enterprise Architecture, Project/Portfolio Management, Operations Management (pp. 497-498).

Q: In Phase G, how does the refined method of measuring work package progress differ from the basic one?

A: The basic method gives each activity an equal share. The refined method weights by duration. With three of four activities done, 75% (basic) versus 62.5% (refined, more realistic) (pp. 523-524).

Q: How is goal realization progress calculated?

A: Average the progress of the work packages linked to the goal. E.g., WP A at 100% and WP B at 75% gives 87.5% for Goal A (pp. 526-527).

Q: In Phase H, what distinguishes a minor from a major change?

A: Minor changes update the framework or principles (and so the metamodel and Content Framework). Major changes require a new Request for Architecture Work and a new ADM cycle (p. 535).

Q: Why is Requirements Management at the centre of the ADM rather than a phase?

A: It is a continuous activity that captures, manages, and refines requirements and makes them available to every phase throughout the cycle (pp. 299, 540).

Q: Which Phase B artifact would you use to show which capabilities each value stream stage needs?

A: The Value Stream/Capability Matrix (or Diagram), a key tool for capability-based planning (pp. 381-382).

Chapter 13: Applying the ADM

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 13 (pp. 561-587)

Overview

Chapter 13 (pp. 561-587) shows how the ADM is adapted to real situations, following the Standard's "Applying the ADM" document. Section 13.1 introduces three scenarios (p. 562). Section 13.2 explains using TOGAF with different **architecture styles** (SOA, microservices, zero-trust) (pp. 563-564). Section 13.3, the core of the chapter, explains **iteration**: the three ways to iterate, the four iteration groups, and baseline-first vs. target-first cycles (pp. 564-580). Section 13.4 covers **architecture partitioning**: dividing the architecture among teams (pp. 580-583). Section 13.5 summarizes (pp. 583-587). Leave with this idea: the ADM graphic is a picture of *information flows*, not an activity sequence. "The Architecture Development Method should not be understood as a process model" (pp. 565-566). Chapter 13 is part of the GREEN and BLUE reading tours (p. 561).

> **Reading note:** The three ways of iterating, the four iteration groups, Table 13-1, and partitioning follow the Standard. The Initiative A/B budget scenario (pp. 569-571) is Jager's example. Keep two lists apart: three *ways* to iterate (whole cycle, within a cycle, Architecture Capability) and four iteration *groups* (Architecture Capability, Architecture Development, Transition Planning, Architecture Governance).

Detailed Section Breakdown

Chapter Opener (p. 561)

- The ADM is step-by-step but "not to be confused with a waterfall approach". It addresses each sub-area in a focused way, works with many architecture styles, and its iterative nature lets it respond to different situations (p. 561).

13.1 Various Architecture Development Method Approaches (p. 562)

- **Tailorability** suits the ADM to different process styles and specialized architectures such as Security Architecture, for "almost any situation" (p. 562).
- **Three scenarios covered:** different architecture styles, iteration, and partitioning. They cover most applications (p. 562).

13.2 Using the TOGAF Framework with Different Architecture Styles (pp. 563-564)

- **Versatile across styles.** E.g., SOA, Microservices, Zero-Trust, which differ in focus, form, techniques, and history. Several styles can coexist in one landscape (p. 563).
- **Approach.** (1) Identify the style's unique characteristics. (2) Decide how to address them, usually by adapting models, views, and tools, not by changing the framework (p. 563).
- **Where it shows up.** In B, C, and D, choose resources and viewpoints for the style. Styles may add elements, emphasize others, change notation, or focus on certain stakeholders, which often means extending the Enterprise Metamodel and Content Framework (p. 563).
- **Preliminary revisited.** A predominant style may require revisiting the Preliminary Phase to adjust the Architecture Capability. Style-specific reference and maturity models help (p. 564).

13.3 Applying Iteration to the Architecture Development Method (pp. 564-580)

- **Iterative by nature.** Phases can be revisited as requirements, feedback, and context change (p. 564).
- **The graphic is misread.** Figure 13-1 is "frequently misinterpreted as a linear, waterfall-style process". It is "a logical method that groups key activities to clarify their relationships and information flow" (p. 565).
- **Any activity is a phase.** E.g., working on a roadmap *is* doing Phase E, with its mandatory inputs and outputs (p. 566).

13.3.1 Different Ways to Apply Iteration (pp. 566-572)

Way to iterate	What it means	Used for	Page
Across the whole ADM cycle	Run the entire cycle repeatedly for different levels or scopes (enterprise vs. project), or layer by layer	Building a comprehensive Architecture Landscape. Moving from high-level EA to detailed architectures. Agile or incremental	pp. 566-567
Within an ADM cycle	Loop back and forth between two or more phases as insights emerge	Managing B/C/D interdependencies. Converging on a detailed target without higher-level context	pp. 567-569
To manage the Architecture Capability	Re-run the Preliminary Phase and Phase A	Adjusting the capability, e.g., after a Phase H change request	pp. 571-572

- **Common within-cycle groups (Figure 13-2).** Architecture Development (B, C, D baseline and target), Transition Planning (to reach an executable roadmap or migration plan when scope, strategy, or requirements shift), and Architecture Governance (governing change toward the target) (pp. 568-569).
- **Key difference.** Within-cycle iteration deepens work inside phases "without affecting the overall flow" of the ADM (p. 569).

Jager's commentary (author's view): His example has two goals (new regulation, efficient HR use) and two initiatives with equal cost: hire staff (Inflow Support capability) or buy compensation software. Both scenarios are developed in B and C and presented to the CIO in Phase F, who picks one via Architecture Alternatives and Trade-Offs (Figure 13-4). A further Architecture Development iteration then refines the chosen scenario (pp. 569-571).

13.3.2 How to Apply Iteration (pp. 573-580)

Iteration group	Purpose	ADM phases	Page
Architecture Capability	Establish and grow the capability: approach, principles, scope, vision, governance	Preliminary, A	p. 573
Architecture Development	Develop architecture content: baseline via B-D, then feasibility via E-F	B to F	p. 573
Transition Planning	Plan the transition and roadmaps from baseline to target	E, F	p. 574
Architecture Governance	Establish and monitor governance	G, H	p. 574

- **Preliminary and A are not repeated routinely.** Approach, principles, vision, and scope don't need redoing for every piece of work (p. 576, Table 13-1).
- **Architecture Development cycles (Figure 13-8):**
  - *Baseline first.* Iteration 1 defines the baseline across B-D (with E/F to test feasibility). Iteration 2 defines the target and the gaps. Further iterations refine (pp. 576-577).

  - *Target first.* Map the desired state first, then compare with the baseline. "A common starting point in many organizations" (p. 577).
- **Transition Planning cycles (Figure 13-9).** Iteration 1 agrees on improvement opportunities against a provisional Transition Architecture, gains buy-in to a solution portfolio, and produces a provisional migration plan. Iteration 2 agrees on the Transition Architecture and refines the plan, feeding issues back to Phase E (pp. 577-578).
- **Architecture Governance cycles (Figure 13-10).** Iteration 1 mobilizes governance and change management processes, people, and technology. Later iterations run periodic compliance reviews, and change results may feed back to the Preliminary Phase or the development phases (pp. 578-579).
- **Treat the steps as a checklist.** Forcing a single linear pass produces "convoluted, looping phase diagrams" (p. 579).

13.4 Architecture Partitioning (pp. 580-583)

- **Definition.** Dividing an EA into smaller, manageable parts to improve governance, scalability, and efficiency, because large organizations can't be architected as one monolith (p. 580).

13.4.1 Why Use Architecture Partitioning (pp. 580-582)

- **Different governance per area.** Domains have different stakeholders, policies, and compliance needs (p. 580).
- **Parallel teams, different timeframes.** Avoids bottlenecks. Architecture can evolve incrementally, with the ADM applied iteratively per partition (p. 581).
- **Definitions per team** prevent architecture that doesn't integrate. Partitioning resembles agile in breaking work into units (p. 581).
- **Keep the whole in view.** Without the broader context you get disjointed components (Figure 13-11) (pp. 582-583).

13.4.2 Supporting Architecture Partitioning (p. 583)

Decided in the Preliminary Phase	Content
Organizational structure	Which teams create which parts of the architecture
Responsibilities for the teams	An agreed definition: subject matter areas, level of detail, delivery time periods
Relationship between architectures	Work toward one integrated architecture. Resolve overlaps so reusable building blocks remain

(p. 583)

13.5 Summary (pp. 583-587)

- Three scenarios: architecture styles, iteration, partitioning (p. 584).
- Three ways to iterate and four iteration groups. Within-cycle iteration loops between phases to refine the architecture over its lifecycle (pp. 585-586).
- Partitioning enables tailored governance, parallel work, and incremental evolution, but requires a holistic view (pp. 586-587).

Key Terms & Definitions

Term	Definition	Page
Architecture style	A distinctive approach (e.g., SOA, microservices, zero-trust) with its own focus, form, and techniques	p. 563
Iteration across the whole cycle	Repeating the full ADM for different levels, scopes, or layers	p. 566
Iteration within a cycle	Looping between two or more phases	p. 567
Architecture Capability iteration	Re-running the Preliminary Phase and A to adjust the capability	pp. 571-572
Iteration groups	Architecture Capability, Architecture Development, Transition Planning, Architecture Governance	pp. 573-574
Baseline first / Target first	Starting Architecture Development iterations with the current or the desired state	pp. 576-577
Architecture partitioning	Dividing the EA into manageable parts, often across teams and timeframes	p. 580

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 13-1	The ADM graphic	p. 565
Figure 13-2	Three most common iteration groups within a cycle	p. 568
Figures 13-3, 13-4	Goals and initiatives. Two scenarios via Alternatives and Trade-Offs	pp. 569, 571
Figure 13-5	Architecture Capability iteration group	p. 572
Figure 13-6	Iteration groups of the ADM	p. 574
Table 13-1	Activity by iteration cycle	p. 575
Figures 13-7 to 13-10	Iteration cycles for Architecture Development, Transition Planning, Governance	pp. 576-578
Figure 13-11	Architecture created without an overall picture	p. 582

Self-Check Questions

Q: Why is the ADM graphic often misunderstood?

A: People read it as a linear, waterfall process. It actually shows information flows between logically grouped activities, not a sequence (p. 565).

Q: Name the three ways to apply iteration to the ADM.

A: Iteration across the whole ADM cycle, iteration within an ADM cycle, and iteration to manage the Architecture Capability (p. 566).

Q: Name the four iteration groups and their phases.

A: Architecture Capability (Preliminary, A); Architecture Development (B-F); Transition Planning (E, F); Architecture Governance (G, H) (pp. 573-574).

Q: Why are the Preliminary Phase and Phase A not part of the common iterations?

A: Establishing the approach, principles, vision, and scope does not need repeating for every piece of architecture work (p. 576).

Q: Contrast "baseline first" and "target first".

A: Baseline first defines the current state in iteration 1, then the target and gaps in iteration 2. Target first maps the desired state first, then compares it with the baseline. Target first is common in many organizations (pp. 576-577).

Q: How should TOGAF be adapted to a specific architecture style?

A: Identify the style's unique characteristics, then address them mainly by adapting models, views, and tools (possibly extending the metamodel and Content Framework), not by changing the framework (p. 563).

Q: What triggers an Architecture Capability iteration?

A: Projects wanting to (re)establish aspects of the capability, or new or changed capability requirements, for example from a Phase H change request (p. 572).

Q: Why partition an architecture?

A: Different areas have different stakeholders, policies, and compliance requirements. Partitioning allows targeted governance, parallel teams, and incremental evolution (pp. 580-581).

Q: What three things must be settled in the Preliminary Phase to support partitioning?

A: The organizational structure (teams), each team's responsibilities (subject matter, detail, time periods), and the relationships between architectures (integration and reuse) (p. 583).

Q: What risk does partitioning carry, and how is it avoided?

A: Disjointed components that don't integrate. Avoid it by keeping a holistic view and agreeing definitions and relationships up front (pp. 581-583).

Chapter 14: ADM Techniques

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 14 (pp. 589-668)

Overview

Chapter 14 (pp. 589-668) covers the eight "ADM Techniques" that the Standard has split into their own document because they are used at many points in the ADM (p. 589). Section 14.1 asks which of them are really techniques (pp. 589-592). The sections then cover Principles, Enterprise vs. Architecture (14.2, pp. 592-601); Stakeholder Management, with analysis, mapping, Communications Plan, and engagement (14.3, pp. 601-615); Architecture Patterns (14.4, pp. 616-617); Gap Analysis (14.5, pp. 617-624); Interoperability Requirements (14.6, pp. 624-631); the Business Transformation Readiness Assessment (14.7, pp. 631-640); Risk Management, with classification, identification, assessment, mitigation, monitoring, and visualization (14.8, pp. 640-659); and Architecture Alternatives and Trade-Offs (14.9, pp. 659-666). Section 14.10 summarizes (pp. 666-668). Leave with this idea: these are the reusable tools that the ADM phases keep calling on. In Chapter 12, Phase A uses stakeholder analysis and readiness, B-D use gap analysis and trade-offs, and E-H use risk management (pp. 590-591). Chapter 14 is part of the GREEN and BLUE reading tours (p. 589).

> **Reading note:** The eight techniques, the principle structure and five quality criteria, the Power Grid, the gap matrix, the readiness factors, and the risk approach come from the Standard. Jager's views: several "techniques" are not really techniques (pp. 590-591); the "Enterprise Architecture Principles" term in a Series Guide is an inconsistency (p. 597); a Communications Plan should reach the whole organization (pp. 611-612). Keep apart **Enterprise Principles** (organization-wide, board-approved) and **Architecture Principles** (architecture work, Architecture Board-approved).

Detailed Section Breakdown

Chapter Opener (p. 589)

- **Unbundled topics.** Principles, Stakeholder Management, and Risk Management, plus patterns, gap analysis, interoperability, readiness, and trade-offs, were split out because they are invoked at several points. This adds to the Standard's modularity (p. 589).

14.1 Technique or No Technique (pp. 589-592)

- **What TOGAF means by "technique".** Not always a specific method, but an elaboration of a frequently used topic that needs explanation to be applied properly (p. 590).

The Standard's eight techniques	Short description (p. 591)
Architecture Principles	Rules and guidelines for consistent, coherent architecture and governance
Stakeholder Management	Identify, analyze, and manage stakeholders
Architecture Patterns	Standardized solutions to common problems
Gap Analysis	Differences between Baseline and Target
Interoperability Requirements	Determining interoperability requirements
Business Transformation Readiness Assessment	Identifying business transformation issues

The Standard's eight techniques	Short description (p. 591)
Risk Management	Identify and mitigate issues before they hit the architecture
Architecture Alternatives and Trade-Offs	Identify alternative targets and trade off between them

Jager's commentary (author's view): Stakeholder Management and Risk Management are essential but "not techniques", otherwise roadmapping would be one too. A readiness assessment and a gap analysis *are* techniques, because they complement an existing activity (Phases A and E, and Phases B-D respectively) (pp. 590-591).

14.2 Principles (pp. 592-601)

Aspect	Enterprise Principles	Architecture Principles
Scope	Organization-wide, business and IT	Focused on Enterprise Architecture
Purpose	Guide overall decisions and strategy	Framework for architecture design and execution
Application	Broad, influences policy and culture	More technical, used in architecture development
Examples	Agility, customer focus, security	Reusability, scalability, standardization

(Table 14-1, p. 593)

14.2.1 The Added Value of Principles (pp. 593-594)

- Clarity, consistency, and direction. Strategic alignment of business and IT. Standardized approaches. A structured way to evaluate options and trade-offs. Accountability. Shared understanding. Regulatory compliance. Less complexity and duplication (pp. 593-594).

14.2.2 The Structure of a Principle (pp. 594-597)

Element	What it contains (Table 14-2)	Tips
Name	A short, clear title	Capture the essence. No product names, no ambiguous words
Statement	A concise declaration of intent	Convey the rule without ambiguity
Rationale	Why it matters (business value)	Mention consistency with or precedence over other principles
Implications	Practical impact, trade-offs, required actions	Requirements in resources, time, money, activities: "How does this affect the business?"

(pp. 594-595)

Five quality criteria for a set of principles	Meaning
Understandable	Quickly grasped. Intent clear, minimizing violations
Robust	Enables high-quality decisions and enforceable policies, even in controversial cases
Complete	Covers every potentially important situation
Consistent	Balanced interpretation possible. No principle violates the spirit of another
Stable	Enduring, but with an amendment process

(pp. 595-596)

- **Storage.** Enterprise and Architecture Principles live in the Governance Repository of the Architecture Repository (Figure 14-1, p. 596).

Jager's commentary (author's view): The Series Guide "Using the TOGAF Standard in the Digital Enterprise" uses "Enterprise Architecture Principles", which blurs two distinct types. In practice, most organizations' "principles" come from IT and often look like requirements (p. 597).

14.2.3 Enterprise Principles (pp. 597-599)

- "Provide a basis for decision-making throughout an enterprise and inform how the organization sets about fulfilling its mission". Ideally drafted by the board. In reality drafted by the Enterprise Architect and approved by the board, with stakeholder buy-in. Departments may have their own sets, but these must be aligned (pp. 597-598).
- Example: *Comply with laws and regulations* (Table 14-3). More examples are in the Principles Catalog, Table 12-5 (pp. 598-599).

14.2.4 Architecture Principles (pp. 599-601)

- Guide the architecture process and the development, maintenance, and use of the EA. Drafted by the Enterprise Architect with key stakeholders and approved by the **Architecture Board** (p. 599).
- **Hierarchy (Figure 14-2).** Segment principles (business, data, application, technology) are informed by and bounded by Enterprise Principles (pp. 599-600).
- Example: *Common vocabulary and definitions* (Table 14-4). Each Architecture Principle should link explicitly to business goals and architectural drivers (pp. 600-601).

14.3 Stakeholder Management (pp. 601-615)

- **Definition.** "The process of identifying and understanding the needs, interests, and expectations of an organization's various stakeholders and planning and implementing actions to address those interests" (p. 601).
- **Stakeholder.** "Any individual, group, or organization that can affect, be affected by, or perceive themselves to be affected by" an EA initiative (pp. 602-603).
- **Why it matters.** Stakeholders give input and requirements, raise concerns, influence decisions, assess impacts (their views, Ch. 7.6), and underpin governance. Early identification of powerful stakeholders secures resources and support (pp. 601-603).
- **When.** Start in Phase A and update in every phase, including E, F, and H (p. 603).

Step (Figure 14-3)	Part of
1. Identify stakeholders	Stakeholder analysis
2. Analyze and classify stakeholder positions	Stakeholder analysis
3. Plan communications	Communications Plan
4. Engage stakeholders	Stakeholder engagement
5. Monitor and review	Stakeholder engagement

(pp. 604-605)

14.3.1 Stakeholder Analysis (pp. 605-609)

- **Stakeholder Analysis Model (Table 14-5).** Per stakeholder: ability to disrupt change (power), current and required understanding, current and required commitment, required support (pp. 606-607).
- **Stakeholder Power Grid (Figure 14-4):**

Quadrant	Power / interest	Communication	Example
Key Players	High / high	Most frequent	Data and process owners
Keep Satisfied	High / low	Slightly less often	Most C-level roles
Keep Informed	Low / high	Regular email or meetings	Program manager
Minimal Effort	Low / low	Least, but may still matter	Employees

(pp. 607-608)

- In the example (Table 14-6): CEO is Keep Satisfied, CIO is Key Player, CFO is Keep Informed, HR Manager is Minimal Effort. Each classification sets the communication frequency and method used in the Communications Plan (p. 609).

14.3.2 Stakeholder Mapping (pp. 609-611)

- The **Stakeholder Catalog** (Table 14-7) adds key concerns and the artifacts of interest to each stakeholder (e.g., the CFO is interested in the Application and Technology Portfolio Catalogs) (pp. 609-611).

14.3.3 Communications Plan (pp. 611-614)

- A critical success factor. Per stakeholder group: purpose, information needed, frequency, method (Table 14-8: CxO fortnightly meetings; data and process owners monthly; employees email plus meetings) (pp. 611-614).

Jager's commentary (author's view): The Standard limits the plan to key stakeholders, but it should reach the whole organization. It should explain everyone's role in implementing the EA and schedule organization-wide progress updates, because transparency builds acceptance (pp. 611-612).

14.3.4 Stakeholder Engagement (p. 615)

- Go beyond the static plan with appealing visuals, infographics, posters, and themed rooms. Invite input. Regularly evaluate effectiveness and update the analysis when new stakeholders appear (p. 615).

14.4 Architecture Patterns (pp. 616-617)

- **Patterns put building blocks in context.** "Building blocks are what you use; patterns can tell you how you use them, when, why, and what trade-offs you have to make". A pattern combines proven ABBs and SBBs (Figure 14-5) (pp. 616-617).
- **Not design patterns.** They are often confused with software design patterns (p. 617).
- **Not yet integrated** into the ADM by the Standard, but they can be defined in B-D when reusable blocks are identified (p. 617).

14.5 Gap Analysis (pp. 617-622)

- **Definition.** Identifies the difference between Baseline and Target (capabilities, processes, applications, technology). "Highlight a shortfall... items that have been deliberately omitted, accidentally left out, or not yet defined" (p. 617).
- **Basic steps.** Identify baseline and target; compare; categorize gaps (missing functionality, technology gaps, process inefficiencies); group activities into work packages; prioritize by impact and feasibility (p. 618).
- **Hospital example (Figures 14-6, 14-7).** Application A (baseline) is replaced by Application B (target), which automates the missing Inflow Support capability and also Personnel Information Management. Gap A (baseline to transition) and Gap B (transition to target) each get a work package. Gaps are found mainly in B-D, and the real analysis and work packaging happen in E (pp. 618-620).

Gap Analysis Matrix steps	Instruction
Set up the matrix	Rows = baseline ABBs, columns = target ABBs
Add New and Eliminated	An extra bottom row "New" and an extra right column "Eliminated"

Gap Analysis Matrix steps	Instruction
Identify common blocks	Mark "Included" where an ABB is in both
Review missing blocks	Baseline-only ABBs go in Eliminated. Check whether this is correct or an accidental omission to reinstate
Document new blocks	Target-only ABBs go in the New row, with implementation steps

(pp. 621-622, Table 14-9)

- Everything under New or Eliminated is a gap: either explained as correctly eliminated, or to be addressed by work packages that feed the Architecture Roadmap (Ch. 12.8.3) (p. 622).

14.6 Interoperability Requirements (pp. 623-627)

- **Idea.** Compare entities and decide the form and degree of interoperability needed. Interoperability is "the ability to share information and services" (p. 623).
- **Cross-mapping.** E.g., stakeholder-to-stakeholder (built in Phase B, Table 14-10) or stakeholder-to-application (Table 14-11). The phase in which a mapping is created depends on its entities (pp. 623-625).

Degree	Name	Meaning
1	Unstructured Data Exchange	Human-interpretable free text (estimates, papers)
2	Structured Data Exchange	Human-interpretable structured data, with manual compilation or dispatch
3	Seamless Sharing of Data	Automated exchange between systems based on a common exchange model
4	Seamless Sharing of Information	Universal interpretation of information through cooperating applications

(p. 624)

- **Sub-degrees.** A letter refines a degree (e.g., 2A, 3C, 4D) to express standardization or compatibility (Figure 14-8, p. 625).
- **Appears across the ADM** (Figure 14-9). Benefit: better integration and fewer data silos (pp. 626-627).

> **Text note:** p. 626 describes the 2A cell as containing "the letter B" (Common Data Exchange). The table shows 2A, so treat the letter in the text as a slip.

14.7 Business Transformation Readiness Assessment (pp. 627-632)

- **Why.** Architecture change affects workforce, culture, and skills. A change-averse culture can sink a good design. The assessment is done collaboratively by HR, lines of business, and (IT) planners, for successful implementation in E and F (pp. 627-628).

Step	What to do	Page
Identify key readiness factors	e.g., cultural readiness, workforce skills, leadership alignment, change capability	p. 628
Use maturity models	Assess baseline and target per factor (Ch. 15 format)	p. 628
Determine readiness factor ratings	Current maturity and gaps	p. 628
Identify and mitigate risks	Using Risk Management (14.8)	p. 628
Integrate into the Implementation and Migration Plan	Created in E, finalized in F	p. 628

- **Example (Tables 14-12, 14-13).** Factors are scored 1-5 (poor, fair, acceptable, good, excellent): leadership commitment, employee engagement, organizational culture, skills and competencies, Risk Management, stakeholder alignment. Each is then rated by *urgency*, *readiness status*, and *remediation difficulty* to give management a quick overview (pp. 629-630).
- **Monitor in Phase G** and take corrective action through the Governance Framework (p. 630).

14.8 Risk Management (pp. 632-656)

- **Phases where it is addressed (Figure 14-10).** A (initial high-level risks, recorded in the Statement of Architecture Work); E (implementation risks from gaps, influencing approach choice); F (detailed migration risks from the Consolidated Matrix, plus fallback plans); G (risks during implementation, governance oversight); H (continuous monitoring of changes) (pp. 632-634).
- **Not formally addressed** in Preliminary or B-D, although B-D risks surface via gaps and are recorded in Phase E (p. 634).
- **TOGAF vs. Enterprise Risk Management.** TOGAF's technique is a simplified version focused on *architecture project* risk. ERM (Series Guide G152) also covers business, system, information, privacy, compliance, and change risk (p. 635).
- **Initial vs. residual risk (Figure 14-11).** Initial = new, before mitigation, so apply the technique. Residual = what remains after mitigation, usually already accepted (pp. 635-636).
- **Five steps.** Classify, identify, assess (impact x frequency), mitigate and assess residual, monitor (in G). Mitigation must truly reduce impact, not just shift the risk to a similarly high quadrant (pp. 636-637).

14.8.1 Risk Classification (pp. 638-642)

- **Primary categories.** Time (schedule), cost (budget), scope (p. 638).
- **Additional categories.** Contractual, technology, scope and complexity, environmental (corporate), people, client acceptance (p. 639).
- **By domain.** Business, information, application, technology risks (p. 640).
- **Implementation Factor Catalog + risk classification (Table 14-14).** E.g., replacing Application A with B is an environmental (corporate) risk and a people risk (training needed, headcount reduction) (pp. 641-642).

14.8.2 Risk Identification (pp. 642-643)

- Risks surface from the B-D gaps and from readiness assessments. They are detailed in the Implementation Factor Catalog (E) and addressed in F. Capability maturity models also reveal risks (p. 642).
- **Risk Management Plan** (templates from PMBOK, PRINCE2): document information, approvals, approach (tools, recorded data, classification metrics, risk tolerance) (pp. 642-643).

14.8.3 Initial Risk Assessment (pp. 644-647)

- Takes place in E and F, scoring impact and frequency on 1-5 scales (p. 644).

Impact (Table 14-15)	Frequency (Table 14-16)
5 Catastrophic: could lead to bankruptcy	5 Frequent: very often or continuously
4 Significant: severe loss across more than one line of business	4 Probable: several times per transformation cycle
3 Moderate: small loss to a line of business, reduced ROI	3 Occasional: sporadically
2 Low: very minor impact, no financial loss	2 Seldom: possible, unlikely more than once
1 Negligible: minimal impact	1 Unlikely: not likely in a cycle

(p. 645)

- **Classifications (Table 14-17).** Critical (C): likely failure with serious consequences. High (H): significant failure of parts. Moderate (M): noticeable failure jeopardizing some goals. Low (L): some goals not fully achieved (p. 646).
- **Classification scheme (Table 14-18).** Impact against frequency gives C/H/M/L (e.g., Catastrophic + Frequent = C; Moderate + Probable = M). This is the basis of the Risk Heat Map (p. 647).
- **The risk is to the organization**, not to the architecture work (p. 646).

14.8.4 Risk Mitigation and Residual Risk Assessment (pp. 647-650)

- Strategies range from monitoring or accepting a risk to full redundancy. Prioritize frequent, high-impact risks (p. 647).
- **Risk Identification and Mitigation Assessment Matrix (Table 14-19).** Columns: risk, preliminary impact, frequency, and level (impact x frequency), mitigation, then residual impact, frequency, and level. E.g., "Reduction in headcount" 3x4=12, after automating capabilities 2x2=4 (pp. 648-649).
- A drop from catastrophic to critical is still extreme, so reconsider such a mitigation (p. 649).

14.8.5 Risk Monitoring (pp. 650-651)

- Residual risks need approval through the Governance Framework (possibly corporate governance). The matrix is a governance artifact maintained in Phase G. Unmitigated critical risks may trigger a new ADM cycle. Phase H monitors risks of changes (pp. 650-651).

14.8.6 Visualizing Risks (pp. 651-656)

- **Risk Heat Map (Figure 14-12).** Frequency on x, impact on y. Green is low, yellow medium, orange/red high or critical. A *risk tolerance* line marks the highest acceptable level. Eyes go to the upper right (pp. 651-653).
- **Calculating risk.** "Risk = Impact (effect) x Frequency (likelihood)". There is no scientific scoring, so use common sense and agreed definitions. Risk #1 (3x4=12) is medium. Risk #2 (2x3=6) is low (pp. 653-654).
- **Adding risks to diagrams.** ArchiMate (not TOGAF) supports risk modelling, e.g., risks added to a Plateau Planning Diagram (Figure 14-14) (p. 655).

14.9 Architecture Alternatives and Trade-Offs (pp. 656-662)

- **Why.** A single solution rarely satisfies all stakeholders. Presenting at least two alternatives reveals hidden agendas and principles (p. 656).
- **Method (Figure 14-15).** Use the vision, principles, requirements, and other information to set criteria. Formulate alternatives. Select one or combine elements. It applies at any stage or level (p. 657).
- **No fixed criteria.** Consider principles, requirements, vision, and stakeholder concerns, discuss pros and cons, and add viewpoints if needed to expose dependencies, risks, and uncertainties (p. 658).

Criteria for alternatives	Question
Flexibility	How adaptable to changing business needs?
Time and cost	Timeframes and costs, including transition and stabilization
Benefit realization	Over what period do the benefits come?
Architectural alignment	Does it follow established styles and guidelines?
Delivery method	Reuse, build, or buy?
Business impact	How little disruption to existing capabilities?
Risk profile	What risks and mitigations?

(p. 659)

Jager's commentary (author's view): In his Lemon-A-de example (from *Getting Started with Enterprise Architecture*), scenario 1 (hire staff and run events) raises brand awareness but risks unmet demand. Scenario 2 (training) raises production but risks unsold inventory. Trade-off analysis often leads to a phased roadmap that eventually delivers both (pp. 659-662).

14.10 Summary (pp. 663-668)

- Techniques are recurring, elaborated topics. Some are not true techniques. There are eight (pp. 663-664).
- Principles, Stakeholder Management (Phase A onward), gap analysis, the five E/F matrices and tables, interoperability, readiness, risk (initial vs. residual, heat maps), and alternatives (pp. 664-668).

Jager's commentary (author's view): Why architecture patterns are in the Standard at all "remains a mystery", because the Standard itself says they are not yet integrated into the ADM (p. 665).

Key Terms & Definitions

Term	Definition	Page
ADM technique	An elaboration of a recurring topic used across the ADM	p. 590
Enterprise Principle	High-level guideline shaping the organization's strategy, culture, and decisions. Board-approved	pp. 592, 597
Architecture Principle	Guideline for architecture work. Architecture Board-approved	pp. 592, 599
Principle structure	Name, Statement, Rationale, Implications	p. 594
Stakeholder	Anyone who can affect, be affected by, or perceive themselves affected by the initiative	pp. 602-603
Stakeholder Power Grid	Power vs. interest: Key Players, Keep Satisfied, Keep Informed, Minimal Effort	pp. 607-608
Communications Plan	Per-group communication goals, methods, frequencies	pp. 611-614
Architecture pattern	A proven combination of ABBs and SBBs that solves a recurring problem	pp. 616-617
Gap Analysis Matrix	Baseline ABBs vs. target ABBs with Included, New, and Eliminated	pp. 621-622
Degrees of interoperability	1 Unstructured, 2 Structured data exchange, 3 Seamless data, 4 Seamless information	p. 624
Readiness factor rating	Urgency, readiness status, remediation difficulty	pp. 629-630
Initial / residual risk	Risk before mitigation / risk remaining after mitigation	pp. 635-636
Risk score	Impact x Frequency	p. 653
Risk tolerance	The highest level of risk still considered acceptable	p. 653

Figures & Tables Referenced

Figure / Table	What it shows	Page
Table 14-1	Enterprise vs. Architecture Principles	p. 593
Table 14-2, Figure 14-1	Principle structure. Principles in the Governance Repository	pp. 594, 596
Tables 14-3, 14-4, Figure 14-2	Example principles. Segment principle sets	pp. 598-600
Figure 14-3	Steps of stakeholder analysis and engagement	p. 604
Tables 14-5 to 14-8, Figure 14-4	Stakeholder Analysis Model, Power Grid, Catalog, Communications Plan	pp. 606-614

Figure / Table	What it shows	Page
Figure 14-5	Architecture patterns and building blocks	p. 616
Figures 14-6, 14-7, Table 14-9	Gap analysis example and Gap Analysis Matrix	pp. 619-622
Tables 14-10, 14-11, Figures 14-8, 14-9	Interoperability cross-mappings and degrees	pp. 623-626
Tables 14-12, 14-13	Readiness assessment and ratings	pp. 629-630
Figures 14-10, 14-11	Risk Management in the ADM. Initial vs. residual risk	pp. 634-635
Tables 14-14 to 14-19	Risk classification, impact, frequency, scheme, mitigation matrix	pp. 641-648
Figures 14-12 to 14-14	Risk Heat Maps. Plateau diagram with risks	pp. 652-655
Figures 14-15 to 14-17	Alternatives and Trade-Offs method and Lemon-A-de example	pp. 657-661

Self-Check Questions

Q: List the eight ADM techniques.

A: Architecture Principles, Stakeholder Management, Architecture Patterns, Gap Analysis, Interoperability Requirements, Business Transformation Readiness Assessment, Risk Management, Architecture Alternatives and Trade-Offs (p. 591).

Q: Who approves Enterprise Principles and who approves Architecture Principles?

A: Enterprise Principles are approved by the board of directors or senior management. Architecture Principles are approved by the Architecture Board (pp. 597-599).

Q: Name the four parts of a principle and the five quality criteria for a set of principles.

A: Parts: Name, Statement, Rationale, Implications. Criteria: Understandable, Robust, Complete, Consistent, Stable (pp. 594-596).

Q: Where in the Architecture Repository are principles stored?

A: In the Governance Repository (p. 596).

Q: Classify these on the Power Grid: high power and low interest; low power and high interest.

A: Keep Satisfied; Keep Informed (pp. 607-608).

Q: What extra element does Jager want in a Communications Plan?

A: It should reach the whole organization: explain everyone's role in implementing the EA and schedule organization-wide progress updates (pp. 611-612).

Q: In a Gap Analysis Matrix, what does an ABB in the Eliminated column mean?

A: It exists in the baseline but not the target. Either it was correctly eliminated, or it is an accidental omission to reinstate in the next iteration (pp. 621-622).

Q: What are the four degrees of interoperability?

A: 1 Unstructured Data Exchange, 2 Structured Data Exchange, 3 Seamless Sharing of Data, 4 Seamless Sharing of Information (p. 624).

Q: In which ADM phases is Risk Management explicitly addressed, and where is it not?

A: Addressed in A, E, F, G, and H. Not formally in the Preliminary Phase or in B-D (B-D risks surface via gaps and are recorded in E) (pp. 632-634).

Q: Calculate the risk score and class for impact 3 (moderate) and frequency 4 (probable).

A: 3 x 4 = 12, a medium (moderate) risk (pp. 647, 654).

Q: Why can a mitigation that reduces a risk from catastrophic to critical still be inadequate?

A: The risk remains extremely high. Mitigation must noticeably reduce the impact, not just move the risk to a similarly high quadrant (pp. 637, 649).

Q: Name four criteria for comparing architecture alternatives.

A: Any four of: flexibility, time and cost, benefit realization, architectural alignment, delivery method, business impact, risk profile (p. 659).

Q: What three ratings are used in the readiness factor rating?

A: Urgency, readiness status (from the 1-5 score), and remediation difficulty (pp. 629-630).

Chapter 15: Architecture Maturity Models

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 15 (pp. 669-700)

Overview

Chapter 15 (pp. 669-700) explains how to measure and raise the maturity of an organization's Architecture Capability. A maturity model benchmarks architecture practice against an ideal and gives a structured path to improve (p. 669). Section 15.1 compares the two models named in the "Architecture Maturity Models" Series Guide: the **Architecture Capability Maturity Model (ACMM)**, with nine elements and six levels and an IT focus, and **Capability Maturity Model Integration (CMMI)**, with five elements and five levels and a strategic focus (pp. 670-689). Section 15.2 shows how to move up a level, with worked examples for "Strategy and vision" and "Architecture method and process" (pp. 690-695). Section 15.3 links CMMI elements to the competencies and roles from Chapter 10 (pp. 695-698). Section 15.4 summarizes (pp. 699-700). Leave with this idea: Jager prefers CMMI for EA because EA is "a strategic business management tool, not a technical instrument", and raising maturity takes several roles, not one architect (pp. 670, 698). Chapter 15 is part of the GOLD reading tour (p. 669).

> **Reading note:** Both models are referenced by the Standard through a Series Guide. The Standard has no maturity model of its own (p. 698). The preference for CMMI, the critique of ACMM's IT-only security element, the improvement activities (Tables 15-7, 15-8), and the competency mappings (Tables 15-9, 15-10) are Jager's. Don't confuse ACMM's levels (0 None to 5 *Measured*) with CMMI's levels (1 Ad hoc to 5 *Optimal*).

Detailed Section Breakdown

Chapter Opener (p. 669)

- **What a maturity model does.** It describes an ideal EA implementation, usually tied to a framework, as a benchmark. It shows current maturity and a pathway to better business-IT alignment (p. 669).
- **Two methods** in the Series Guide (ACMM, CMMI). Organizations may also build their own (p. 669).

15.1 Determining Architecture Maturity (pp. 670-689)

- **When.** Typically in the Preliminary Phase, to gauge the internal Architecture Capability (p. 670; see Ch. 12.3).
- **ACMM** measures architecture process components, often relative to IT, to find weaknesses. **CMMI** emphasizes maturing the capability to achieve organizational goals, so it is more appropriate for EA (p. 670).

15.1.1 Architecture Capability Maturity Model (pp. 670-681)

- **Three parts.** The model, the elements grouped by maturity level, and a scoring mechanism (p. 670).

Nine ACMM elements (Table 15-1)	What is assessed
Architecture process	How defined, documented, and consistently followed the process is
Architecture development	Effectiveness of the methods and tools, and adherence to standards
Business linkage	Alignment of IT investments with business objectives
Senior management involvement	Leadership sponsorship and resources

Nine ACMM elements (Table 15-1)	What is assessed
Business unit (operating unit) participation	How involved the business units are
Architecture communication	How well architecture is communicated
Architecture governance	Structures overseeing architecture
IT security	Ability to meet cybersecurity requirements
IT investment and acquisition strategy	Allocation of resources to existing and new capabilities

(pp. 671-673)

ACMM level (Table 15-2)	Meaning
0 None	No architecture at all (often omitted, since there is nothing to assess)
1 Initial	Elements recognized as worth developing. No formal processes. Ad hoc and isolated
2 Under development	Initial methods and tools. Informal, not yet consistent
3 Defined	Standardized practices, aligned with strategy, dedicated team
4 Managed	KPIs monitored. Refinement with executives, focus on ROI
5 Measured	Optimized and a reference for others. Analytics drive continuous improvement

(pp. 673-675)

- **Full framework (Table 15-3).** Describes each element at each level, e.g., Architecture process moves from "ad hoc and localized" (Initial) to "part of the culture, quality metrics captured" (Managed) to "concerted efforts to optimize" (Measured). IT investment moves to "no unplanned IT investment or acquisition activity" (Measured) (pp. 676-680).

Jager's commentary (author's view): ACMM "focuses primarily on IT architecture", so it is less suited to EA. Its "IT security" element contradicts the Standard's own view that security is a cross-cutting concern (Ch. 11.2). CMMI is "a much better fit" (p. 681).

15.1.2 Capability Maturity Model Integration (pp. 681-689)

Five CMMI elements (Table 15-4)	Meaning and practice
Strategy and vision	A clear architecture strategy and vision aligned with goals. Stakeholder involvement at all levels. Regular evaluation
Architecture governance	Competent people with clear roles and responsibilities. Senior management training, onboarding, awareness. Collaboration with similar organizations
Architecture method and process	Standardized, maintained processes improved via metrics. EA is a means to achieve goals, not an end
Architecture deliverables	Suitable tools. Artifacts used, maintained, and pruned. Integrated tools with a repository
Business alignment	Clear controls, KPIs, and quality standards. Regular stakeholder consultation. Awareness of the capability

(pp. 682-685)

CMMI level (Table 15-5)	Meaning
1 Ad hoc	No frameworks or standards. Informal. Relies on individuals. Strategy not yet translated into implementation
2 Repeatable	Formalized, repeatable process and a vision. Templates being built. Senior management recognizes the need. Metrics captured
3 Defined	Framework chosen. Formally defined process. Standards widely used. Roadmap exists

CMMI level (Table 15-5)	Meaning
4 Managed	Measurement data actively improves the process. EA informs strategy
5 Optimal	Mature. Drivers, goals, and objectives aligned with EA. No project proceeds without EA input. Collaboration beyond the organization

(pp. 685-687)

- **Full framework (Table 15-6).** E.g., Strategy and vision: ad hoc activities, then a draft vision, then deployment with defined roles, then evaluation and adjustment, then proactive action plans based on measured data (pp. 688-689).

Jager's commentary (author's view): EA's ultimate goal is to support *strategy* (Ch. 9), though in practice it may only support solution delivery or projects (p. 684).

15.2 Maturing the Architecture Capability (pp. 690-695)

- **Approach.** For each CMMI element, compare the current and next level and identify the actions needed (p. 690).
- **Example 1: Strategy and vision, Repeatable to Defined.** The vision needs governance, roles, responsibilities, and an executive sponsor. Ask: is ownership assigned at executive level? Is there a process owner? Is business management responsible? Is the owner accountable for business contribution? (pp. 691-692).

Activities (Table 15-7)	Description
Obtain mandate for architecture	Senior management commits and explicitly assigns responsibility
Create responsibility matrix	RACI mapping architecture tasks to functions
Establish architecture board	Formally approves products. Escalation platform
Appoint architecture process owner	Responsible for process effectiveness and efficiency
Assign final responsibility for architecture	Senior (especially business) management effectively involved

(p. 692)

- **Example 2: Architecture method and process, Ad hoc to Repeatable.** Ask: is architecture made only when there is a client? Is there an action plan? Is the process described and known? Is it reviewed regularly? (pp. 693-694).

Activities (Table 15-8)	Description
Plan architecture	Decide on needed products with stakeholders. Make an action plan
Describe architecture processes	Creating and maintaining architecture, formalizing products, role in projects
Establish architecture product management process	Procedures to maintain and update the architecture as a whole

(p. 694)

Five steps to raise maturity	
1	Identify the current maturity level per element
2	Determine the next desired level
3	Ask targeted questions
4	Implement the identified activities
5	Review and reassess, and raise the rating if justified

(p. 695)

- Decide actions together with senior management and key stakeholders (p. 695).

15.3 Alignment with Competencies (pp. 695-698)

- **Competencies drive maturity.** Table 15-9 maps the ten competencies from Ch. 10.4 (strategic thinking, decision-making, leadership, governance and compliance, communication, adaptability, analytical thinking, systems thinking, change leadership, business-technology alignment) to the five CMMI elements. Table 15-10 adds the core roles that hold each competency (pp. 696-697).
- **Conclusions.** One architecture role is not enough to significantly raise maturity, and the Enterprise Architect has a significant stake in professionalizing the capability (p. 698).
- **Use.** Turning maturity activities into initiatives raises maturity (p. 698).

Jager's commentary (author's view): The Standard has no maturity model of its own yet. He hopes a future one will focus on the strategic nature of the discipline rather than IT (p. 698).

15.4 Summary (pp. 699-700)

- ACMM: nine elements, IT-related, finds weaknesses. CMMI: five elements, goal-oriented, more suitable for EA (pp. 699-700).
- Competencies and multiple roles, especially the Enterprise Architect, are needed to raise maturity (p. 700).

Key Terms & Definitions

Term	Definition	Page
Maturity model	Framework benchmarking architecture practice against an ideal, with a path to improve	p. 669
ACMM	Architecture Capability Maturity Model: nine elements, levels 0-5, IT-oriented	pp. 670-674
CMMI	Capability Maturity Model Integration: five elements, levels 1-5, goal-oriented	pp. 681-686
Maturity element	An aspect of the practice assessed per level (e.g., Strategy and vision)	pp. 671, 682
Scoring mechanism	The ACMM part used to determine and report the current level	pp. 670-671
Architecture process owner	Role responsible for the effectiveness and efficiency of architecture processes	p. 692

Figures & Tables Referenced

Figure / Table	What it shows	Page
Tables 15-1, 15-2	ACMM elements and maturity levels	pp. 671, 674
Table 15-3	Full ACMM framework	pp. 676-680
Tables 15-4, 15-5	CMMI elements and maturity levels	pp. 682, 685
Table 15-6	Full CMMI framework	pp. 688-689
Figure 15-1, Table 15-7	Strategy and vision excerpt and activities	pp. 691-692
Figure 15-2, Table 15-8	Architecture method and process excerpt and activities	pp. 693-694
Tables 15-9, 15-10	CMMI elements mapped to competencies and roles	pp. 696-697

Self-Check Questions

Q: Which two maturity models does the "Architecture Maturity Models" Series Guide name?

A: The Architecture Capability Maturity Model (ACMM) and Capability Maturity Model Integration (CMMI) (p. 669).

Q: In which ADM phase is a maturity model typically used?

A: The Preliminary Phase, to gauge the Architecture Capability's maturity (p. 670).

Q: Name at least six of the nine ACMM elements.

A: Architecture process; architecture development; business linkage; senior management involvement; business unit participation; architecture communication; architecture governance; IT security; IT investment and acquisition strategy (p. 671).

Q: Name the five CMMI elements.

A: Strategy and vision; architecture governance; architecture method and process; architecture deliverables; business alignment (p. 682).

Q: Give the ACMM and CMMI level names.

A: ACMM: 0 None, 1 Initial, 2 Under development, 3 Defined, 4 Managed, 5 Measured. CMMI: 1 Ad hoc, 2 Repeatable, 3 Defined, 4 Managed, 5 Optimal (pp. 674, 685).

Q: Why does Jager prefer CMMI for Enterprise Architecture?

A: ACMM focuses on IT architecture (even treating security as IT-only), while CMMI focuses on maturing the capability to achieve organizational goals, which fits EA as a strategic discipline (pp. 670, 681).

Q: What activities move "Strategy and vision" from Repeatable to Defined?

A: Obtain a mandate, create a RACI, establish an architecture board, appoint a process owner, and assign final responsibility to (business) senior management (p. 692).

Q: What are the five steps for raising a maturity level?

A: Identify the current level; determine the next level; ask targeted questions; implement the activities; review and reassess (p. 695).

Q: What does mapping competencies to CMMI elements reveal?

A: Several architecture roles are needed to raise maturity, and the Enterprise Architect has a significant stake (p. 698).

Q: What characterizes CMMI level 5 (Optimal)?

A: A mature process aligned with drivers, goals, and objectives. Continuous improvement. No projects proceed without EA input. Collaboration with similar organizations (p. 687).

Chapter 16: Alignment with Other Frameworks

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 16 (pp. 701-711)

Overview

Chapter 16 (pp. 701-711) explains how TOGAF fits alongside the other frameworks an organization already uses for planning and execution, IS governance and operations, management and measurement, and industry-specific matters (p. 701). Section 16.1 gives three questions and four steps for mapping existing frameworks (pp. 701-703). Section 16.2 presents Jager's own **Enterprise Architecture Implementation Wheel** (Document, Define, Execute, Control) as an example of a method that connects to TOGAF (pp. 703-706). The following section maps every Wheel stage to ADM phases (pp. 707-710). Section 16.4 summarizes (p. 711). Leave with this idea: alignment works in both directions. You can map existing frameworks onto TOGAF, or map other methods into TOGAF, and the ADM's tailorability makes both possible (pp. 703, 710). Chapter 16 is part of the BLUE reading tour (p. 701).

> **Reading note:** The Implementation Wheel is Jager's own method, from his book *Getting Started with Enterprise Architecture*, not part of the TOGAF Standard (p. 703). The mapping section is numbered "16.3.3" in the book although it directly follows 16.2 (p. 707). Treat it as the third section of the chapter.

Detailed Section Breakdown

Chapter Opener (p. 701)

- Organizations use many frameworks besides TOGAF. Aligning them with TOGAF ensures solid governance and outcomes (p. 701).

16.1 Mapping Existing Frameworks and Methods (pp. 701-703)

- **Three questions.** Which framework takes precedence? What is the organization's level of commitment to it? How does TOGAF fit in? These reveal how the organization views planning, execution, and governance (pp. 701-702).

Step	What to do
Build the framework catalog	List all frameworks used (planning and execution, IS governance and operations, management and measurement, industry-specific)
Organize by key models	Group them by models such as risk, accounting, planning
Determine what's needed from each	Identify the elements, methods, or techniques needed to implement the Architecture Capability's recommendations
Assess organizational commitment	Focus on frameworks the organization is truly committed to

(p. 702)

- **Reverse direction.** You can also examine how other frameworks connect *into* TOGAF (p. 703).

16.2 Enterprise Architecture Implementation Wheel (pp. 703-706)

- **What it is.** A practical, pragmatic method for developing a *fundamental* (not complete) EA, based on TOGAF but reorganized into four stages, each with steps, key focus areas, and artifacts. A streamlined entry point and "a simplified, tailored version of the TOGAF Standard" (pp. 703, 706).
- **Structure (Figure 16-1).** A core with three rings: stages (ring 1), steps (ring 2), key focus areas with deliverables (ring 3). Read clockwise from the inside out, but iterative, so you can start anywhere, e.g., with strategy if the inventory is already done (pp. 704-705).

Stage	Steps	Content
1 Document	Information; Stakeholders	Capture entities of interest across Business, Information, Application, Technology. Classify and map stakeholders. Create a Communications Plan
2 Define	Maturity; Framework; Strategy	Current and desired capability maturity. Principles, requirements, standards. Drivers, goals, objectives, initiatives
3 Execute	Strategy; Roadmap	Execute the strategy against goals and objectives. A roadmap of objectives, initiatives, activities
4 Control	Measure progress	Measure and visualize progress on goals and objectives. Monitor initiatives

(pp. 705-706)

- **"Standard" here** means specific SBBs that detail products or services meeting requirements, e.g., Azure or AWS web services for "use cloud-based services for commoditized services" (p. 706, footnote).
- **Why reorder?** For organizations new to architecture, TOGAF works "even more effectively" with a slightly changed sequence (p. 705).

Mapping to the Architecture Development Method (pp. 707-710)

Wheel stage	Step: key focus areas, then ADM phase
Document	Information: Organization (B), Processes (B), Concepts (C), Applications (C). Stakeholders: Classification, Mapping, Communication (all A)
Define	Maturity: Model, Level (A). Framework: Principles (Preliminary), Requirements and Standards (B, C, D). Strategy: Drivers, Goals, Objectives, Initiatives (Preliminary)
Execute	Strategy: Objectives (F). Roadmap: Initiatives (F), Activities (E, F)
Control	Measure progress: Objectives (G, H)

(Tables 16-1 to 16-4, pp. 707-710)

- **Document vs. ADM.** B covers organizational structure and processes. C and D capture applications and technology, and the Wheel adds classification during inventory. A covers stakeholders, mapping, and communication (pp. 707-708).
- **Define vs. ADM.** Phase A's maturity assessment is supplemented with maturity model guidance. TOGAF defines principles once the context is understood, so the Wheel inventories first. Strategy elements follow BIZBOK guidance, which TOGAF refers to (pp. 708-709).
- **Execute vs. ADM.** Phase F assigns business value to work packages. The Wheel turns strategy objectives into initiatives and links roadmap initiatives (E, F) to goals and objectives for visual governance (p. 709).
- **Control vs. ADM.** G ensures conformance and H tracks change. The Wheel adds tools to measure transition progress and to prioritize by business value (p. 710).
- **Conclusion.** The Wheel shows "how easy it is to hook into the TOGAF Standard" (p. 710).

16.4 Summary (p. 711)

- Determine precedence, commitment, and fit. Align in both directions. The Wheel follows TOGAF but adapts the sequence, and every element maps to ADM phases (p. 711).

Key Terms & Definitions

Term	Definition	Page
Framework catalog	A list of all frameworks used, grouped by key models	p. 702
Organizational commitment	How fully a framework is adopted in operations	p. 702
EA Implementation Wheel	Jager's four-stage method (Document, Define, Execute, Control) based on TOGAF	p. 703
Key focus area	Third-ring item in the Wheel, detailing deliverables per step	p. 705
Fundamental EA	A basic EA to build on over time, as opposed to a complete one	p. 706

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 16-1	The Enterprise Architecture Implementation Wheel	p. 704
Tables 16-1 to 16-4	Mapping of the four Wheel stages to ADM phases	pp. 707-710

Self-Check Questions

Q: What three questions guide the alignment of existing frameworks with TOGAF?

A: Which framework takes precedence? What is the organization's commitment to it? How does TOGAF fit in? (pp. 701-702).

Q: List the four steps for mapping existing frameworks.

A: Build the framework catalog; organize it by key models; determine what is needed from each; assess organizational commitment (p. 702).

Q: Name the four stages of the Implementation Wheel.

A: Document, Define, Execute, Control (p. 703).

Q: To which ADM phase do the Wheel's stakeholder classification, mapping, and communication map?

A: Phase A (Architecture Vision) (pp. 707-708).

Q: Why does the Wheel inventory the organization before defining principles?

A: TOGAF says principles should be defined once the organizational context is understood (p. 708).

Q: Which ADM phases correspond to the Control stage?

A: Phases G and H (p. 710).

Q: Is the Wheel part of the TOGAF Standard?

A: No. It is Jager's method based on TOGAF, a simplified, tailored version for getting started (pp. 703, 706).

Q: What does "alignment in the reverse direction" mean?

A: Examining how other frameworks or methods can connect into TOGAF, rather than mapping existing frameworks onto TOGAF (p. 703).

Chapter 17: Tailoring the TOGAF Standard

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 17 (pp. 713-723)

Overview

Chapter 17 (pp. 713-723) brings together the book's recurring message that TOGAF must be adapted to the organization. Factors that drive tailoring include industry regulation and compliance, project scope and complexity, the organization's maturity, and preferred modelling languages and tools (p. 713). Section 17.1 lists five ways to customize the Standard, with a nine-step checklist (pp. 714-718). Section 17.2 applies them to a healthcare case to rebut the claim that TOGAF lacks real-world examples (pp. 719-722). Section 17.3 summarizes (pp. 722-723). Leave with this idea: "The essential scaffolding of the TOGAF Framework is the concepts. Everything else... is either an example or a starter set", so substitute what doesn't fit (p. 715). Chapter 17 is part of the GREEN, BLUE, and GOLD reading tours (p. 713).

> **Reading note:** The quotations about customization come from the Standard. The healthcare case (HIPAA, GDPR, HL7 FHIR, ISO 13485, phase prioritization, regulatory decision gates) is Jager's illustration (pp. 719-722). The metamodel and Content Framework tailoring was covered in Ch. 5.2 and Ch. 6.4. This chapter summarizes all tailoring options in one place.

Detailed Section Breakdown

Chapter Opener (p. 713)

- TOGAF states early on that it should be adapted. Drivers: industry regulations, best practice and compliance, project scope and complexity, current maturity, modelling languages, tools, and techniques (p. 713).

17.1 Designed for Customization (pp. 714-718)

- "The TOGAF Framework is designed to require interpretation or customization" (p. 714).

Way to tailor	What can be adapted	Page
Organizational context	Entities to fit strategy and operating model. Metamodel to industry regulation and compliance. Size and structure (small, medium, large; centralized or decentralized). Governance and processes to culture and maturity	pp. 714-715
The ADM	Modify, omit, reorder, or combine phases (e.g., B, C, D together for a high-level definition). Skip unneeded steps (no new principles or team needed). Integrate with agile (e.g., SAFe). Add governance checkpoints. The Implementation Wheel is an example	pp. 715-716
Content Framework and deliverables	Align the metamodel and Content Framework with proprietary data models. Simplify or expand deliverables to match maturity. Align with ITIL, COBIT, Zachman, ArchiMate, agile (e.g., ITIL processes as functions or capabilities, a stakeholder entity from ArchiMate)	p. 716
Governance and compliance	Add organization-specific policies to the governance structure. Tailor techniques such as Risk and Stakeholder Management	p. 716
Tools and techniques	Choose ArchiMate, UML, BPMN, or custom notations. Integrate EA tools (e.g., Bizzdesign, LeanIX)	p. 716

- **Nine-step checklist (pp. 717-718):** align with organizational goals; adapt to industry standards; scale to organization size (large may use the full breadth, small only the parts that add value); reflect

maturity; adjust the ADM; customize the Content Framework and deliverables; change governance and compliance structures; select tools and techniques.

- **Answering critics.** Practitioners abandon TOGAF because it describes *what*, not *how*, and lacks real examples. Section 17.2 provides one (p. 718; see Ch. 1.1).

17.2 Tailoring the TOGAF Standard for the Healthcare Industry (pp. 719-722)

- **Healthcare challenges.** Regulatory compliance, interoperability, patient data security, fast-moving technology (p. 719).
- **Licence to adapt.** Architects "may choose not to include entities and relationships... and/or add additional entities and relationships" (p. 719).

Tailoring area	Healthcare adaptation	Page
Healthcare-specific needs	Align with HIPAA, GDPR, HL7 FHIR, ISO 13485. Add patient-centric entities to the metamodel. Modify artifacts for patient experience, EHRs, digital health. Align clinical and operational work. Strengthen security architecture	pp. 719-720
ADM	Agile for fast tech adoption (telemedicine, AI). *Phase prioritization*: emphasize B (patient journeys, workflows), strengthen D (cloud, data lakes, AI diagnostics), enhance F (legacy and EHR modernization)	p. 720
Content Framework and deliverables	FHIR-based data models. Capability models (telemedicine, remote monitoring, AI diagnostics; Figure 17-1). Interoperability maps (EHR, lab, pharmacy, PACS/RIS). Risk and compliance models (breaches, AI misdiagnosis, downtime)	pp. 720-721
Governance and compliance	An Architecture Board with clinical, IT security, compliance, and medical informatics members. Regulatory decision gates (e.g., HIPAA risk assessments, AI diagnostic approvals)	p. 721
Tools and techniques	ArchiMate with healthcare extensions. FHIR interoperability testing tools. AI analytics. Predictive modelling of patient flow	pp. 721-722

- **Result.** Better patient outcomes, compliance, and interoperability, with focus on security, interoperability, AI, and cloud ecosystems (p. 722).

17.3 Summary (pp. 722-723)

- Tailoring is essential to get value. Five customization areas: organizational context, ADM, Content Framework and deliverables, governance and compliance, tools and techniques. TOGAF can be tailored to "virtually any situation, scenario, organization, or industry" (pp. 722-723).

Key Terms & Definitions

Term	Definition	Page
Tailoring	Adapting the TOGAF Standard to the organization's needs, context, and maturity	pp. 713-714
Essential scaffolding	TOGAF's concepts. Everything else is an example or starter set that can be substituted	p. 715
Phase prioritization	Emphasizing certain ADM phases (e.g., B, D, F) to fit industry needs	p. 720
Regulatory decision gate	A governance checkpoint for regulatory approval (e.g., a HIPAA risk assessment)	p. 721
HL7 FHIR	Fast Healthcare Interoperability Resources: rules for secure exchange of electronic health data	pp. 719-720

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 17-1	Tailored industry-specific Business Capability Diagram (healthcare)	p. 721

Self-Check Questions

Q: Name the five ways the TOGAF Standard can be customized.

A: Tailoring for organizational context; tailoring the ADM; tailoring the Content Framework and deliverables; tailoring governance and compliance; tailoring tools and techniques (p. 714).

Q: What does the Standard call its "essential scaffolding", and what does that imply?

A: Its concepts. Everything else is an example or starter set that can be replaced using the modular structure (p. 715).

Q: Give two examples of tailoring the ADM.

A: Skip Preliminary steps when no new principles or team are needed; combine B, C, and D for a high-level architecture; reorder phases; integrate SAFe; add governance checkpoints (p. 715).

Q: How can ITIL or COBIT be combined with TOGAF content?

A: By adding ITIL processes or COBIT parts to the Content Framework as business functions or capabilities (p. 716).

Q: In the healthcare case, which ADM phases are prioritized, and why?

A: B (patient journeys and hospital workflows), D (cloud, data lakes, AI diagnostics), and F (legacy and EHR modernization) (p. 720).

Q: How might an Architecture Board be tailored for healthcare?

A: Include clinical representatives, IT security, compliance officers, and medical informatics and health IT experts, and add regulatory decision gates (p. 721).

Q: Which factors drive how much TOGAF should be tailored?

A: Industry regulations, best practices, and compliance; project scope and complexity; the organization's maturity level; preferred modelling languages, tools, and techniques (p. 713).

Q: Why should a small organization not adopt the whole framework?

A: Only the parts that add value should be used. Large organizations may implement the full breadth, small ones only parts (p. 717).

Chapter 18: Agile Architecture

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 18 (pp. 725-756)

Overview

Chapter 18 (pp. 725-756) argues that "agile architecture is not an oxymoron" and shows how TOGAF supports agile delivery through iteration, partitioning, and "just enough" architecture (p. 725). Section 18.1 defines agility and the basic concepts: partitions, sprints, Minimum Viable Architecture, and parallel work (pp. 726-731). Section 18.2 revisits the three levels of granularity: Strategic, Segment, Capability (pp. 731-735). Section 18.3 shows how partitions, including partitioned Transition Architectures, map to sprints (pp. 735-739). Section 18.4 maps ADM phases to agile and SAFe concepts and adapts each phase for sprints (pp. 739-754). Section 18.5 summarizes (pp. 754-756). Leave with this idea: partition the architecture into Minimum Viable Architectures that teams can build in parallel sprints, while keeping the strategic big picture (pp. 728-729, 736). Chapter 18 is part of the BLUE reading tour (p. 725).

> **Reading note:** The definitions of agile, Minimum Viable Architecture, Transition Architecture, and Minimum Viable Product, and the granularity-to-agile mapping, come from the Standard and its Series Guides (G20F "Enabling Enterprise Agility", G210 "Applying the TOGAF ADM using Agile Sprints"). The painting analogy (p. 731) and the plain-language summaries are Jager's. Don't confuse **Minimum Viable Architecture** (the least architecture that is realizable and adds value) with **Minimum Viable Product** (the least output meeting minimum requirements in live operation).

Detailed Section Breakdown

Chapter Opener (p. 725)

- Agile architecture embraces change, collaboration, and incremental value. TOGAF adapts to it through iterative cycles, continuous stakeholder engagement, and incremental delivery, and even *enhances* agile by managing complexity and change in a structured way (p. 725).

18.1 What Is Agile Architecture (pp. 726-731)

- **Agility.** Adaptability. Short iterative cycles. Reprioritization toward high-value outcomes with minimal intermediate documentation. Experimentation ("fail fast"). Autonomous multidisciplinary teams. "A journey, not a destination" (p. 726).
- **Definition.** "Agile means to move/change quickly and easily, often to provide value-generating outcomes." Organizational agility is broader than the Agile Software Development Manifesto (p. 726).

18.1.1 The Basic Concepts Explained (pp. 727-731)

Concept	Meaning	Page
Architecture partition	A logical scope of work that can be independently architected and managed (Ch. 13.4)	p. 727
Sprint	A short, time-bound iteration that delivers working functionality. Each partition maps to one or more sprints	p. 728
Minimum Viable Architecture	"The minimum architecture that is realizable and add[s] business value". A partition contains at least this	p. 728
Parallel work	Agile does not require sequential completion, but the big picture must be kept	p. 729

- **Regular approach (Figure 18-3) mapped to the ADM.** Define Problem (A), Define Baseline and Define Target (B-D), Deploy Target (E-G), Govern and Manage Change (H) (pp. 729-730).
- **With sprints (Figure 18-4).** Each component runs several, possibly parallel, sprints. Define *enough* of the problem to give context and refine it as you go, but no part of the target can be implemented before its baseline and target are sufficiently defined (pp. 730-731).
- **TOGAF's route to agile** is architecture partitioning across teams with clear definitions, delivered via sprints (p. 731).

Jager's commentary (author's view): "It's like creating a painting. If you don't know what you want the painting to look like, you cannot start painting" (p. 731).

18.2 Three Levels of Architecture Granularity (pp. 731-735)

Level	Description (Ch. 7.2)	Agile role	Page
Strategic Architecture	Long-term summary of the whole organization, for executive direction	A dynamic iteration, not a static blueprint. Supported by A (and detailed in B). Identifies needed architecture and delivery capabilities. The problem is defined here and at segment level	pp. 731-733
Segment Architecture	Detailed operating models for areas. Program or portfolio level ("Domain Architecture")	"Just enough": key features plus functional and non-functional requirements, without every ADM step. Parallel iterations by teams, who help define it	pp. 731-733
Capability Architecture	Detail on supporting a capability, with increments, grouping work packages	Link between strategy and implementation. Increments map to one or more sprints. Builds on Segment Architectures. The target is deployed here by agile teams	pp. 732, 734-735

- **Example (Figures 18-6, 18-7).** A Capability Architecture within *Information Management* is partitioned into three components (a logical application component, an application service, a data entity), each built in its own sprint (pp. 734-735).

18.3 Architecture Partitions and Agile Sprints (pp. 735-739)

- **No waiting.** Segment work starts once the areas of interest are known, and Capability work doesn't wait for Segments to finish (pp. 735-736).
- **Balance detail.** Provide enough detail at each level, but not too much too early without feedback from earlier sprints. Partitions are "the agile equivalent of a Minimum Viable Architecture" (p. 736).
- **Flow.** Identify segments, refine into Capability Architectures, deliver incrementally (in parallel where dependencies allow). The output is a solution specification built by agile teams under architecture guidelines, metrics, and compliance (p. 736).
- **Partitioned transitions (Figures 18-8 to 18-11).** A single Transition Architecture plateau can be split into Transitions A, B, C, each developed in sprints. A Transition Architecture is "a formal description of one state of the architecture at an architecturally significant point in time, usually including a number of capability increments" (pp. 737-739).

18.4 Developing an Agile Architecture (pp. 739-754)

- The three levels and their ADM phases map well to agile, and each phase supports agility (pp. 739-740).

18.4.1 Mapping the ADM to Agile Concepts (pp. 740-744)

Granularity	ADM phases	SAFe / Agile / Scrum layer
Strategic	Preliminary, A	SAFe Strategic Themes. Enterprise Portfolio
Segment	B, C, D	Product/Portfolio Backlog. Change Plans. Epics
Capability	E, F, G, H	Product/Solution Backlog (capabilities). Program Backlog (capability increments, i.e., features). Program Team and Scrum Sprint Team Backlogs (deliverable user stories)

(Table 18-1, pp. 740-741)

- **Strategic level.** Preliminary (if capability changes are needed) and A produce strategic plans called *courses of action*. Agile equivalents are strategic themes and the top portfolio backlog. Cross-functional teams are needed (p. 741).
- **Segment level.** Partition courses of action across units. Explore B-D further if A is insufficient. Self-organizing teams and a high-level C/D iteration produce *Epics* and an initial segment backlog, which can feed prototypes (pp. 741-742).
- **Capability level.** B-D specify ABBs (functional and non-functional). E-F develop SBBs and their integration. The smallest deliverable unit is the *Minimum Viable Product*: "an output that satisfies a minimum set of functional and non-functional requirements and can be realized when implemented in a live operational environment" (p. 742).
- **Backlogs are refined into sprint backlogs** deliverable within weeks or a month. G confirms that contracted capabilities were delivered and that operating information is stored. H confirms that benefits were realized and users are satisfied (pp. 743-744).

18.4.2 ADM Phases and Agile Sprints (pp. 744-754)

- **Each iteration should yield a Minimum Viable Architecture.** Figure 18-13 shows the per-phase sprint adaptation (the Preliminary Phase is omitted) (pp. 744-745).
- **Phase A.** A high-level vision of outcomes, capabilities, and value. It evolves gradually ("evolutionary, not revolutionary"), is a stable foundation for growth, and gives teams clear direction (p. 745).
- **Phases B-D.** Break the work into Minimum Viable Architectures delivered by parallel teams. Business Architects, Enterprise Architects, and solution teams collaborate on Minimum Viable Business Designs. B-E need no strict order (pp. 745-747).

Phase E adapted for sprints (Table 18-2)	Phase F adapted for sprints (Table 18-3)
Understand stakeholder readiness (may need doing only once)	Confirm the management framework (before the agile journey)
Identify business constraints before each sprint, and review after a series	Assign business value to work packages per sprint
Review the gap analysis regularly (Ch. 14.5)	Estimate resources and timelines during sprint planning
Manage cross-functional requirements each sprint	Prioritize by cost/benefit and risk per sprint
Ensure interoperability of sprint solutions (Ch. 14.6)	Update the roadmap and ADD after each sprint
Refine dependencies between teams	Complete the migration plan for the next sprint (update the backlog)
Assess readiness and risks for the MVP (Ch. 14.8)	Document lessons learned each sprint

Phase E adapted for sprints (Table 18-2)	Phase F adapted for sprints (Table 18-3)
Develop an implementation strategy, define work packages for backlogs, plan transitions, create the roadmap and migration plan with product owners (Ch. 12.8.3)	

(pp. 746-748)

- **Phase G for sprints (Table 18-4).** Compliance is less formal and *ongoing*, so short cycles catch non-compliance early. Scope and resources are confirmed at sprint start, guidance is informal, an increment is deployed after each approved demo, and the post-implementation review summarizes earlier reviews (pp. 749-750).
- **Phase H for sprints (Table 18-5).** Separate change management into a parallel process: establish value realization and deploy monitoring tools. Collect risks from demos and retrospectives. Move change requests into a business change backlog. Governance cascades (business governs the EA team, the EA team governs solution teams, and vice versa) (pp. 750-751).
- **Sprint rhythm.** Sprint zero lays the groundwork. Each sprint delivers a Minimum Viable Architecture. A demo shows stakeholders the progress. A retrospective reviews quality and process (velocity, estimate accuracy, requirement allocation). Plan from the highest-value backlog items. Parallel EA teams are possible. Decoupled design lets building start before the architecture is finished (pp. 751-753).
- **Sprint planning** is also the moment to sharpen outcomes, success measures, and functional and non-functional requirements (p. 753).

18.5 Summary (pp. 754-756)

- Agility is adaptability and results first. Strategic, Segment, and Capability levels. Partitioning into Minimum Viable Architectures delivered by sprints in parallel, respecting dependencies (pp. 754-755).
- Strategic: Preliminary and A (courses of action). Segment: organize plans across units (B-D if needed). Capability: B-D detailed, then E-F into SBBs. G catches non-compliance early. H runs in parallel to confirm value (pp. 755-756).

Key Terms & Definitions

Term	Definition	Page
Agile	To move or change quickly and easily, often to provide value-generating outcomes	p. 726
Architecture partition	A logical scope of work that can be independently architected and managed	p. 727
Sprint	A short, time-bound iteration that delivers working functionality	p. 728
Minimum Viable Architecture	The minimum architecture that is realizable and adds business value	p. 728
Minimum Viable Product	An output meeting a minimum set of functional and non-functional requirements, realizable in live operation	p. 742
Transition Architecture	A formal description of one architecture state at an architecturally significant point in time, usually including capability increments	p. 738
Courses of action	High-level strategic plans produced at the strategic level	p. 741
Epic	A large or long-running user story, produced at the segment level	p. 742
Sprint zero	The initial preparation sprint that lays the groundwork	p. 751

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figures 18-1, 18-2	Architecture partitions. Partitions mapped to sprints	pp. 727-728
Figures 18-3, 18-4	Regular approach components, without and with sprints	pp. 729-730
Figures 18-5, 18-6, 18-7	Three granularity levels. A partitioned Capability Architecture built in three sprints	pp. 732-735
Figures 18-8 to 18-11	Baseline, Transition, Target plateaus. Partitioned transitions and sprints	pp. 737-739
Table 18-1, Figure 18-12	Granularity and ADM phases mapped to agile and SAFe	pp. 740-743
Figure 18-13	ADM phases and sprints	p. 745
Tables 18-2 to 18-5	Adapting Phases E, F, G, H for sprints	pp. 746-751

Self-Check Questions

Q: What is a Minimum Viable Architecture, and how does it relate to partitions?

A: The minimum architecture that is realizable and adds business value. Partitions are the agile equivalent of Minimum Viable Architectures (pp. 728, 736).

Q: Map the five components of the regular approach to ADM phases.

A: Define Problem: A. Define Baseline and Define Target: B-D. Deploy Target: E, F, G. Govern and Manage Change: H (pp. 729-730).

Q: Must the problem definition be complete before other work starts?

A: No. Define enough to give context and keep refining. But you can't implement a part of the target until its baseline and target are sufficiently defined (pp. 730-731).

Q: Which ADM phases and agile layers belong to the Segment level?

A: Phases B, C, D, mapped to the Product/Portfolio Backlog, Change Plans, and Epics (pp. 740-742).

Q: What is produced at the strategic level in agile terms?

A: Courses of action (high-level strategic plans), corresponding to strategic themes and the top portfolio backlog (p. 741).

Q: How does Phase G change in a sprint-based approach?

A: Compliance becomes informal and ongoing rather than a gate, so short cycles reveal non-compliant solutions early. Increments deploy after approved demos (pp. 749-750).

Q: How is Phase H adapted for agile?

A: Change management runs as a separate parallel process: value realization, monitoring tools, risks from demos and retrospectives, and change requests into a business change backlog (pp. 750-751).

Q: How can a single Transition Architecture be made agile?

A: Partition it into several Transition Architectures (A, B, C), each developed in one or more sprints (pp. 737-738).

Q: Why does decoupled design matter?

A: It lets solution teams start building before the whole architecture is finished, without waiting on each other (p. 752).

Q: Distinguish a Minimum Viable Architecture from a Minimum Viable Product.

A: A Minimum Viable Architecture is the least *architecture* that is realizable and adds value. A Minimum Viable Product is the least *output* meeting minimum functional and non-functional requirements that can run in live operation (pp. 728, 742).

Chapter 19: Digital Transformation

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 19 (pp. 757-779)

Overview

Chapter 19 (pp. 757-779) shows how the ADM structures a digital transformation: Phase A for the strategic vision, B for process redesign, C and D for the digital ecosystem and infrastructure, E-G for prioritizing, planning, and governing, and H for continuous innovation (p. 757). Section 19.1 defines digital transformation and its components (pp. 758-759). Section 19.2 relates it to Enterprise Architecture and distinguishes **digitization**, **digitalization**, and **digital transformation** (pp. 759-763). Section 19.3 gives nine ADM-based steps and a five-phase real hospital case (pp. 763-777). Section 19.4 summarizes (pp. 777-779). Leave with this idea: digital transformation starts with a change in the *business model*, not in IT. EA and transformation should be integrated in a hybrid approach (pp. 761-762, 769). Chapter 19 is part of the BLUE reading tour (p. 757).

> **Reading note:** The TOGAF view of digital transformation as the "strategic and systematic evolution of an organization's capabilities, processes, and technologies" is attributed to the Standard (p. 759). The nine steps are Jager's ADM-based approach. The hospital case (government policy "the right care in the right place", a 30% cost reduction) is his personal experience (pp. 768-777). Note that Ch. 3 (p. 48) points to "Chapter 18" for digital transformation, but it is covered here in Chapter 19.

Detailed Section Breakdown

Chapter Opener (p. 757)

- **ADM as the transformation vehicle.** A: strategic vision linking goals and digital capabilities. B: optimize processes for digital readiness. C and D: data, applications, and infrastructure for the digital ecosystem. E-G: continuous evaluation and governance. H: refine strategy as technology and needs evolve (p. 757).

19.1 What Is Digital Transformation? (pp. 758-759)

- **Definition.** Integrating digital technologies into all aspects of the organization to improve efficiency, customer experience, and innovation. A fundamental shift in how it operates and delivers value, "primarily based on a major change in the business and operating models" (p. 758).

Key component	Examples
Technology adoption	Cloud, AI, automation, analytics
Process optimization	Automated workflows, collaboration tools, AI insights
Customer experience enhancement	Digital platforms, self-service portals, personalization
Cultural change	Digital-first mindset, upskilling, agile
Data-driven decision-making	Big data, real-time analytics, BI

(pp. 758-759)

- **An ongoing journey**, not a one-time project (p. 759).
- **TOGAF's view.** "The strategic and systematic evolution of an organization's capabilities, processes, and technologies to enhance business outcomes through digital innovation", aligned with objectives,

governance, and efficiency (p. 759).

19.2 Digital Transformation and Enterprise Architecture (pp. 759-763)

- **Not just digitizing processes.** It requires a cultural shift. EA provides the framework that aligns technology, processes, and people with business objectives (p. 759).
- **Both are strategic, business-driven initiatives** that may require reorganizing internal and external information environments (p. 760).

Term	Meaning (Figure 19-1)
Digitization	Converting physical assets into digital formats (scanning, data conversion)
Digitalization	Using digital technology to modify existing processes and improve workflows
Digital transformation	Reimagining the business model and value propositions, from physical to predominantly digital engagement and delivery

(pp. 760-761)

- **Business-led.** "True transformation must come from within the business itself". Isolated IT or data strategies must align with business objectives. A strategy without the business structures is "a house without a solid foundation" (p. 761).
- **Mutual influence.** Transformation reshapes the Target Architecture and often starts with use cases scaled via the EA roadmap. EA provides the target, operating model, baseline, and gaps to set priorities and guide investment (p. 762).
- **Hybrid approach.** Integrating both gives immediate value plus a long-term foundation (p. 762).

19.3 A Structured Approach to Digital Transformation (pp. 763-777)

Step (Figure 19-2)	ADM phase	Key activities	Page
1. Prepare for transformation	Preliminary	Principles, governance, capabilities. Identify stakeholders (CxO, executives, architects). Set boundaries via transformation principles (customer centricity, data-driven, cloud-first). Digital maturity and readiness assessment. EA team and governance	p. 764
2. Define the digital strategy	A	Drivers (customer experience, efficiency, compliance) linked to SMART goals and KPIs. Enablers (AI, automation, cloud, IoT) in a high-level target. Stakeholder buy-in	pp. 764-765
3. Re-engineer business automation	B	Gap analysis of current vs. target business models. Redesign processes with automation. New roles. Customer-centric platforms. Business transformation roadmap (Work Package Portfolio Map)	p. 765
4. Design the digital ecosystem	C	Data: governance (GDPR, HIPAA), centralized strategy (data lakes, real-time analytics), AI insights. Applications: modernization (Application Portfolio Catalog), cloud and API integration	p. 766
5. Develop digital infrastructure	D	Cloud strategy (public, private, hybrid). Microservices and containers. Zero-trust and AI threat detection. High availability and DR. DevOps and CI/CD	p. 766
6. Create the roadmap	E	Prioritize by value and feasibility using the Business Value and Risk Matrix. Quick wins (high value, low risk). Short- and long-term initiatives	p. 767
7. Manage the implementation process	F	Incremental deployment (pilots, phased rollouts as Transitions). Risks and mitigation. Change management plan. Digital-first training	p. 767
8. Measure compliance and success	G	Governance frameworks, feedback loops, EA performance dashboards (Figure 12-54). Security, legal, and compliance risks	pp. 767-768

Step (Figure 19-2)	ADM phase	Key activities	Page
9. Ensure continuous improvement	H	Monitor trends, regular EA reviews, business-IT alignment, continuous innovation strategy	p. 768

Jager's hospital case (pp. 768-777)

- **Driver.** Government policy, "the right care in the right place", plus cost savings. The answer was home-based care, which required changing the business and operating model (telemedicine, remote monitoring, self-medication) (pp. 768-770).
- **Biggest challenge: people.** "Every change meets resistance". Many conversations were held to surface and address employee concerns (p. 769).

Case phase (Figure 19-3)	Months	ADM phases	Main activities
1. Assessment and strategy development	0-3	Preliminary, A	Assess IT (EHR, telemedicine), inefficiencies, a cybersecurity/HIPAA risk assessment, staff and patient feedback. Goals and KPIs (wait times, telehealth adoption). Board alignment (Table 19-1)
2. Technology selection and process optimization	3-6	B, C, D	Automate scheduling and billing, AI chatbots, digital HR, EHR enhancement, AI diagnostics, network security. New capabilities (Figure 19-4) (Table 19-2)
3. Implementation and staff training	6-12	E, F	Digital literacy and cybersecurity training, change management, telemedicine launch, wearables, AI clinical decision support (Table 19-3)
4. Patient experience and engagement	12-18	G	Patient portals, 24/7 AI virtual assistants, mobile apps, IoT smart beds, predictive analytics, RPA for claims (Table 19-4)
5. Continuous improvement and future innovations	18+	H	Track KPIs, gather feedback, update security, explore blockchain, AI drug discovery, AR/VR (Table 19-5)

(pp. 770-776)

- **Outcomes.** Virtual assistants handled non-acute consultations, freeing physicians for acute care (Figure 19-5). Healthcare costs were cut by 30% by moving non-acute care home. Staff came to see technology as an enabler (pp. 774-777).

19.4 Summary (pp. 777-779)

- Nine summarized steps: outline the digital vision; get ready for change; rethink business automation; build the digital framework; deploy the right technology; plan the journey; manage the transition; monitor progress and compliance; continue to improve. The ADM provides "a robust approach" for systematic transformation (pp. 777-779).

Key Terms & Definitions

Term	Definition	Page
Digital transformation	Integrating digital technology across the organization, based on a major change in business and operating models	p. 758
Digital transformation (TOGAF view)	Strategic, systematic evolution of capabilities, processes, and technologies to enhance outcomes through digital innovation	p. 759
Digitization	Converting physical assets to digital format	p. 760
Digitalization	Applying digital technology to modify existing processes	p. 760

Term	Definition	Page
Digital maturity assessment	Evaluation of customer engagement, workforce skills, processes, and infrastructure	p. 764
Hybrid approach	Integrating EA and digital transformation for immediate and long-term value	p. 762

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 19-1	Focus areas of digitization, digitalization, digital transformation	p. 761
Figure 19-2	Nine steps for digital transformation using the ADM	p. 763
Figure 19-3	Five phases of the hospital transformation	p. 769
Tables 19-1 to 19-5	Activities per case phase	pp. 770-776
Figure 19-4	Introduction of new capabilities	p. 771
Figure 19-5	New capability removing the need for a physician role in non-acute cases	p. 774

Self-Check Questions

Q: Distinguish digitization, digitalization, and digital transformation.

A: Digitization converts physical assets to digital. Digitalization uses digital technology to improve existing processes. Digital transformation reimagines the business model and value propositions toward digital delivery (p. 760).

Q: Where must true digital transformation originate?

A: Within the business itself, as a change in the business model, not in IT (pp. 761, 769).

Q: How do EA and digital transformation influence each other?

A: Transformation insights reshape the Target Architecture and start from use cases scaled via the EA roadmap. EA supplies the target, operating model, baseline, and gaps for priorities and investment. A hybrid approach is best (p. 762).

Q: Which ADM phase defines the digital strategy, and what does it include?

A: Phase A: drivers, SMART goals and KPIs, enablers (AI, cloud, IoT) in a high-level target, and stakeholder buy-in (pp. 764-765).

Q: Which artifact helps identify quick-win transformation projects in Phase E?

A: The Business Value and Risk Matrix, which highlights high-value, low-risk projects (p. 767).

Q: Name the five key components of digital transformation.

A: Technology adoption, process optimization, customer experience enhancement, cultural change, data-driven decision-making (pp. 758-759).

Q: What was the biggest challenge in Jager's hospital case, and how was it handled?

A: Getting employees on board. Many conversations surfaced and addressed their concerns (p. 769).

Q: What result did the hospital achieve?

A: About 30% lower healthcare costs by moving non-acute care to home care through telehealth, remote monitoring, and self-medication (p. 777).

Chapter 20: Putting the TOGAF Standard into Practice

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 20 (pp. 781-815)

Overview

Chapter 20 (pp. 781-815) is a single end-to-end case that shows the whole book working together. Jim, the Enterprise Architect of the hospital **Academic Diagnostic Medicine**, turns a 2028-2030 strategy into an implemented architecture using a tailored ADM (p. 781). Section 20.1 follows him through seven time-boxed stages, from March 2027 to August 2030: scoping the work, envisioning, starting development, setting direction, getting ready for implementation, overseeing, and maintaining control. Each stage produces specific artifacts (pp. 782-808). Section 20.2 then maps the stages back to ADM phases and the steps actually used (pp. 808-815). Leave with this idea: Jim used only the ADM steps that added value, iterated and jumped between phases, and linked every requirement, work package, and roadmap item to goals. That is TOGAF "as intended" (p. 815). Chapter 20 is part of the GREEN and GOLD reading tours (p. 781).

> **Reading note:** The whole chapter is Jager's fictional case, not Standard text. It reuses artifacts and techniques from Chapters 5, 7, 12, and 14 (Driver/Goal/Objective Diagram, Requirements Catalog, Stakeholder Catalog, Plateau Planning, Work Package Portfolio Map, Implementation Factor Catalog, Business Value and Risk Matrix). It is the best chapter for seeing how the parts connect. Note the tailoring choices: Jim *skipped* a maturity assessment and *added* a risk entity and Risk Catalog (p. 809).

Detailed Section Breakdown

Chapter Opener (p. 781)

- TOGAF richly describes metamodel and Content Framework tailoring, the repository, governance, and techniques, but applying them is a challenge. This case shows it done (p. 781).

20.1 A Case Scenario (pp. 782-808)

- **Setting.** A patient-centred hospital with a 2028-2030 agenda: patient satisfaction, workflow optimization, financial health. It has fairly high architecture maturity and TOGAF has long been its standard. Jim's role is to translate drivers into capabilities, align business and IT, and keep change structured, traceable, and value-driven (p. 782).

Stage	Period	What Jim did	Key artifacts	Page
20.1.1 Scoping the Work	Mar-Apr 2027	Two board strategy meetings. Visualized the strategy and proposed three projects (improve patient satisfaction, improve clinical workflows, reduce operating costs). The board approved and gave requirements. Planned to add a *risk* entity to the metamodel	Driver/Goal/Objective Diagram (Fig. 20-1); Requirements Catalog linked to goals (Table 20-1)	pp. 782-784

Stage	Period	What Jim did	Key artifacts	Page
20.1.2 Envisioning the Architecture	May-Jun 2027	Interviewed board, executives, clinical and operational staff. Classified stakeholders. Recorded what communication worked. Captured concerns and constraints (e.g., low digital literacy of older patients, resistance to standardization, retraining, speed of savings)	Stakeholder Catalog (Table 20-2); Communications Plan; concerns and constraints (Table 20-3); Statement of Architecture Work	pp. 784-787
20.1.3 Starting the Development	Jul-Oct 2027	Reused repository building blocks. Linked strategy to capabilities. Got the Statement of Architecture Work approved. Built baselines and found missing capabilities (Feedback Analysis, Supply Chain Management), processes, and roles (Patient Safety Officer, Financial Analyst). Heat-mapped capabilities. Baseline and target application views. Accepted the COO's no-infrastructure-change constraint (cloud already in place). First ADD	Business Capability Map (Tables 20-4, 20-6); Capability/Business Process Matrix (Tables 20-5, 20-7); Strategy/Capability Diagram (Fig. 20-2); Organization/Business Roles Catalog (Table 20-8); Capability Heat Map (Fig. 20-3); Process/Application Realization Diagrams (Figs. 20-4, 20-5)	pp. 787-793
20.1.4 Setting the Direction	Nov-Dec 2027	Two Transition Architectures. Gaps A, B, C closed by work packages linked to goals. Draft roadmap with work packages colour-coded by project. HR to fill missing roles. Checked impact on running projects (none). Stakeholder sessions. Requirements agreed as criteria. Rationale documented. ADD finalized after two feedback iterations	Plateau Planning Diagram (Fig. 20-6); Work Package Portfolio Map (Table 20-9); Draft Architecture Roadmap (Fig. 20-7); ADD	pp. 794-798
20.1.5 Getting Ready for Implementation	Jan-May 2028	Identified three implementation factors (automating discharge planning, training the Planner role, feedback analysis distinguishing positive and negative). Revisited concerns and constraints. Three-phase migration plan respecting dependencies (new capabilities first). Readiness assessment flagged employee engagement and skills (score 3). Plan approved. Roles filled. Project manager adopts the plan. Risks plotted per project and per building block. Formal approval	Implementation Factor Catalog (Table 20-10); Implementation and Migration Plan (Table 20-11); BTRA (Table 20-12); Business Value and Risk Matrix (Fig. 20-8); Risk Catalog; Risk/Process/Application Realization Diagram (Fig. 20-9)	pp. 798-805
20.1.6 Overseeing the Architecture Work	Jun 2028-Aug 2029	Work carried out under the Architecture Contract. Random compliance checks, with reports shared. Transition A delivered. Capability map and heat map updated (new capabilities working, Patient Feedback Management improved)	Architecture Contract; Architecture Compliance Report; updated Capability Heat Map (Fig. 20-10)	pp. 805-806

Stage	Period	What Jim did	Key artifacts	Page
20.1.7 Maintaining Control	Sep 2029-Aug 2030	Stakeholder questionnaires and performance reviews to manage change. Heat-mapped the strategy for the board. Investigated lagging objectives: the Planning Application's learning curve delayed the throughput target, and the cost focus had been on recording before reducing. Target Architecture nearly reached	Driver/Goal/Objective Diagram Heat Map (Fig. 20-11)	pp. 806-808

- **Why linking matters.** Requirements tied to goals keep them clear to everyone (p. 784). Work packages mapped to objectives' timeframes make progress traceable (pp. 795-796).
- **Risk becomes first-class.** By adding a risk entity early, Jim could capture risks in a Risk Catalog and later visualize them on projects (Business Value and Risk Matrix: optimize clinical workflows is the highest risk because of automation and retraining) and on building blocks (traffic-light scores across five levels) (pp. 784, 799-805).

20.2 Summary (pp. 808-815)

- Jim covered all ADM phases, iterating and sometimes jumping ahead and back, which the Standard's adaptability allows (p. 808).

Period	ADM phase(s)	Steps actually performed	Page
Mar-Apr 2027	Preliminary	Identify key drivers and context; assess EA maturity (skipped in practice, since already mature); choose and tailor the framework (added the risk entity and Risk Catalog)	p. 809
May-Jun 2027	A	Establish the project; identify stakeholders and requirements; high-level vision (Strategy/Capability Diagram); Statement of Architecture Work approved	p. 810
Jul-Oct 2027	B, C, D	Select models and tools (repository reuse); Baseline and Target; gap analysis; candidate roadmap components; stakeholder reviews; ADD	pp. 811-812
Nov-Dec 2027	E	Corporate change attributes (Implementation Factor Catalog); interoperability; group work packages; Transition Architectures; roadmap and migration plan	pp. 812-813
Jan-May 2028	F	Align with project management; assign business value (Business Value and Risk Matrix); prioritize; confirm roadmap; complete plan	pp. 813-814
Jun 2028-Aug 2029	G	Confirm scope and priorities (Work Package Portfolio Map); compliance reviews; post-implementation review after each transition	pp. 814-815
Sep 2029-Jul 2030	H	Monitoring tools (dashboards); manage risks; analysis (strategy heat map). New risks feed the next ADM iteration	p. 815

- **The lesson.** "Not all of the method's steps were necessary... he only used the parts that added value... He used the TOGAF Standard as intended" (p. 815).

Key Terms & Definitions

Term	Definition (as used in the case)	Page
Strategy/Capability Diagram	Combines the Driver/Goal/Objective Diagram with the Business Capability Map to show where strategy hits capabilities	p. 788
Capability/Business Process Matrix	A tailored artifact relating capabilities to existing (or missing) processes	pp. 787-790

Term	Definition (as used in the case)	Page
Organization/Business Roles Catalog	Roles per unit with status (existing or non-existent)	p. 790
Risk Catalog	A tailored artifact made possible by adding a risk entity to the metamodel	pp. 784, 804
Risk/Process/Application Realization Diagram	Risks attached to building blocks with traffic-light scores	p. 805
Driver/Goal/Objective Heat Map	Strategy diagram coloured by achievement, used for board reporting	p. 807

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 20-1, Table 20-1	The hospital's strategy. Requirements Catalog	pp. 783-784
Tables 20-2, 20-3	Stakeholder Catalog. Concerns and constraints	pp. 785-786
Tables 20-4 to 20-8, Figure 20-2	Capability map, capability/process matrix, missing capabilities, processes, roles. Strategy/Capability Diagram	pp. 787-790
Figures 20-3 to 20-5	Capability Heat Map. Baseline and target Process/Application Realization	pp. 791-793
Figure 20-6, Table 20-9, Figure 20-7	Plateau Planning. Work Package Portfolio Map. Draft roadmap	pp. 794-796
Tables 20-10 to 20-12	Implementation Factor Catalog. Migration plan. Readiness assessment	pp. 799-802
Figures 20-8, 20-9	Business Value and Risk Matrix. Risk/Process/Application Realization	pp. 804-805
Figures 20-10, 20-11	Updated Capability Heat Map. Driver/Goal/Objective Heat Map	pp. 806-807
Tables 20-13 to 20-19	Steps performed per ADM phase	pp. 809-815

Self-Check Questions

Q: What artifact did Jim use first to visualize the strategy, and what did it lead to?

A: A Driver/Goal/Objective Diagram, which suggested three projects: improve patient satisfaction, improve clinical workflows, and reduce operating costs (pp. 782-783).

Q: Why did Jim link each requirement to a goal?

A: So that the requirements for achieving each goal were clear to everyone at all times (p. 784).

Q: Which tailoring decisions did Jim make in the Preliminary Phase?

A: He skipped the maturity measurement (the organization was already mature with TOGAF) and added a risk entity to the metamodel and a Risk Catalog to the Content Framework (p. 809).

Q: What gaps did the baseline analysis reveal?

A: Missing capabilities (Feedback Analysis, Supply Chain Management), missing processes (feedback trend analysis, infection prevention audit, supply chain rationalization), and missing roles (Patient Safety Officer, Financial Analyst) (pp. 789-790).

Q: How did Jim handle the COO's constraint on infrastructure?

A: He adapted the architecture to the existing cloud infrastructure, since there was no budget for changes (p. 793).

Q: Which readiness factors needed attention, and what was the remedy?

A: Employee engagement and skills and competencies (both scored 3), linked to automating discharge planning. The remedy was training staff (pp. 802-803).

Q: Which project carried the highest risk on the Business Value and Risk Matrix, and why?

A: Optimize clinical workflows, because it automates manual tasks and requires the Planner role to be reinterpreted and retrained (p. 804).

Q: Why was the 10% throughput objective late?

A: The new Planning Application had a steeper learning curve than assumed, which delayed adoption (p. 807).

Q: In which ADM phase did Jim use the Work Package Portfolio Map to prioritize execution?

A: Phase G, to confirm scope and priorities and keep activities in the right order (p. 814).

Q: What is the overall lesson of the case?

A: Tailor the ADM: use only the steps that add value, iterate as needed, and keep everything traceable to goals. That is how TOGAF is intended to be used (pp. 808, 815).

Chapter 21: The TOGAF Standard on a Page

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 21 (pp. 817-822)

Overview

Chapter 21 (pp. 817-822) is a one-picture summary of how TOGAF's core concepts connect: the Governance Framework, Architecture Repository, ADM, Enterprise Metamodel, and Content Framework, plus building blocks, artifacts, domains, and roles (p. 817). Section 21.1 lists 13 numbered relationships behind Figure 21-1, "The TOGAF Standard on a page" (pp. 817-822). Leave with this idea: TOGAF "is not a random collection of components but rather the result of careful design". Each concept feeds or depends on another (p. 822). Chapter 21 is part of the GOLD reading tour (p. 817).

> **Reading note:** Figure 21-1 and the numbering of the 13 relationships are Jager's synthesis, built from concepts the Standard defines (covered in Chapters 5-12). The figure is simplified. "The TOGAF Standard is much more complex than shown" (p. 822). This chapter is the best quick revision sheet in the book. Pair it with the separate "Connecting the Concepts" guide.

Detailed Section Breakdown

Chapter Opener (p. 817)

- The key concepts are the Governance Framework, Architecture Repository, ADM, Enterprise Metamodel, and Content Framework. They are closely linked, form the foundation of the Standard, and enable structured development with mutual dependencies (p. 817).

21.1 Connecting the Concepts (pp. 817-822)

#	Relationship	Meaning	Where covered
1	Enterprise Metamodel is created during execution of the ADM	A tailored metamodel with all relevant entities is built in the Preliminary Phase	Ch. 5.2, 12.3
2	Enterprise Metamodel specifies entities to be used as building blocks	Metamodel entities are the basis of building blocks	Ch. 5, 7.5.2
3	Enterprise Metamodel realizes the Content Framework	The metamodel is the foundation of the (also tailored) Content Framework, whose categories structure the metamodel	Ch. 5.2, 6
4	Content Framework complements the ADM	Running the ADM may add or remove artifacts in the Content Framework	Ch. 6.4
5	ADM develops content for architecture domains	Phases B, C, D develop the four core domains	Ch. 11, 12.5-12.7
6	Architecture domains need corresponding architecture roles	A specialist per domain: the Segment Architect	Ch. 10, 11.3
7	Governance Framework governs architecture roles	Roles are governed by a central Architecture Governance Framework	Ch. 8.1
8	Governance Framework enables the use of the ADM	The repository, as part of governance, holds the processes needed for architecture activities	Ch. 8, 7
9	Governance Framework enables the Architecture Repository	Governance oversees management of the repository, where artifacts and deliverables are stored	Ch. 7, 8

#	Relationship	Meaning	Where covered
10	Architecture Repository stores artifacts and building blocks	Developed artifacts and reusable building blocks live in the repository	Ch. 7.2, 7.5
11	Architecture Repository contains the Enterprise Metamodel	The tailored metamodel is stored there (Architecture Metamodel)	Ch. 7.2
12	Artifacts are created out of building blocks	Artifacts use reusable building blocks	Ch. 7.4-7.5
13	Building blocks are used to create catalogs, diagrams, maps, and matrices	These forms are made from repository building blocks	Ch. 7.4

(pp. 818-820, Figure 21-1 on p. 821)

- **A simplified view.** It shows the essential interrelationships clearly, demonstrating careful design (p. 822).

Key Terms & Definitions

Term	Definition	Page
Key components (core concepts)	Governance Framework, Architecture Repository, ADM, Enterprise Metamodel, Content Framework	p. 817
TOGAF on a page	Figure 21-1: the core concepts and their 13 relationships	p. 821

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 21-1	The TOGAF Standard on a page	p. 821

Self-Check Questions

Q: Name the five key components the chapter connects.

A: Governance Framework, Architecture Repository, Architecture Development Method, Enterprise Metamodel, Content Framework (p. 817).

Q: When is the tailored Enterprise Metamodel created, and where is it stored?

A: During the Preliminary Phase of the ADM. It is stored in the Architecture Repository (pp. 818, 820).

Q: How do the Enterprise Metamodel and the Content Framework relate?

A: The metamodel realizes, i.e., is the foundation of, the Content Framework. The Content Framework's categories structure a representation of the metamodel (p. 818).

Q: How can the ADM change the Content Framework?

A: Running the ADM may require new artifacts, adding to or removing parts of the Content Framework (p. 818).

Q: Which ADM phases develop the four domains, and which role pairs with each domain?

A: Phases B, C, D. Each domain needs a Segment Architect specialization (p. 819).

Q: Complete the chain: entities, then building blocks, then...?

A: Building blocks are used to create catalogs, diagrams, maps, and matrices, which form artifacts. Artifacts and blocks are stored in the repository (pp. 818-820).

Q: What two things does the Governance Framework enable or govern?

A: It governs the architecture roles, and it enables the use of the ADM and the management of the Architecture Repository (p. 819).

Chapter 22: Closing Remarks

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 22 (pp. 823-829)

Overview

Chapter 22 (pp. 823-829) closes the book by restating Jager's purpose and dispelling **seven common myths** about the TOGAF Standard (pp. 823-828). The chapter has no numbered sections. Leave with this idea: most criticism of TOGAF comes from misunderstanding it, from not reading it carefully, from outdated exam knowledge, or from hearsay. The Standard is a flexible, tailorable, business-wide EA framework, not a rigid IT documentation process (p. 828). Chapter 22 is part of the PINK reading tour (p. 823).

> **Reading note:** The myths and rebuttals are Jager's. He supports them with quotations from the Standard: the "essential scaffolding" quote (p. 825), the generic-deliverables quote (p. 826), and "The Architecture Development Method should not be understood as a process model" (p. 828). The closing reprises his four reasons for writing from Chapter 1. Small inconsistency: p. 828 mentions "four different types of iteration", while Chapter 13 describes three *ways* to iterate and four iteration *groups*.

Detailed Section Breakdown

Opening (p. 823)

- TOGAF is still highly relevant and complete, with methods and techniques for an effective approach. Yet many practitioners don't apply it as intended, because of the volume of information and its sometimes vague or academic explanation. This creates room for myths (p. 823).

The Seven Myths (pp. 824-828)

Myth	Jager's rebuttal	Links	Page
1. TOGAF is only for large organizations	Frameworks are size-independent, scalable, and adaptable. Every organization benefits from translating goals into operational impact	Ch. 17	p. 824
2. TOGAF is rigid and prescriptive	It is highly flexible and meant to be tailored. "The essential scaffolding... is the concepts. Everything else... is either an example or a starter set"	Ch. 17	p. 825
3. TOGAF is only about IT Architecture	Technology is one of four domains (Business, Data, Application, Technology). It is an EA framework with a holistic view. It refers to BIZBOK rather than duplicating it	Ch. 4.2, 11	pp. 825-826
4. TOGAF replaces other frameworks	It doesn't compete with Zachman, SAFe, ITIL, or COBIT. Its generic deliverables "may be replaced or extended by a more specific set" from other frameworks	Ch. 16	pp. 826-827
5. TOGAF is outdated	It keeps evolving (digital transformation, cloud, agile). New editions every few years, while the core stays stable	Ch. 2, 18, 19	p. 827
6. TOGAF is just about documentation	Nowhere does it say so. It needs four key documents with set structures, but not endless documentation. The repository should hold building blocks, not elaborate solution diagrams	Ch. 7.1, 12.2	pp. 827-828

Myth	Jager's rebuttal	Links	Page
7. The ADM must be followed step by step	The graphic shows information flows, not a sequence. "The ADM should not be understood as a process model." It is tailorable and iterative	Ch. 13.3, 12.13	p. 828

- **Why the myths persist.** Not reading the content carefully, relying on outdated knowledge (e.g., from an old TOGAF exam), and accepting hearsay (p. 828).

Closing (pp. 828-829)

- **What the book did.** Refuted the myths with practical examples, presented a practical approach, and covered the most important parts plus extra context (pp. 828-829).
- **Four reasons for writing (restated).** Explain and clarify the methodology; present a practical approach; show how to correctly apply and tailor the framework; and so refute unfounded criticism (p. 829; see Ch. 1).
- **Delivered.** Fundamental concepts explained, actionable steps given, tailoring of the metamodel, Content Framework, and ADM demonstrated, with examples of implementing EA (p. 829).
- **Final message.** Use TOGAF "as it was intended: in a practical and pragmatic way that can be tailored to the situation" (p. 829).

Key Terms & Definitions

Term	Definition	Page
Myth (about TOGAF)	A common misconception that stems from not understanding how the theory is applied	p. 823
Generic deliverables	TOGAF's default deliverables, which may be replaced or extended by other frameworks' specific sets	pp. 826-827

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 22-1	Four core architecture domains	p. 826

Self-Check Questions

Q: List the seven myths about TOGAF.

A: Only for large organizations; rigid and prescriptive; only about IT Architecture; replaces other frameworks; outdated; just about documentation; the ADM must be followed step by step (p. 824).

Q: How does Jager refute "TOGAF is only about IT"?

A: Technology is only one of four core domains. TOGAF covers Business, Data, Application, and Technology, giving a holistic EA view (pp. 825-826).

Q: What should, and should not, be stored in the Architecture Repository according to the documentation myth?

A: Architecture and Solution Building Blocks should be stored, not elaborate solution diagrams, matrices, or catalogs (p. 827).

Q: What does the Standard say about using other frameworks' deliverables?

A: TOGAF may be used in its own right, or its generic deliverables may be replaced or extended by a more specific set from any relevant framework (p. 826).

Q: Why do the myths persist, according to Jager?

A: Careless reading of the framework, outdated knowledge (e.g., from old exams), and accepting hearsay without thinking (p. 828).

Q: What are Jager's four reasons for writing the book?

A: To explain and clarify TOGAF's methodology; to present a practical approach; to show how to correctly apply and tailor it; and thereby to refute unfounded criticism (p. 829).